

EXERCISES

Level I**(Problems based on Fundamentals)**

Find the order and the degree of each of the following differential equations:

1. $\frac{d^2y}{dx^2} + 4y = 0$
2. $\left(\frac{dy}{dx}\right)^2 + \frac{dx}{dy} = 2$
3. $\frac{d^2y}{dx^2} + \left(\frac{dy}{dx}\right)^2 + xy = 0$
4. $\sqrt{1 + \left(\frac{dy}{dx}\right)^2} = \sqrt[3]{c \cdot \frac{d^2y}{dx^2}}$
5. $x + \frac{dy}{dx} = \sqrt{1 + \left(\frac{dy}{dx}\right)^2}$
6. $y = x \frac{dy}{dx} + a \sqrt{1 + \left(\frac{dy}{dx}\right)^2}$
7. $\frac{d^4y}{dx^4} + \sin\left(\frac{d^3y}{dx^3}\right) = 0$
8. $\left(\frac{d^2y}{dx^2}\right)^2 + \left(\frac{dy}{dx}\right)^6 = x \sin\left(\frac{d^2y}{dx^2}\right)$
9. $\frac{dy}{e^{dx}} = (x + 1)$
10. $\sin\left(\frac{dy}{dx}\right) + \cos\left(\frac{dy}{dx}\right) = x$

Formation of a Differential Equation

11. Find the differential equation of all non-vertical lines in a plane.
12. Find the differential equation of the family of parabolas having vertex at the origin and the axis along positive x -axis.
13. Find the differential equation of the family of all circles, whose centre lies on x -axis and touches the y -axis at the origin.
14. Find the differential equation of all parabolas whose axes are parallel to the x -axis and have latus rectum $4a$.
15. Find the differential equation corresponding to the family of curves $y = c(x - c)^2$, where c is an arbitrary constant.
16. Find the differential equation of the system of ellipses $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$, where a and b are arbitrary constants.

17. $y = mx$, where m is a parameter.
18. $y = A \cos x + B \sin x$, where A and B are parameters.
19. $y = ae^x + be^{-x}$, where a and b are parameters.
20. $y = \sin(bx + c)$, b and c being parameters.
21. $y = ae^{2x} + be^{3x}$, where a and b are parameters.
22. Find the differential equation of all circles touching the
 - (a) x -axis at the origin.
 - (b) y -axis at the origin.
23. Obtain the differential equation of all circles of the radius r .
24. Find the differential equation of all the circles in the first quadrant which touch the co-ordinate axes.
25. Find the differential equation of all the circles which pass through the origin and the centre lies on x -axis.
26. Find the differential equation of all non-vertical lines in a plane.
27. Find the differential equation of all the parabolas with the latus rectum $4a$ and whose axes are parallel to the x -axis.
28. Find the differential equation of all the ellipses having foci on the x -axis and the centre at the origin.
29. Find the differential equation of the family of parabolas having vertex at the origin and the axis along positive y -axis.
30. Find the differential equation of the family of curves $y^2 = 2c(x + \sqrt{c})$.
31. Find the order and the degree of the differential equation of all parabolas whose axis of symmetry is parallel to x -axis.

Differential equation of first order and first degree

32. Solve: $\frac{dy}{dx} = \frac{x^2 - 1}{x^2 + 1}$.
33. Solve: $\frac{dy}{dx} = \frac{1}{e^x + 1}$.
34. Solve: $\frac{dy}{dx} = \frac{\cos 2x - \cos x}{1 - \cos x}$.
35. Solve: $\frac{dy}{dx} = \frac{1}{x(x^4 + 1)}$.
36. Solve: $\frac{dy}{dx} = \frac{\sqrt{\sin x}}{\sin x \cos x}$.
37. Solve: $\frac{dy}{dx} = \frac{x^2}{(3 + 2x)^2}$.

38. Solve: $\frac{dy}{dx} = \sqrt{\frac{1-\sqrt{x}}{1+\sqrt{x}}}$.

39. Solve: $\frac{dy}{dx} = (\sqrt{\tan x} + \sqrt{\cot x})$.

40. Solve: $\frac{dy}{dx} = \frac{1}{\sin^3 x + \cos^3 x}$.

41. Solve: $\frac{dy}{dx} = \frac{1}{(\sqrt{\sin x} + \sqrt{\cos x})^4}$.

Variable Separable Form

42. Solve: $\frac{dy}{dx} = 1 + x + y + xy$.

43. Solve: $\frac{dy}{dx} = e^{x-y} + x^2 \cdot e^{-y}$.

44. Solve: $3e^x \tan y dx + (1 - e^x) \sec^2 y dy = 0$.

45. Solve: $\sqrt{1+x^2+y^2+x^2y^2} + xy \frac{dy}{dx} = 0$.

46. Solve: $\frac{dy}{dx} = \frac{\sin y + \cos y}{x(2\log x + 1)}$.

47. Solve: $x \frac{dy}{dx} = a \left(y^2 + \frac{dy}{dx} \right)$.

48. Solve: $xy^2 \frac{dy}{dx} = 1 - x^2 + y^2 - x^2y^2$.

49. Solve: $(x+1) \frac{dy}{dx} = 2xy$.

50. Solve: $\sec^2 x \tan y dx + \sec^2 y \tan x dy = 0$.

51. Solve: $(1 + e^{2x}) dy + (1 + y^2) e^x dx = 0$.

52. Solve: $\frac{dy}{dx} = e^{x+y} + x^2 e^x$.

53. Solve: $y\sqrt{1+x^2} + x\sqrt{1+y^2} \frac{dy}{dx} = 0$.

54. Solve: $\sqrt{1+x^2} dy + \sqrt{1+y^2} dx = 0$.

55. Solve: $\sqrt{1+x^2+y^2+x^2y^2} + xy = 0$.

56. Solve: $xy \frac{dy}{dx} + 1 + x + y + xy = 0$.

57. Solve: $(1+x)(1+y^2) dx + (1+y)(1+x^2) dy = 0$.

58. Solve: $x\sqrt{1-y^2} dx + y\sqrt{1-x^2} dy = 0$.

59. Solve: $\left(y - x \frac{dy}{dx} \right) = a \left(y^2 + \frac{dy}{dx} \right)$.

60. Solve: $xy \frac{dy}{dx} = \frac{1+y^2}{1+x^2} (1+x+x^2)$.

61. Solve: $(x^2 - yx^2) dy + (y^2 + xy^2) dx = 0$.

Reducible to Variable Separable Form

62. Solve: $\frac{dy}{dx} = (x+y+1)^2$.

63. Solve: $\frac{dy}{dx} = \sin(x+y) + \cos(x+y)$.

64. Solve: $(x+y)(dx-dy) = (dx+dy)$.

65. Solve: $\tan y \frac{dy}{dx} = \sin(x+y) + \sin(x-y)$.

66. Solve: $\frac{dy}{dx} - x \tan y (y-x) = 1$.

67. Solve: $\frac{dy}{dx} = \frac{(x-y)+3}{2(x-y)+5}$.

68. Solve: $(x+y)(dx-dy) = dx+dy$.

69. Solve: $\frac{dy}{dx} = \sec(x+y)$.

70. Solve: $\sin^{-1} \left(\frac{dy}{dx} \right) = x+y$.

71. Solve: $\frac{dy}{dx} = \cos(x+y+1)$.

72. Solve: $(x^2 + 2xy + y^2 + 1) \frac{dy}{dx} = 2(x+y)$.

73. Solve: $\frac{dy}{dx} = \sin(10x+6y)$.

74. Solve: $\frac{y}{x} \frac{dy}{dx} + \frac{2(x^2+y^2)-1}{x^2+y^2+1} = 0$.

Homogeneous Differential Equation

75. Solve: $(x^3 + y^3) dy - x^2 y dx = 0$.

76. Solve: $x dy - y dx = \sqrt{x^2 + y^2} dx$.

77. Solve: $(x^2 - y^2) dx = 2xy dy$.

78. Solve: $(1 + 2e^{xy}) dx + 2e^{xy} (1 - x/y) dy = 0$.

79. Solve: $x \left(\frac{dy}{dx} + 1 \right) = y \left(1 - \frac{dy}{dx} \right)$.

80. Solve: $x \frac{dy}{dx} = x+y$.

81. Solve: $\frac{dy}{dx} = \frac{y-x}{y+x}$.

82. Solve: $2xy \frac{dy}{dx} = x^2 + y^2$.

83. Solve: $(y^2 - 2xy) dx = (x^2 - 2xy) dy$.

84. Solve: $x^2y dx - (x^3 + y^3) dy = 0$.

85. Solve: $(x^3 - 3xy^2) dx = (y^3 - 3x^2y) dy$.

86. Solve: $\frac{dy}{dx} = \frac{y}{x} - \sqrt{\frac{y^2}{x^2} - 1}$.

87. Solve: $\frac{dy}{dx} = \frac{y}{x} + \sin\left(\frac{y}{x}\right)$.

88. Solve: $x \frac{dy}{dx} = y - x \tan\left(\frac{y}{x}\right)$.

89. Solve: $(1 + e^{\frac{x}{y}}) dx + e^{\frac{x}{y}} \left(1 - \frac{x}{y}\right) dy = 0$.

Reducible to Homogeneous Differential Equation

90. Solve: $\frac{dy}{dx} = \frac{x + 2y + 3}{2x + 3y + 4}$.

91. Solve: $\frac{dy}{dx} = \frac{(x + y)^2}{(x + 2)(y - 2)}$.

93. Solve: $(x + 2y + 3) dx = (2x + 3y + 4) dy$.

94. Solve: $\frac{dy}{dx} = \frac{x + 2y - 5}{2x + y - 4}$.

95. Solve: $\frac{dy}{dx} = \frac{4x + 6y - 5}{6x + 9y + 7}$.

96. Solve: $\frac{dy}{dx} = \frac{6x - 2y - 7}{3x - y + 4}$.

Linear Differential Equation

97. Solve: $y dx - x dy + \ln x dx = 0$.

98. Solve: $x^2 \frac{dy}{dx} - 3xy = 4x^4 + 2x^2$.

99. Solve: $\frac{dy}{dx} = y \tan x - \sin x$.

100. Solve: $x \log x \frac{dy}{dx} + y = \frac{2}{x} \log x$.

101. Solve: $x dx = \left(\frac{x}{y^2} - y\right) dy$.

102. Solve: $(x + 3y + 2) \frac{dy}{dx} = 1$.

103. Solve: $\frac{dy}{dx} + 2y = e^{3x}$.

104. Solve: $x \frac{dy}{dx} = x + y$.

105. Solve: $x \frac{dy}{dx} + y = x e^x$.

106. Solve: $x \frac{dy}{dx} + y = x \log x$.

107. Solve: $\frac{dy}{dx} + \frac{y}{x} = x^3$.

108. Solve: $\frac{dy}{dx} = y \tan x - 2 \sin x$.

109. Solve: $\frac{dy}{dx} - \frac{y}{x} = 2x^2$.

110. Solve: $x \log x \frac{dy}{dx} + y = \frac{2}{x} \log x$.

111. Solve: $y dx - (x + 2y^2) dy = 0$.

112. Solve: $y dx + (x - y^3) dy = 0$.

113. Solve: $x \frac{dy}{dx} - ay = x + 1$.

114. Solve: $(x + 1) \frac{dy}{dx} - ny = e^x (x + 1)^{n+1}$.

115. Solve: $(1 + y^2) dx = (\tan^{-1} y - x) dx$.

116. Solve: $(x + 3y + 2) \frac{dy}{dx} = 1$.

117. Solve: $(1 + y^2) dx = (xy + y^3 + y) dy$.

Bernoulli's Differential Equation

118. Solve: $\frac{dy}{dx} + \frac{y}{x} \log x = \frac{y}{x^2} (\log y)^2$.

119. Solve: $\frac{dy}{dx} + xy = x^2 y^6$.

120. Solve: $\frac{dy}{dx} + \frac{\sin^2 y}{x} = x^3 \cos^2 y$.

121. Solve: $\frac{dy}{dx} - x^3 y^2 + xy = 0$.

122. Solve: $(1 - x^2) \frac{dy}{dx} + xy = xy^2$.

123. Solve: $x \frac{dy}{dx} + y = x^3 y^6$.

124. Solve: $(1 + x^2) \frac{dy}{dx} + y = e^{\tan^{-1} x}$.

125. Solve: $\frac{dy}{dx} (x^2 y^3 + xy) = 1$.

126. Solve: $(y \log x - 1) y dx = x dy$.

127. Solve: $\frac{dy}{dx} = x^2 y^3 - xy$.

128. Solve: $\frac{dy}{dx} + \frac{xy}{1 - x^2} = x \sqrt{y}$.

129. Solve: $\frac{dy}{dx} + \frac{y}{x} \cdot \log y = \frac{y}{x^2} (\log y)^2$.

130. Solve: $\frac{dy}{dx} + \frac{1}{x} \cdot \sin 2y = x^3 \cdot \cos 2y$.

131. Solve: $(xy^2 - e^{1/x^2})dx - xy^2dy = 0$.

132. Solve: $\frac{dy}{dx} + x(x+y) = x^3(x+y)^3 - 1$.

Exact Differential Equation

133. Solve: $x dx + y dy = x dy - y dx$.

134. Solve: $x dy - y dx = x^4 dx$.

135. Solve: $x dy + y dx = \sin y dy$.

136. Solve: $x dy + y dx + y^2(x dy - y dx) = 0$.

137. Solve: $x dy + y dx + xy^2 dx - x^2 y dy = 0$.

138. Solve: $(4x - 3y)dx + (2y - 3x)dy = 0$.

139. Solve: $\left(\sin y + y \sin x + \frac{1}{x}\right)dx + \left(x \cos y - \cos x + \frac{1}{y}\right)dy = 0$.

140. Solve: $\frac{x dx + y dy}{\sqrt{x^2 + y^2}} = \frac{y dx - x dy}{x^2}$.

141. Solve: $\left(\frac{\sin 2x}{y} + x\right)dx + \left(y - \frac{\sin^2 x}{y^2}\right)dy = 0$.

142. Solve: $\frac{x + y \frac{dy}{dx}}{y - x \frac{dy}{dx}} = x^2 + 2y^2 + \frac{y^4}{x^2}$.

143. Solve: $x + y \frac{dy}{dx} = \frac{a^2 \left(x \frac{dy}{dx} - y\right)}{x^2 + y^2}$.

144. Solve: $x dx + y dy = (x^2 + y^2)y dy$.

145. Solve: $(x + y)dx + (x - y)dy = 0$.

146. Solve: $y dx + x(x - 1)dy = 0$.

147. Solve: $y dx + x(1 - xy)dy = 0$.

148. Solve: $(x + y)(dx - dy) = (dx + dy)$.

149. Solve: $dx + dy = x dy + y dx$.

150. Solve: $x dy - y dx = (x^2 + y^2)dx$.

151. Solve: $\left(\frac{1}{x} + \frac{1}{y}\right)(x dy + y dx) = dx + dy$.

152. Solve: $(x dy + y dx)\sqrt{x^2 + y^2} = x^2 y dx + xy^2 dy$.

153. Solve: $(x - x^3)dy = y(dx + x^2 dy)$.

154. Solve: $(x + 2y)dy + y dx = 0$.

155. Solve: $(x + \sin y)dy + y dx = 0$.

156. Solve: $e^y dx + (xe^y - 2y)dy = 0$.

157. Solve: $2y dy + (\cos x \cdot \cot y - y^2)dx$.

$= (2xy + \sin x \cdot \operatorname{cosec}^2 y)$

158. Solve: $\frac{dy}{dx} = \left(\frac{2x - 3y + 1}{3x - 2y - 2}\right)$.

159. Solve: $x(dy + dx) = y(dx - dy)$.

160. Solve: $x(\tan^{-1} x - \log x)dy + y dx = \left(\frac{xy}{1 + x^2}\right)dx$.

161. Solve: $x(dy - x dx) + y dx = 0$.

162. Solve: $\frac{y - x \frac{dy}{dx}}{y + x \frac{dy}{dx}} = \frac{1}{x^2} + \frac{1}{y^2}$.

163. Solve: $x^2 y dy - xy^2 dx = x^4 dy + x^3 y dx$.

164. Solve: $dx + x(y dx + x dy) = e^{-ey} dx$.

165. Solve: $y(y^2 dx + (x^2 dy - xy dx)) = e^{-\frac{x}{y}} dy$.

166. Solve: $y dx - x dy + xy^2 dx = 0$.

167. Solve: $2\left(x - y \frac{dy}{dx}\right)(x^2 + y^2) = (x^2 - y^2)\left(y - x \frac{dy}{dx}\right)$.

168. Solve: $(x^2 + y^2 + a^2)y dy = (x^2 + y^2 - a^2)x dx$.

169. Solve: $(1 + xy)y dx + x(1 - xy)dy = 0$.

170. Solve: $\frac{x dy}{x^2 + y^2} = \left(\frac{y}{x^2 + y^2} - 1\right)dx$.

171. Solve: $\frac{x + y \frac{dy}{dx}}{y - x \frac{dy}{dx}} = x^2 + 2y^2 + \frac{y^4}{x^2}$.

172. Solve: $x dx + y dy = m(x dy - y dx)$.

173. Solve: $\frac{x + y \frac{dy}{dx}}{y - x \frac{dy}{dx}} = \frac{x \sin^2(x^2 + y^2)}{y^3}$.

174. Solve: $x + y \frac{dy}{dx} = \frac{a^2 \left(x \frac{dy}{dx} - y\right)}{x^2 + y^2}$.

175. Solve: $\frac{x dx + y dy}{x dy - y dx} = \sqrt{\frac{a^2 - x^2 - y^2}{x^2 + y^2}}$.

176. Solve: $\frac{x dx - y dy}{x dy - y dx} = \sqrt{\frac{1 + x^2 - y^2}{x^2 - y^2}}$.

177. Solve: $\sin\left(\frac{x}{y}\right)(y dx - x dy) = xy^2(x dy + y dx)$.

178. Solve: $x dy + y dx + y^2(x dy - y dx) = 0$.

179. Solve: $(4x^3 + e^x \sin y)dx + e^x \cos y dy = 0$.

Orthogonal Trajectories

180. Find the orthogonal trajectories of the family of the straight lines which are passing through the origin.
181. Determine the 45° trajectories of the family of concentric circles $x^2 + y^2 = a^2$.
182. Find the orthogonal trajectories of the family of the curves $ax^2 + y^2 = 1$.
183. Find the orthogonal trajectories of the circles $x^2 + y^2 - ay = 0$, where a is a parameter.
184. Find the orthogonal trajectories of the family of parabolas $y^2 = 4ax$, where a is a parameter.
185. Find the orthogonal trajectories of the family of rectangular hyperbola $xy = c^2$.
186. Find the orthogonal trajectories of the family of curves $x^2 - y^2 = c^2$.

First Order Higher Degree Differential Equation. Equation Solvable for p

187. Solve: $(x + y + p)(2x + p) = 0$.
188. Solve: $p^2 - p(e^x + e^{-x}) + 1 = 0$.
189. Solve: $p^2 + 2p \cot x = y^2$.
190. Solve: $p^2 - px - xy - y^2 = 0$.
191. Solve: $p^2 y + (x - y)p - x = 0$.
192. Solve: $(p^2 - 1)xy = (x^2 - y^2)p$.
193. Solve: $xy p^2 - (x^2 + y^2)p + xy = 0$.

Equation Solvable for x

194. Solve: $px - yp^2 = ap$.
195. Solve: $y = 2px + y^2 p^3$.
196. Solve: $y^2 \log y = xyp + p^2$.
197. Solve: $p^2 y + 2px = y$.
198. Solve: $p^3 - 4xyp + 8p^2 = 0$.
199. Solve: $yp = 2p^2 x + y^2 p^4$.
200. Solve: $xp^2 - yp - p + 1 = 0$.

Equations Solvable for y

201. Solve: $y = (1 + p)x + ap^2$.
202. Solve: $y = yp^2 + 2px$.
203. $y = 2px + \tan^{-1}(xp^2)$.
204. $x^2 p^4 + 2xp = y$.
205. $y + px = x^4 p^2$.
206. $y = p \sin p + \cos p$.

Clairaut Differential Equation

207. Solve: $(y + 1)P - xP^2 + 2 = 0$.
208. Solve: $P^3 x - P^2 y - 1 = 0$.

209. Solve: $(y + 1)P - xP^2 + 2 = 0$.

210. Solve: $\sin y \cdot \cos Px - \cos y \cdot \sin Px - P = 0$.

211. Solve: $(x - a)P^2 + (x - y)P - y = 0$.

212. Solve: $y = px + a\sqrt{1 + p^2}$.

213. Solve: $y = P(x - b) + \frac{a}{P}$.

Higher Order Differential Equation

Differential equation is the form $\frac{d^2 y}{dx^2} = f(x)$

214. Solve: $\frac{d^2 y}{dx^2} = x + \sin x$.

215. Solve: $\frac{d^2 y}{dx^2} = e^{2x} + e^x + 2014$.

216. Solve: $\frac{d^2 y}{dx^2} = \sin^2 x$.

217. Solve: $\frac{d^2 y}{dx^2} = \cos^3 x$.

218. Solve: $\frac{d^2 y}{dx^2} = \frac{1}{\sin^2 x \cos^2 x}$.

219. Solve: $\frac{d^2 y}{dx^2} = \sin^4 x + \cos^4 x$.

220. Solve: $\frac{d^2 y}{dx^2} = xe^x$.

Differential equation of the form $\frac{d^2 y}{dx^2} = f(y)$

221. Solve: $\frac{d^2 y}{dx^2} + y = 0$.

222. Solve: $\frac{d^2 y}{dx^2} = \frac{1}{y^3}$.

223. Solve: $\frac{d^2 y}{dx^2} = \frac{1}{4\sqrt{y}}$.

224. Solve: $a^2 \frac{d^2 y}{dx^2} - y = 0$.

225. Solve: $\frac{d^2 y}{dx^2} = e^{2y}$.

226. Solve: $2 \frac{d^2 y}{dx^2} = 3y^2$,

$y(-2) = -1, \left(\frac{dy}{dx}\right)_{x=-2} = -1,$

227. Solve: $y^3 \left(\frac{d^2y}{dx^2} \right) = -1$,

$$y(1) = -1, \left(\frac{dy}{dx} \right)_{x=-1} = 0.$$

Differential equation is the form $\phi \left(\frac{d^2y}{dx^2}, \frac{dy}{dx} \right) = 0$

228. Solve: $x \frac{d^2y}{dx^2} + \frac{dy}{dx} + x = 0$.

229. Solve: $\frac{d^2y}{dx^2} = \frac{1}{x} \frac{dy}{dx} + x$.

230. Solve: $y \frac{d^2y}{dx^2} + \left(\frac{dy}{dx} \right)^2 = 1$.

231. Solve: $y \frac{d^2y}{dx^2} - \left(\frac{dy}{dx} \right)^2 = y^2 \left(\frac{dy}{dx} \right)$.

232. Solve: $(x + a) \frac{d^2y}{dx^2} + x \left(\frac{dy}{dx} \right)^2 = \left(\frac{dy}{dx} \right)$.

233. Solve: $x \frac{d^2y}{dx^2} - \frac{1}{4} \left(\frac{dy}{dx} \right)^2 = \left(\frac{dy}{dx} \right)$.

234. Solve: $y \frac{d^2y}{dx^2} = 1 - \left(\frac{dy}{dx} \right)^2$.

235. Solve: $\frac{d^2y}{dx^2} = 2y \left(\frac{dy}{dx} \right)$.

236. Solve: $y \frac{d^2y}{dx^2} - \left(\frac{dy}{dx} \right)^2 = y^2 \left(\frac{dy}{dx} \right)$.

237. Solve: $\frac{d^2y}{dx^2} + \frac{1}{x} \left(\frac{dy}{dx} \right) - \frac{a}{x^2} = 0$.

238. Solve: $y \frac{d^2y}{dx^2} - y \left(\frac{dy}{dx} \right) \ln y = \left(\frac{dy}{dx} \right)^2$.

Geometrical Applications

239. The slope of the curve at any point is the reciprocal of twice the ordinate at the point. The curve passes through the point (4, 3).

Find the equation of the curve.

240. Find the equation of the curve whose slope at any point is $y + 2x$ and which passes through the origin.

241. Find the equation of the curve which passes through the origin and the tangent to which at every point (x, y) has slope equal to

$$\frac{x^4 + 2xy - 1}{x^2 + 1}.$$

242. A curve passes through (2, 0) and the slope of the tangent at $P(x, y)$ is $\frac{(x+1)^2 + y - 3}{(x+1)}$.

Find the equation of the curve.

243. A curve C has the property that if the tangent drawn at any point P on C meets the co-ordinate axes at A and B , P is the mid-point of AB . The curve passes through the point (1, 1). Determine the equation of the curve.

244. A curve passing through the point (1, 1) has the property that the perpendicular distance of the origin from the normal at any point P of the curve is equal to the distance of P from the x -axis. Determine the equation of the curve.

245. If the length of the tangent at any point on the curve $y = f(x)$ intercepted between the point and the x -axis is of length 1. Find the equation of the curve.

246. At any point (x, y) of a curve, the slope of the tangent is twice the slope of the line segment joining the point of contact to the point (-4, -3). Find the equation of the curve, given that it passes through (-2, 1).

247. The ordinate and the normal at any point P on the curve meet the x -axis at points A and B respectively. Find the equation of the family of the curves satisfying the condition AB is the AM of abscissa and the ordinate of P .

248. Find the nature of the curve for which the length of the normal at any point P is equal to the radius vector.

8. We have,

$$\left(\frac{d^2y}{dx^2}\right)^2 + \left(\frac{dy}{dx}\right)^2 = x \sin\left(\frac{d^2y}{dx^2}\right)$$

So, the order is 2 but the degree is not defined.

9. We have,

$$\frac{dy}{e^{dx}} = (x + 1)$$

$$1 + \left(\frac{dy}{dx}\right) + \frac{1}{2!} \left(\frac{dy}{dx}\right)^2 + \frac{1}{3!} \left(\frac{dy}{dx}\right)^3 + \dots$$

So, the order is 1 and the degree is not defined.

10. We have,

$$\begin{aligned} & \sin\left(\frac{dy}{dx}\right) + \cos\left(\frac{dy}{dx}\right) \\ &= x \left(\left(\frac{dy}{dx}\right) - \frac{1}{3!} \left(\frac{dy}{dx}\right)^3 + \frac{1}{5!} \left(\frac{dy}{dx}\right)^5 - \dots \right) \\ &+ \left(1 - \frac{1}{2!} \left(\frac{dy}{dx}\right)^2 + \frac{1}{4!} \left(\frac{dy}{dx}\right)^4 - \frac{1}{6!} \left(\frac{dy}{dx}\right)^6 + \dots \right) = x \end{aligned}$$

So, the order is 1 and degree is not defined.

11. Let the equation of the family of non-vertical lines in a plane is

$$y = mx + c,$$

where m and c are arbitrary constants.

$$\text{Now } \frac{dy}{dx} = m \text{ and } \frac{d^2y}{dx^2} = 0.$$

From the above three equations, we should eliminate m .

Thus, the required differential equation is

$$\frac{d^2y}{dx^2} = 0.$$

12. Let the equation of the parabola be

$$y^2 = 4ax.$$

where a is an arbitrary constant.

Differentiating w.r.t. x , we get

$$2y \frac{dy}{dx} = 4a$$

$$\Rightarrow \frac{y}{2} \frac{dy}{dx} = a$$

Thus, the required differential equation is

$$y^2 = 2xy \frac{dy}{dx}.$$

13. The equation of the circle be

$$(x - a)^2 + y^2 = a^2,$$

where a is an arbitrary constant.

$$\Rightarrow x^2 + y^2 - 2ax = 0$$

$$\Rightarrow 2x + 2y \frac{dy}{dx} - 2a = 0$$

$$\Rightarrow x + y \frac{dy}{dx} = a.$$

Thus, the required differential equation is

$$\left(y \frac{dy}{dx}\right)^2 + y^2 = \left(x + y \frac{dy}{dx}\right)^2$$

$$\Rightarrow 2xy \frac{dy}{dx} = y^2 - x^2.$$

14. The equation of all parabolas whose axes parallel to x -axis is

$$(y - k)^2 = 4a(x - h)$$

Here, h and k are two arbitrary constants.

Differentiating both sides, we get

$$(y - k) \frac{dy}{dx} = 2a \quad \dots(i)$$

Differentiating again w.r.t. x , we get

$$(y - k) \frac{d^2y}{dx^2} + \left(\frac{dy}{dx}\right)^2 = 0 \quad \dots(ii)$$

Eliminating k from Eq. (i) and (ii), we get

$$2a \left(\frac{d^2y}{dx^2}\right) + \left(\frac{dy}{dx}\right)^3 = 0$$

which is the required differential equation.

15. The given curve is

$$y = c(x - c)^2 \quad \dots(i)$$

Differentiating w.r.t. x , we get

$$\frac{dy}{dx} = 2c(x - c) \quad \dots(ii)$$

$$\text{From Eq. (i) and (ii), we get, } c = \frac{1}{4y} \left(\frac{dy}{dx}\right)^2$$

Put the value of c in Eq. (ii), we get

$$\frac{dy}{dx} = 2 \cdot \frac{1}{4y} \left(\frac{dy}{dx}\right)^2 \left(x - \frac{1}{4y} \left(\frac{dy}{dx}\right)^2\right)$$

$$\Rightarrow 2y = \frac{dy}{dx} \left(x - \frac{1}{4y} \left(\frac{dy}{dx}\right)^2\right)$$

$$\Rightarrow 8y^2 = 4xy \frac{dy}{dx} - \left(\frac{dy}{dx}\right)^3$$

which is the required differential equation.

16. The given curve is

$$\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1 \quad \dots(i)$$

Differentiating w.r.t. x , we get

$$\frac{2x}{a^2} + \frac{2yy''}{b^2} = 0$$

$$\Rightarrow \frac{x}{a^2} + \frac{yy'}{b^2} = 0$$

Again differentiating w.r.t. x , we get

$$\begin{aligned} \frac{1}{a^2} + \frac{yy''}{b^2} + \frac{(y')^2}{b^2} &= 0 \\ \Rightarrow \frac{1}{a^2} + \frac{xyy''}{b^2} + \frac{x(y')^2}{b^2} &= 0 \\ \Rightarrow \frac{yy'}{b^2} + \frac{xyy''}{b^2} + \frac{x(y')^2}{b^2} &= 0 \\ \Rightarrow xy \frac{d^2y}{dx^2} + x \left(\frac{dy}{dx} \right)^2 - y \left(\frac{dy}{dx} \right) &= 0 \end{aligned}$$

which is the required differential equation.

17. Given curve is

$$\begin{aligned} y &= mx \\ \Rightarrow \frac{dy}{dx} &= m \\ &= \frac{y}{x} \end{aligned}$$

which is the required differential equation.

18. Given curve is

$$\begin{aligned} y &= A \cos x + B \sin x \\ \Rightarrow \frac{dy}{dx} &= -A \sin x + B \cos x \\ \Rightarrow \frac{d^2y}{dx^2} &= -A \cos x - B \sin x = -y \end{aligned}$$

Hence, the required differential equation is

$$\frac{d^2y}{dx^2} + y = 0$$

19. We have,

$$\begin{aligned} y &= ae^x + be^{-x} \\ \Rightarrow \frac{dy}{dx} &= ae^x - be^{-x} \\ \Rightarrow \frac{d^2y}{dx^2} &= ae^x + be^{-x} = y \end{aligned}$$

Hence, the required differential equation is

$$\frac{d^2y}{dx^2} - y = 0$$

20. We have,

$$\begin{aligned} y &= \sin(bx + c) \\ \Rightarrow y' &= b \cos(bx + c) \\ \Rightarrow y'' &= -b^2 \sin(bx + c) \\ &= -b^2 y \\ &= - \left(\frac{y'}{\sqrt{1-y^2}} \right)^2 y \end{aligned}$$

$$\Rightarrow y''(1-y^2) + y(y')^2 = 0$$

$$\Rightarrow (1-y^2) \frac{d^2y}{dx^2} + y \left(\frac{dy}{dx} \right)^2 = 0$$

which is the required differential equation.

21. We have,

$$\begin{aligned} y &= ae^{2x} + be^{3x} \\ \Rightarrow \frac{dy}{dx} &= 2ae^{2x} + 3be^{3x} \\ \Rightarrow \frac{d^2y}{dx^2} &= 4ae^{2x} + 9be^{3x} \\ &= 2(2ae^{2x} + 3be^{3x}) + 3be^{3x} \end{aligned}$$

$$\begin{aligned} \frac{d^2y}{dx^2} &= 2 \frac{dy}{dx} + 3be^{3x} \\ &= 2 \frac{dy}{dx} + 3 \left(\frac{dy}{dx} - 2y \right) \end{aligned}$$

$$\Rightarrow 1 + \left(\frac{dy}{dx} \right)^2 + (y-a) \left(\frac{d^2y}{dx^2} \right) = 0$$

$$\Rightarrow \frac{d^2y}{dx^2} = 5 \frac{dy}{dx} - 6y$$

$$\Rightarrow \frac{d^2y}{dx^2} - 5 \frac{dy}{dx} + 6y = 0$$

22. Let the equation of the circle be

$$\begin{aligned} x^2 + (y-a)^2 &= a^2 \\ \Rightarrow x^2 + y^2 - 2ay &= 0 \\ \Rightarrow 2x + 2y \frac{dy}{dx} - 2a \frac{dy}{dx} &= 0 \\ \Rightarrow x + y - \frac{dy}{dx} &= a \frac{dy}{dx} \\ \Rightarrow a = \frac{x + y \frac{dy}{dx}}{\frac{dy}{dx}} &= \frac{x + yy_1}{y_1} \end{aligned}$$

Hence, the required differential equation is

$$\begin{aligned} \Rightarrow x^2 + y^2 &= 2 \left(\frac{x + yy_1}{y_1} \right) y \\ \Rightarrow \frac{dy}{dx} (x^2 + y^2 - 2y^2) &= 2xy \\ \Rightarrow \frac{dy}{dx} &= \frac{2xy}{(x^2 - y^2)} \end{aligned}$$

(b) Do yourself.

23. Let the equation of the circle be

$$(x - h)^2 + (y - k)^2 = r^2$$

where h and k are parameters but not r .

$$\Rightarrow 2(x - h) + 2(y - k) \frac{dy}{dx} = 0$$

$$\Rightarrow (x - h) + (y - k) \frac{dy}{dx} = 0$$

$$\Rightarrow 1 + \left(\frac{dy}{dx}\right)^2 + (y - k) \frac{d^2y}{dx^2} = 0$$

24. Let the equation of the circle be

$$(x - a)^2 + (y - a)^2 = a^2$$

$$\Rightarrow x^2 + y^2 - 2a(x + y) + a^2 = 0$$

$$\Rightarrow 2x + 2y - 2a(1 + y_1) = 0$$

$$\Rightarrow x + y - a(1 + y_1) = 0$$

$$\Rightarrow a = \frac{x + y}{1 + y_1}$$

Hence, the required differential equation is

$$x^2 + y^2 - 2\left(\frac{x + y}{1 + y_1}\right)(x + y) + \left(\frac{x + y}{1 + y_1}\right)^2 = 0$$

$$\Rightarrow (1 + y_1)^2(x^2 + y^2) - 2(x + y)^2(1 + y_1) + (x + y)^2 = 0$$

25. Do yourself

26. Let the line be

$$y = mx + c$$

$$\Rightarrow \frac{dy}{dx} = m$$

$$\Rightarrow \frac{d^2y}{dx^2} = 0$$

which is the required differential equation.

27. Let the equation of the parabola be

$$(y - k)^2 = 4a(x - h),$$

where h and k are parameters.

$$\Rightarrow 2(y - k) \frac{dy}{dx} = 4a$$

$$\Rightarrow (y - k) \frac{dy}{dx} = 2a$$

$$\Rightarrow (y - k) \frac{d^2y}{dx^2} + \left(\frac{dy}{dx}\right)^2 = 0$$

$$\Rightarrow 2a \frac{d^2y}{dx^2} + \left(\frac{dy}{dx}\right)^3 = 0$$

which is the required differential equation

28. Let the equation of the ellipse be

$$\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$$

$$\Rightarrow \frac{2x}{a^2} + \frac{2y}{b^2} \frac{dy}{dx} = 0$$

$$\Rightarrow \frac{x}{a^2} + \frac{y}{b^2} \frac{dy}{dx} = 0$$

$$\Rightarrow \frac{1}{a^2} + \frac{1}{b^2} \left(\frac{dy}{dx}\right)^2 + \frac{y}{b^2} \frac{d^2y}{dx^2} = 0$$

$$\Rightarrow -\frac{y}{b^2x} \frac{dy}{dx} + \frac{1}{b^2} \left(\frac{dy}{dx}\right)^2 + \frac{y}{b^2} \frac{d^2y}{dx^2} = 0$$

$$\Rightarrow -\frac{y}{x} \frac{dy}{dx} + \left(\frac{dy}{dx}\right)^2 + y \frac{d^2y}{dx^2} = 0$$

which is the required differential equation.

29. Let the equation of the parabola be

$$(x - h)^2 = 4a(y - k)$$

where h and k are parameters

$$\Rightarrow 2(x - h) = 4a \frac{dy}{dx}$$

$$\Rightarrow (x - h) = 2a \frac{dy}{dx}$$

$$\Rightarrow 1 = 2a \frac{d^2y}{dx^2}$$

$$\Rightarrow \frac{d^2y}{dx^2} = \frac{1}{2a}$$

which is the required differential equation.

30. Given curve is

$$y^2 = 2c(x + \sqrt{c})$$

$$\Rightarrow 2y \frac{dy}{dx} = 2c$$

$$\Rightarrow y \frac{dy}{dx} = c$$

$$\Rightarrow y \frac{d^2y}{dx^2} + \left(\frac{dy}{dx}\right)^2 = 0$$

which is the required differential equation.

31. Do yourself.

32. Given differential equation is

$$dy = \frac{x^2 - 1}{x^2 + 1} dx.$$

Integrating, we get

$$\int dy = \int \frac{x^2 - 1}{x^2 + 1} dx$$

$$\Rightarrow y = \int \left(\frac{1 - 2}{x^2 + 1} \right) dx$$

$$= x - 2 \tan^{-1} x + c$$

which is the required solution.

33. Given differential equation is

$$dy = \frac{1}{e^x + 1} dx$$

Integrating, we get

$$\int dy = \frac{1}{e^x + 1} dx$$

$$\Rightarrow y = \int \left((e^x + 1) - \frac{e^x}{e^x + 1} + 1 \right) dx$$

$$\Rightarrow y = \int \left(\frac{1 - e^x}{1 + e^x} \right) dx$$

$$= x - \log|e^x + 1| + c,$$

which is the required solution.

34. The given differential equation is

$$\frac{dy}{dx} = \frac{\cos 2x - \cos x}{1 - \cos x}$$

$$\Rightarrow dy = \left(\frac{\cos 2x - \cos x}{1 - \cos x} \right) dx$$

Integrating, we get

$$\int dy = \int \frac{\cos 2x - \cos x}{1 - \cos x} dx$$

$$\Rightarrow y = - \int \frac{(2\cos x + 1)(1 - \cos x)}{(1 - \cos x)} dx$$

$$= - \int (2\cos x + 1) dx$$

$$= -(2\sin x + x) + c$$

which is the required solution of the given differential equation.

35. The given differential equation is

$$\frac{dy}{dx} = \frac{1}{x}(x^4 + 1)$$

$$\Rightarrow dy = \frac{dx}{x(x^4 + 1)}$$

Integrating, we get

$$\int dy = \int \frac{dx}{x(x^4 + 1)}$$

$$\Rightarrow y = \int \frac{x^3 dx}{x^4(x^4 + 1)}$$

$$= \frac{1}{4} \int \frac{dt}{t(t + 1)}$$

$$= \frac{1}{4} \int \left(\frac{1}{t} - \frac{1}{t + 1} \right) dt$$

$$= \frac{1}{4} \log \left| \frac{t}{t + 1} \right| + c$$

$$= \frac{1}{4} \log \left| \frac{x^4}{x^4 + 1} \right| + c$$

which is the required solution.

36. The given differential equation is

$$\frac{dy}{dx} = \frac{\sqrt{\sin x}}{\sin x \cos x} dx$$

$$\Rightarrow dy = \frac{\sqrt{\sin x}}{\sin x \cos x} dx$$

Integrating, we get

$$\int dy = \int \frac{\sqrt{\sin x}}{\sin x \cos x} dx$$

$$\Rightarrow y = \int \frac{\sqrt{\sin x}}{\sin x \cos x} dx$$

$$= \int \frac{\sec^2 x}{\sqrt{\tan x}} dx$$

(Dividing the numerator and the denominator by $\cos^2 x$)

$$\Rightarrow y = 2\sqrt{\tan x} + C$$

which is the general solution.

37. The given differential equation is

$$\frac{dy}{dx} = x^2(3x + 2x)^2$$

$$\Rightarrow dy = \frac{x^2}{(3 + 2x)^2} dx$$

Integrating, we get

$$\int dy = \int \frac{x^2}{(3 + 2x)^2} dx$$

$$\Rightarrow y = \int \frac{x^2 dx}{(3 + 2x)^2}$$

Let $3 + 2x = t$

$$= \frac{1}{2} \int \left(\frac{t-3}{2} \right)^2 \cdot \frac{1}{t^2} dt$$

$$= \frac{1}{8} \int \left(\frac{t-3}{t} \right)^2 dt$$

$$= \frac{1}{8} \int \left(1 - \frac{3}{t} \right)^2 dt$$

$$= \frac{1}{8} \int \left(1 - \frac{6}{t} + \frac{9}{t^2} \right) dt$$

$$= \frac{1}{8} \left(1 - 6 \log t - \frac{1}{t} \right) + c$$

$$= \frac{1}{8} \left((3 + 2x) - 6 \log(3 + 2x) - \frac{1}{(3 + 2x)} \right) + c$$

which is the required differential equation.

38. The given differential equation is

$$\frac{dy}{dx} = \sqrt{\frac{1-\sqrt{x}}{1+\sqrt{x}}}$$

$$\Rightarrow dy = \sqrt{\frac{1-\sqrt{x}}{1+\sqrt{x}}} dx$$

Integrating, we get

$$\int dy = \int \sqrt{\frac{1-\sqrt{x}}{1+\sqrt{x}}} dx$$

$$\Rightarrow y = \int \sqrt{\frac{1-\sqrt{x}}{1+\sqrt{x}}} dx$$

Let $x = \cos^2 2\theta$

$$\Rightarrow dx = -2\sin 4\theta d\theta$$

Integrating, we get

$$\begin{aligned} \Rightarrow y &= \int \sqrt{\frac{1-\cos 2\theta}{1+\cos 2\theta}} \times -2\sin 4\theta d\theta \\ &= \int \tan \theta \times -2\sin 4\theta d\theta \\ &= -2 \int \cos 2\theta (1 - \cos 2\theta) d\theta \\ &= -2 \int (2\cos 2\theta - 1 - \cos 4\theta) d\theta \\ &= -2 \left(\sin 2\theta - \theta - \frac{\sin 4\theta}{4} \right) + c \\ &= (-2\sin 2\theta + 2\theta + \sin 2\theta \cos 2\theta) + c \\ &= -2\sqrt{1-x} + \cos^{-1} \sqrt{x} + \sqrt{x}(1-x) + c \end{aligned}$$

which is the required solution of the given differential equation.

39. The given differential equation is

$$\frac{dy}{dx} = (\sqrt{\tan x} + \sqrt{\cot x})$$

$$\Rightarrow dy = (\sqrt{\tan x} + \sqrt{\cot x}) dx$$

Integrating, we get

$$\begin{aligned} \int dy &= \int (\sqrt{\tan x} + \sqrt{\cot x}) dx \\ \Rightarrow y &= \int (\sqrt{\tan x} + \sqrt{\cot x}) dx \\ &= \int \left(\frac{\sin x + \cos x}{\sqrt{\sin x \cos x}} \right) dx \\ &= \sqrt{2} \int \left(\frac{\sin x + \cos x}{\sqrt{2\sin x \cos x}} \right) dx \\ &= \sqrt{2} \int \left(\frac{\sin x + \cos x}{\sqrt{\sin 2x}} \right) dx \end{aligned}$$

Let $\sin x - \cos x = t$

$$\Rightarrow (\cos x + \sin x) dx = dt$$

Also, $1 - \sin 2x = t^2$

$$\begin{aligned} \Rightarrow \sin 2x &= 1 - t^2 \\ &= \sqrt{2} \int \frac{dt}{\sqrt{1-t^2}} \\ &= \sqrt{2} \sin^{-1}(t) + c \\ &= \sqrt{2} \sin^{-1}(\sin x - \cos x) + c \end{aligned}$$

which is the required differential equation.

40. The given differential equation is

$$\frac{dy}{dx} = \frac{1}{\sin^3 x + \cos^3 x}$$

$$\Rightarrow dy = \frac{dx}{\sin^3 x + \cos^3 x}$$

Integrating, we get

$$\begin{aligned} \int dy &= \int \frac{dx}{\sin^3 x + \cos^3 x} \\ &= \int \frac{dx}{\sin^3 x + \cos^3 x} \\ &= \int \frac{dx}{(\sin x + \cos x)(1 - \sin x \cos x)} \\ &= \frac{1}{3} \int \left(\frac{2}{(\sin x + \cos x)} + \frac{\sin x + \cos x}{(1 - \sin x \cos x)} \right) dx \\ &= \frac{2}{3} \int \frac{dx}{(\sin x + \cos x)} \\ &\quad + \frac{1}{3} \int \frac{\sin x + \cos x}{(1 - \sin x \cos x)} dx \\ &= \frac{2}{3\sqrt{2}} \int \frac{dx}{\sin \left(x + \frac{\pi}{4} \right)} \\ &\quad + \frac{2}{3} \int \frac{\sin x + \cos x}{(2 - \sin 2x)} dx \\ &= \frac{2}{3\sqrt{2}} \log \left| \frac{x}{2} + \frac{\pi}{8} \right| \\ &\quad + \frac{2}{3} \tan^{-1}(\sin x - \cos x) + c \end{aligned}$$

which is the required solution of the given differential equation.

41. The given differential equation is

$$\frac{dy}{dx} = \frac{1}{(\sqrt{\sin x} + \sqrt{\cos x})^4}$$

$$\Rightarrow dy = \frac{dx}{(\sqrt{\sin x} + \sqrt{\cos x})^4}$$

Integrating, we get

$$\begin{aligned} \int dy &= \int \frac{dx}{(\sqrt{\sin x} + \sqrt{\cos x})^4} \\ \Rightarrow y &= \int \frac{dx}{(\sqrt{\sin x} + \sqrt{\cos x})^4} \\ &= \int \frac{\sec^2 x dx}{(\sqrt{\tan x} + 1)^4} \\ &= \frac{2t dt}{(t+1)^4}, \text{ (Let } \tan x = t^2) \\ &= 2 \int \frac{(t+1) - 1}{(t+1)^4} dt \\ &= 2 \int \left(\frac{1}{(t+1)^3} - \frac{1}{(t+1)^4} \right) dt \\ &= 2 \left(\frac{1}{-2(t+1)^2} + \frac{1}{3(t+1)^3} \right) + c \\ &= 2 \left(\frac{1}{-2(\sqrt{\tan x} + 1)^2} + \frac{1}{3(\sqrt{\tan x} + 1)^3} \right) + c \end{aligned}$$

which is the required solution of the given differential equation.

42. Given differential equation is

$$\int \frac{dy}{(1+y)} = \int (1+x) dx$$

Integrating, we get

$$\begin{aligned} \frac{dy}{(1+y)} &= (1+x) dx \\ \Rightarrow \log|1+y| &= x + \frac{x^2}{2} + c, \end{aligned}$$

which is the required solution.

43. Given differential equation is

$$\begin{aligned} \frac{dy}{dx} &= e^{-y}(e^x + x^2) \\ \Rightarrow \frac{dy}{e^{-y}} &= (e^x + x^2) dx \end{aligned}$$

Integrating, we get

$$\begin{aligned} \int \frac{dy}{e^{-y}} &= \int (e^x + x^2) dx \\ \Rightarrow \int e^y dy &= \int (e^x + x^2) dx \\ \Rightarrow e^y &= e^x + \frac{x^3}{3} + c, \end{aligned}$$

which is the required solution.

44. The given differential equation is

$$3e^x \tan y dx + (1 - e^x) \sec^2 y dy = 0$$

$$\Rightarrow \frac{3e^x}{(1 - e^x)} dx = \frac{\sec^2 y}{\tan y} dy$$

On Integrating, we get

$$\begin{aligned} \int \frac{3e^x}{(1 - e^x)} dx &= \int \frac{\sec^2 y}{\tan y} dy \\ \Rightarrow -3 \log|1 - e^x| &= \log|\tan y| + c \end{aligned}$$

which is the required solution of the given differential equation.

45. The given differential equation is

$$\begin{aligned} \sqrt{1+x^2+y^2+x^2y^2} + xy \frac{dy}{dx} &= 0 \\ \Rightarrow \sqrt{(1+x^2)(1+y^2)} + xy \frac{dy}{dx} &= 0 \\ \Rightarrow \sqrt{1+x^2} \sqrt{1+y^2} &= -xy \frac{dy}{dx} \\ \frac{\sqrt{1+x^2}}{x} dx &= -\frac{y}{\sqrt{1+y^2}} dy \end{aligned}$$

Integrating, we get

$$\begin{aligned} \int \frac{\sqrt{1+x^2}}{x} dx &= -\int \frac{y}{\sqrt{1+y^2}} dy \\ \Rightarrow \sqrt{1+x^2} + \frac{1}{2} \log \left| \frac{\sqrt{1+x^2}-1}{\sqrt{1+x^2}+1} \right| &+ \sqrt{1+y^2} = c \end{aligned}$$

which is the required solution of the given differential equation.

46. The given differential equation is

$$\begin{aligned} \frac{dy}{dx} &= \frac{\sin y + \cos y}{x(2 \log x + 1)} \\ \Rightarrow \frac{dy}{\sin y + \cos y} &= \frac{dx}{x(2 \log x + 1)} \end{aligned}$$

Integrating, we get

$$\begin{aligned} \int \frac{dy}{\sin y + \cos y} &= \int \frac{dx}{x(2 \log x + 1)} \\ \Rightarrow \frac{1}{\sqrt{2}} \int \frac{dy}{\sin\left(y + \frac{\pi}{4}\right)} &= \int \frac{dx}{x(2 \log x + 1)} \\ \Rightarrow \frac{1}{\sqrt{2}} \log \left| \tan\left(\frac{y}{2} + \frac{\pi}{8}\right) \right| &= \frac{1}{2} \log|2 \log x + 1| \end{aligned}$$

which is the required solution of the given differential equation.

47. The given differential equation is

$$x \frac{dy}{dx} = a \left(y^2 + \frac{dy}{dx} \right)$$

$$\Rightarrow (x-a) \frac{dy}{dx} = ay^2$$

$$\Rightarrow y^{-2} dy = \frac{a}{(x-a)} dx$$

Integrating, we get

$$\Rightarrow -\frac{1}{y} = a \log|x-a| + C$$

which is the required solution of the given differential equation.

48. The given differential equation is

$$xy^2 \frac{dy}{dx} = 1 - x^2 + y^2 - x^2 y^2$$

$$\Rightarrow xy^2 \frac{dy}{dx} = (1-x^2)(1+y^2)$$

$$\Rightarrow \frac{y^2}{(1+y^2)} dy = \left(\frac{1-x^2}{x} \right) dx$$

$$\Rightarrow \left(1 - \frac{1}{1+y^2} \right) dy = \left(\frac{1}{x} - x \right) dx$$

Integrating, we get

$$y - \tan^{-1}y = \log|x| - \frac{x^2}{2} + c$$

which is the required solution of the given differential equation.

49. Given differential equation is

$$(x+1) \frac{dy}{dx} = 2xy$$

$$\Rightarrow \frac{dy}{y} = \left(\frac{2x}{x+1} \right) dx$$

$$\Rightarrow \frac{dy}{y} = 2 \left(1 - \frac{1}{x+1} \right) dx$$

Integrating, we get

$$\log|y| = 2x - 2\log|x+1| + c$$

which is the required solution.

50. Given differential equation is

$$\sec^2 x \tan y dx + \sec^2 y \tan x dy = 0$$

$$\Rightarrow \frac{\sec^2 x}{\tan x} dx + \frac{\sec^2 y}{\tan y} dy = 0$$

Integrating, we get

$$\log|\tan x| + \log|\tan y| = \log c$$

$$\Rightarrow \tan x \tan y = c$$

which is the required solution.

51. Given differential equation is

$$(1+e^{2x}) dy + (1+y^2) e^x dx = 0$$

$$\Rightarrow \frac{dy}{1+y^2} + \frac{e^x}{1+e^{2x}} dx = 0$$

Integrating, we get

$$\tan^{-1}(y) + \tan^{-1}(e^x) = \tan^{-1}c$$

$$\Rightarrow \tan^{-1} \left(\frac{y+e^x}{1-e^x y} \right) = \tan^{-1}c$$

$$\Rightarrow \left(\frac{y+e^x}{1-e^x y} \right) = c$$

which is the required solution.

52. Given differential equation is

$$\frac{dy}{dx} = e^{x+y} + x^2 e^y$$

$$\Rightarrow \frac{dy}{dx} = (e^x + x^2) e^y$$

$$\Rightarrow e^{-y} dy = (e^x + x^2) dx$$

Integrating, we get

$$-e^{-y} = e^x + \frac{x^3}{3} + c$$

which is the required differential equation.

53. Given differential equation is

$$y \sqrt{1+x^2} + x \sqrt{1+y^2} \frac{dy}{dx} = 0$$

$$\Rightarrow y + x \frac{dy}{dx} = 0$$

$$\Rightarrow \frac{dx}{x} + \frac{dy}{y} = 0$$

Integrating, we get

$$\log|x| + \log|y| = \log c$$

$$\Rightarrow xy = c$$

which is the required solution.

54. Given differential equation is

$$\sqrt{1+x^2} dy + \sqrt{1+y^2} dx = 0$$

$$\Rightarrow \frac{dy}{\sqrt{1+y^2}} + \frac{dx}{\sqrt{1+x^2}} = 0$$

Integrating, we get

$$\log|y + \sqrt{y^2+1}| + \log|x + \sqrt{x^2+1}| = \log c$$

$$\Rightarrow (y + \sqrt{y^2+1})(x + \sqrt{x^2+1}) = c$$

which is the required solution.

55. Given differential equation is

$$\sqrt{1+x^2+y^2+x^2 y^2} + xy \frac{dy}{dx} = 0$$

$$\Rightarrow \sqrt{(1+x^2)(1+y^2)} + xy \frac{dy}{dx} = 0$$

$$\Rightarrow xy \frac{dy}{dx} = -\sqrt{(1+x^2)(1+y^2)}$$

$$\Rightarrow \frac{y dy}{\sqrt{1+y^2}} = -\frac{\sqrt{1+x^2}}{x} dx$$

Integrating, we get

$$\sqrt{1+y^2} = \sqrt{x^2+1} + \frac{1}{2} \log \left| \frac{\sqrt{x^2+1}-1}{\sqrt{x^2+1}+1} \right| + c$$

which is the required solution.

56. Given differential equation is

$$xy \frac{dy}{dx} + 1 + x + y + xy = 0$$

$$\Rightarrow xy \frac{dy}{dx} + (1+x)(1+y) = 0$$

$$\Rightarrow \left(\frac{y}{1+y} \right) dy + \left(\frac{1+x}{x} \right) dx = 0$$

$$\Rightarrow \left(1 - \frac{1}{1+y} \right) dy + \left(1 + \frac{1}{x} \right) dx = 0$$

Integrating, we get

$$y - \log|1+y| + x + \log|x| = c$$

$$\Rightarrow (x+y) - \log \left| \frac{x}{1+y} \right| = c$$

57. Given differential equation is

$$(1+x)(1+y^2)dx + (1+y)(1+x^2)dy = 0$$

$$\Rightarrow \left(\frac{1+x}{1+x^2} \right) dx + \left(\frac{1+y}{1+y^2} \right) dy = 0$$

Integrating, we get

$$\tan^{-1}x + \tan^{-1}y + \frac{1}{2} \log|(x^2+1)(y^2+1)| = c$$

58. Given differential equation is

$$x\sqrt{1-y^2}dx + y\sqrt{1-x^2}dy = 0$$

$$\Rightarrow \frac{x}{\sqrt{1-x^2}}dx + \frac{y}{\sqrt{1-y^2}}dy = 0$$

Integrating, we get

$$\sqrt{1-x^2} + \sqrt{1-y^2} = c$$

which is the required solution.

59. Given differential equation is

$$\left(y - x \frac{dy}{dx} \right) = a \left(y^2 + \frac{dy}{dx} \right)$$

$$\Rightarrow y(1-ay) = (x+a) \frac{dy}{dx}$$

$$\Rightarrow \frac{dx}{x+a} = \frac{dy}{y(1-ay)}$$

$$\Rightarrow \frac{dx}{x+a} = \left(\frac{1}{y} + \frac{a}{1-ay} \right)$$

Integrating, we get

$$\log|x+a| + \log c = \log|y| - \log|1-ay|$$

$$\Rightarrow c(x+a) = \frac{y}{1-ay}$$

which is the required solution.

60. We have,

$$xy \frac{dx}{dy} = \frac{1+y^2}{1+x^2} (1+x+x^2)$$

$$\Rightarrow \frac{y dy}{1+y^2} = \frac{(1+x+x^2)}{(1+x^2)x} dx$$

$$= \frac{(1+x+x^2)}{(1+x^2)x} dx$$

$$= \frac{dx}{x} + \frac{dx}{1+x^2}$$

$$\Rightarrow \frac{1}{2} \log|1+y^2| = \log|x| + \tan^{-1}x + c$$

which is the required solution.

61. We have,

$$(x^2 - yx^2)dy + (y^2 + xy^2)dx = 0$$

$$\Rightarrow x^2(1-y)dy + y^2(1+x)dx = 0$$

$$\Rightarrow \frac{(1-y)}{y^2} dy + \frac{(1+x)}{x^2} dx = 0$$

Integrating, we get

$$-\frac{1}{y} - \log|y| - \frac{1}{x} + \log|x| = c$$

$$\Rightarrow \log \left| \frac{x}{y} \right| = c + \left(\frac{1}{x} + \frac{1}{y} \right)$$

which is the required solution.

62. The given differential equation is

$$\frac{dy}{dx} = (x+y+1)^2 \quad \dots(i)$$

$$\text{Let } x+y+1 = v$$

$$\Rightarrow \frac{dy}{dx} = \frac{dv}{dx} - 1$$

$$\frac{dv}{dx} - 1 = v^2$$

$$\Rightarrow \frac{dv}{v^2+1} = dx$$

Integrating, we get

$$\int \frac{dv}{v^2+1} = \int dx$$

$$\Rightarrow \tan^{-1}(v) = x + c$$

$$\Rightarrow \tan^{-1}(x+y+1) = x + c$$

is the required solution.

63. The given differential equation is

$$\frac{dy}{dx} = \sin(x+y) + \cos(x+y) \quad \dots(i)$$

$$\text{Put } x+y=v \Rightarrow 1 + \frac{dy}{dx} = \frac{dv}{dx}$$

$$\Rightarrow \frac{dy}{dx} = \frac{dv}{dx} - 1$$

$$\frac{dv}{dx} - 1 = \sin v + \cos v$$

$$\Rightarrow \frac{dv}{dx} = 1 + \sin v + \cos v$$

$$\Rightarrow \frac{dv}{1 + \sin v + \cos v} = dx$$

Integrating, we get

$$\int \frac{dv}{1 + \sin v + \cos v} = x + c$$

$$\Rightarrow \int \frac{dv}{1 + \left(\frac{1 - \tan^2(v/2)}{1 + \tan^2(v/2)} \right) + \left(\frac{2 \tan(v/2)}{1 + \tan^2(v/2)} \right)} = x + c$$

$$\Rightarrow \int \frac{\sec^2(v/2) dv}{2(1 + \tan(v/2))} = x + c$$

$$\Rightarrow \log |1 + \tan(v/2)| = x + c$$

$$\Rightarrow \log |1 + \tan(v/2)| = x + c$$

$$\Rightarrow \log \left| 1 + \tan \left(\frac{x+y}{2} \right) \right| = x + c$$

which is the required solution of the given differential equation.

64. The given differential equation is

$$(x+y)(dx-dy) = (dx+dy)$$

$$\Rightarrow (x+y-1)dx = (x+y+1)dy$$

$$\Rightarrow \frac{dy}{dx} = \frac{(x+y-1)}{(x+y+1)} \quad \dots(i)$$

$$\text{Let } x+y=v \Rightarrow \frac{dy}{dx} = \frac{dv}{dx} - 1$$

$$\Rightarrow \frac{dv}{dx} - 1 = \frac{v-1}{v+1}$$

$$\Rightarrow \frac{dv}{dx} = \frac{v-1}{v+1} + 1 = \frac{2v}{v+1}$$

$$\Rightarrow \left(\frac{v+1}{v} \right) dv = 2 dx$$

Integrating, we get

$$\int \left(1 + \frac{1}{v} \right) dv = 2x + c$$

$$\Rightarrow (v + \log |v|) = 2x + c$$

$$\Rightarrow (x+y + \log |x+y|) = 2x + c$$

$$\Rightarrow (y + \log |x+y|) = x + c$$

which is the required solution of the given differential equation.

65. The given differential equation is

$$\tan y \frac{dy}{dx} = \sin(x+y) + \sin(x-y)$$

$$\Rightarrow \frac{dy}{dx} = \frac{\sin(x+y) + \sin(x-y)}{\tan y}$$

$$= \frac{2 \sin x \cos y}{\tan y}$$

$$\Rightarrow \frac{\tan y}{\cos y} dy = 2 \sin x dx$$

$$\Rightarrow \frac{\sin y}{\cos^2 y} dy = 2 \sin x dx$$

Integrating, we get

$$\int \frac{\sin y}{\cos^2 y} dy = \int 2 \sin x dx$$

$$\Rightarrow \frac{1}{\cos y} = c - 2 \cos x$$

which is the required solution of the given differential equation.

66. The given differential equation is

$$\frac{dy}{dx} - x \tan(y-x) = 1$$

$$\Rightarrow \frac{dy}{dx} = 1 + x \tan(y-x)$$

$$\text{Let } y-x=v \Rightarrow \frac{dy}{dx} = \frac{dv}{dx} + 1$$

$$\Rightarrow \frac{dv}{dx} + 1 = x \tan v + 1$$

$$\Rightarrow \frac{dv}{dx} = x \tan v$$

$$\Rightarrow \cot v dv = x dx$$

Integrating, we get

$$\int \cot v dv = x + c$$

$$\Rightarrow \log |\sin v| = \frac{x^2}{2} + c$$

$$\Rightarrow \log |\sin(x-y)| = \frac{x^2}{2} + c$$

which is the required solution of the given differential equation.

67. We have,

$$\frac{dy}{dx} = \frac{(x-y)+3}{2(x-y)+5} \quad \dots(i)$$

$$\text{Let } x - y = v$$

$$\Rightarrow 1 - \frac{dy}{dx} = \frac{dv}{dx}$$

$$\Rightarrow \frac{dy}{dx} = 1 - \frac{dv}{dx}$$

$$\Rightarrow \left(1 - \frac{dv}{dx}\right) = \frac{v+3}{2v+5}$$

$$\Rightarrow \frac{dv}{dx} = 1 - \frac{v+3}{2v+5}$$

$$= \frac{2v+5-v-3}{2v+5} = \frac{v+2}{2v+5}$$

$$\Rightarrow \left(\frac{2v+5}{v+2}\right)dv = dx$$

$$\Rightarrow \left(2 + \frac{1}{v+2}\right)dv = dx$$

Integrating, we get

$$(2v + \log|v+2|) = x + c$$

$$\Rightarrow (2(x-y) + \log|x-y+2|) = x + c$$

$$\Rightarrow (x - 2y + \log|x-y+2|) = c$$

which is the required solution.

68. We have,

$$(x+y)(dx-dy) = dx+dy$$

$$\Rightarrow \frac{d(x+y)}{(x+y)} = d(x-y)$$

Integrating, we get

$$\log|x+y| = (x-y) + c$$

69. We have $\frac{dy}{dx} = \sec(x+y)$

$$\Rightarrow \frac{dv}{dx} - 1 = \sec v \quad (\text{Let } v = x+y)$$

$$= 1 + \sec v$$

$$\Rightarrow \frac{dv}{1 + \sec v} = dx$$

$$\Rightarrow \frac{\cos v dv}{1 + \cos v} = dx$$

$$\Rightarrow \frac{\cos v(1 - \cos v)dv}{\sin^2 v} = dx$$

$$\Rightarrow [\operatorname{cosec} v \cot v - (\operatorname{cosec} v - 1)]dv = dx$$

Integrating, we get

$$-\operatorname{cosec} v + \cot v - v = x + c$$

$$\Rightarrow -\operatorname{cosec}(x+y) + \cot(x+y) = 2x + y + c$$

which is the required solution.

70. We have,

$$\sin^{-1}\left(\frac{dy}{dx}\right) = x+y$$

$$\Rightarrow \frac{dy}{dx} = \sin(x+y)$$

$$\Rightarrow \frac{dv}{dx} - 1 = \sin v, \quad (\text{Let } v = x+y)$$

$$\Rightarrow \frac{dv}{dx} = 1 + \sin v$$

$$\Rightarrow \frac{dv}{(1 + \sin v)} = dx$$

$$\Rightarrow \frac{(1 - \sin v)dv}{\cos^2 v} = dx$$

$$\Rightarrow (\sec^2 v - \sec v \tan v)dv = dx$$

Integrating, we get

$$\tan v - \sec v = x + c$$

$$\Rightarrow \tan(x+y) - \sec(x+y) = x + c$$

which is the required solution.

71. We have

$$\frac{dy}{dx} = \cos(x+y+1)$$

$$\Rightarrow \frac{dv}{dx} - 1 = \cos v, \quad (\text{Let } x+y+1 = v)$$

$$\Rightarrow \frac{dv}{1 + \cos v} = dx$$

$$\Rightarrow \frac{1 - \cos v}{\sin^2 v} dv = dx$$

$$\Rightarrow (\operatorname{cosec}^2 v - \operatorname{cosec} v \cot v)dv = dx$$

Integrating, we get

$$\operatorname{cosec} v - \cot v = x + c$$

$$\Rightarrow \operatorname{cosec}(x+y+1) - \cot(x+y+1)$$

72. We have,

$$\frac{dy}{dx} = \frac{2(x+y)}{1+(x+y)^2}$$

$$\Rightarrow \frac{dv}{dx} - 1 = \frac{2v}{1+v^2}, \quad v = x+y$$

$$\Rightarrow \frac{dv}{dx} = 1 + \frac{2v}{1+v^2}$$

$$= 1 + \frac{2v}{1+v^2} = \frac{(1+v)^2}{1+v^2}$$

$$\Rightarrow \frac{1+v^2}{(1+v)^2} dv = dx$$

$$\Rightarrow \left(1 - \frac{2}{(1+v)^2}\right) dv = dx$$

$$\Rightarrow \left(1 - \frac{2}{1+v} + \frac{2}{(1+v)^2}\right) dv = dx$$

Integrating, we get

$$v - 2\log|1+v| + \frac{2}{(1+v)} = x + c$$

$$\Rightarrow y - 2\log|1+x+y| - \frac{2}{(1+x+y)} = c$$

which is the required solution.

73. We have,

$$\frac{dy}{dx} = \sin(10x + 6y)$$

$$\Rightarrow \frac{1}{6} \left(\frac{dv}{dx} - 10 \right) = \sin v$$

$$\Rightarrow \frac{dv}{dx} = 6\sin v + 10$$

$$\Rightarrow \frac{dv}{6\sin v + 10} = dx$$

$$\Rightarrow \frac{dv}{6 \left(\frac{2 \tan(v/2)}{1 + \tan^2(v/2)} \right) + 10} = dx$$

$$\Rightarrow \frac{\sec^2(v/2) dv}{10(1 + \tan^2(v/2)) + 12 \tan(v/2)} = dx$$

$$\Rightarrow \frac{2 dt}{10(1 + t^2) + 12t} = dx, \quad t = \tan(v/2)$$

$$\Rightarrow \frac{dt}{5(1 + t^2) + 6t} = dx$$

$$\Rightarrow \frac{dt}{5t^2 + 6t + 5} = dx$$

$$\Rightarrow \frac{dt}{t^2 + \frac{6}{5}t + 1} = 5dx$$

$$\Rightarrow \frac{dt}{\left(t + \frac{3}{5}\right)^2 + \left(\frac{4}{5}\right)^2} = 5dx$$

Integrating, we get

$$\frac{5}{4} \tan^{-1} \left(\frac{t + 3/5}{4/5} \right) = 5x + c$$

$$\Rightarrow \frac{5}{4} \tan^{-1} \left(\frac{5t + 3}{4} \right) = 5x + c$$

$$\Rightarrow \frac{5}{4} \tan^{-1} \left(\frac{5 \tan(v/2) + 3}{4} \right) = 5x + c$$

$$\Rightarrow \frac{5}{4} \tan^{-1} \left(\frac{5 \tan(5x + 3y) + 3}{4} \right) = 5x + c$$

which is the required solution.

74. We have,

$$\frac{y}{x} \frac{dy}{dx} + \frac{2(x^2 + y^2) - 1}{x^2 + y^2 + 1} = 0$$

$$\Rightarrow \frac{1}{2} \frac{dv}{dx} - 1 + \frac{2v - 1}{v + 1} = 0$$

$$\Rightarrow \frac{1}{2} \frac{dv}{dx} = 1 - \frac{2v - 1}{v + 1}$$

$$= \frac{2 - v}{v + 1}$$

$$\Rightarrow \left(\frac{v + 1}{2 - v} \right) dv = 2dx$$

$$\Rightarrow \left(\frac{v + 1}{v - 1} \right) dv = -2dx$$

$$\Rightarrow \left(\frac{v - 2 + 3}{v - 2} \right) dv = -2dx$$

$$\Rightarrow 1 + \left(\frac{3}{v - 2} \right) dv = -2dx$$

Integrating, we get

$$(x^2 + y^2) + \log|(x^2 + y^2) - 2| = c - 2x$$

which is the required solution.

75. The given differential equation can be written as

$$\frac{dy}{dx} = \frac{x^2 y}{x^3 + y^3}$$

$$\Rightarrow \frac{dy}{dx} = \frac{\left(\frac{y}{x}\right)}{1 + \left(\frac{y}{x}\right)^3}$$

$$\text{Let } \frac{y}{x} = v$$

$$\Rightarrow y = vx$$

$$\Rightarrow \frac{dy}{dx} = v + x \frac{dv}{dx}$$

$$\Rightarrow v + x \frac{dv}{dx} = \frac{v}{1 + v^3}$$

$$\Rightarrow x \frac{dv}{dx} = \frac{v}{1 + v^3} - v$$

$$\Rightarrow \left(\frac{v^3 + 1}{v^4} \right) dv = -\frac{dx}{x}$$

$$\Rightarrow \left(\frac{1}{v} + v^{-4} \right) dv = -\frac{dx}{x}$$

Integrating, we get

$$\log |v| - \frac{1}{3v^3} = c - \log |x|$$

$$\Rightarrow \log \left| \frac{y}{x} \right| - \frac{x^3}{3y^3} = c - \log |x|$$

$$\Rightarrow \log |v| - \frac{1}{3v^3} = c,$$

which is the required

solution of the given differential equation.

76. The given differential equation is

$$x dy - y dx = \sqrt{x^2 + y^2} dx$$

$$\Rightarrow x \frac{dy}{dx} - y = \sqrt{x^2 + y^2}$$

$$\Rightarrow \frac{dy}{dx} - \frac{y}{x} = \frac{\sqrt{x^2 + y^2}}{x}$$

$$\Rightarrow \frac{dy}{dx} - \frac{y}{x} = \sqrt{1 + \left(\frac{y}{x}\right)^2} \quad \dots(i)$$

$$\text{Let } \frac{y}{x} = v \Rightarrow y = vx$$

$$\Rightarrow \frac{dy}{dx} = v + x \frac{dv}{dx}$$

$$\Rightarrow v + x \frac{dv}{dx} - v = \sqrt{1 + v^2}$$

$$\Rightarrow x \frac{dv}{dx} = \sqrt{1 + v^2}$$

$$\Rightarrow \frac{dv}{\sqrt{1 + v^2}} = \frac{dx}{x}$$

Integrating, we get

$$\int \frac{dv}{\sqrt{1 + v^2}} = \int \frac{dx}{x}$$

$$\Rightarrow \log |v + \sqrt{1 + v^2}| = \log |x| + \log c$$

$$\Rightarrow (v + \sqrt{1 + v^2}) = cx$$

$$\Rightarrow \left(\frac{y}{x} + \sqrt{1 + \left(\frac{y}{x}\right)^2} \right) = cx$$

$$\Rightarrow (y + \sqrt{x^2 + y^2}) = cx^2$$

which is the required solution of the given differential equation.

77. The given differential equation is

$$(x^2 - y^2) dx = 2xy dy$$

$$\Rightarrow x \frac{dy}{dx} - y = \sqrt{x^2 + y^2} \quad \dots(i)$$

$$\Rightarrow \frac{dy}{dx} = \frac{1}{2} \left(\frac{x}{y} - \frac{y}{x} \right)$$

$$\text{Let } \frac{y}{x} = v \Rightarrow y = vx$$

$$\Rightarrow \frac{dy}{dx} = v + x \frac{dv}{dx}$$

$$\Rightarrow v + x \frac{dv}{dx} = \frac{1}{2} \left(\frac{1}{v} - v \right)$$

$$\Rightarrow x \frac{dv}{dx} = \frac{1}{2v} - \frac{3v}{2} = \frac{1 - 3v^2}{2v}$$

$$\Rightarrow \frac{2v}{1 - 3v^2} dv = \frac{dx}{x}$$

Integrating, we get

$$\int \frac{2v}{1 - 3v^2} dv = \int \frac{dx}{x}$$

$$\Rightarrow -\frac{1}{3} \log |1 - 3v^2| = \log c + \log |x|$$

$$\Rightarrow -\frac{1}{3} \log \left| 1 - 3 \left(\frac{y}{x} \right)^2 \right| = \log c + \log |x|$$

which is the required solution of the given differential equation.

78. The given differential equation is

$$(1 + 2e^{xy}) dx + 2e^{xy} (1 - x/y) dy = 0$$

$$\Rightarrow \frac{dy}{dx} = \frac{(1 + 2e^{xy})}{2e^{xy} \left(\frac{x}{y} - 1 \right)}$$

$$\Rightarrow \frac{dy}{dx} = \frac{2e^{xy} \left(\frac{x}{y} - 1 \right)}{(1 + 2e^{xy})} \quad \dots(i)$$

$$\text{Let } \frac{x}{y} = v$$

$$\Rightarrow x = vy$$

$$\Rightarrow \frac{dx}{dy} = v + y \frac{dv}{dy}$$

$$\Rightarrow v + y \frac{dv}{dy} = \frac{2e^v(v-1)}{(1 + 2e^v)}$$

$$\Rightarrow y \frac{dv}{dy} = \frac{2e^v(v-1)}{(1 + 2e^v)} - v$$

$$= \frac{2ve^v - 2e^v - v - 2ve^v}{1 + 2e^v}$$

$$= \frac{2e^v + v}{1 + 2e^v}$$

$$\Rightarrow \frac{1 + 2e^v}{v + 2e^v} dv = -\frac{dy}{y}$$

$$\Rightarrow \log|v + 2e^v| + \log|y| = \log c$$

$$\Rightarrow y(v + 2e^v) = c$$

$$\Rightarrow y\left(\frac{x}{y} + 2e^{xy}\right) = c$$

$$\Rightarrow (x + 2ye^{xy}) = c$$

which is the required solution of the given differential equation.

79. The given differential equation can be written as

$$\frac{dy}{dx} = \frac{y-x}{y+x} \quad \dots(i)$$

$$(\text{Let } \frac{y}{x} = v \Rightarrow y = vx)$$

$$\Rightarrow \frac{dy}{dx} = v + x \frac{dv}{dx}$$

$$\begin{aligned} \Rightarrow v + x \frac{dv}{dx} &= \frac{v-1}{v+1} \\ &= \frac{v-1}{v+1} - v \\ &= \frac{v-1-v^2-v}{v+1} \\ &= \frac{1+v^2}{v+1} \end{aligned}$$

$$\Rightarrow \frac{v+1}{v^2+1} dv = -\frac{dx}{x}$$

$$\Rightarrow \frac{1}{2} \log|v^2+1| + \tan^{-1}(v) = c - \log|x|$$

$$\Rightarrow \frac{1}{2} \log|x^2+y^2| + \tan^{-1}\left(\frac{y}{x}\right) = c$$

which is the required solution of the given differential equation.

80. We have,

$$x \frac{dy}{dx} = x + y$$

$$\Rightarrow \frac{dy}{dx} = \frac{x+y}{x} = 1 + \frac{y}{x}$$

$$\Rightarrow v + x \frac{dv}{dx} = 1 + v, v = \frac{y}{x}$$

$$\Rightarrow x \frac{dv}{dx} = 1$$

$$\Rightarrow dv = \frac{dx}{x}$$

Integrating, we get

$$\log|v^2-1| + \log|x| = \log c$$

$$\Rightarrow \frac{y}{x} = \log|x| + c$$

which is the required solution.

81. We have,

$$\frac{dy}{dx} = \frac{y-x}{y+x} = \frac{(y/x)-1}{(y/x)+1}$$

$$\Rightarrow v + x \frac{dv}{dx} = \frac{v-1}{v+1}$$

$$\begin{aligned} \Rightarrow x \frac{dv}{dx} &= \frac{v-1}{v+1} - v \\ &= \frac{v-1}{v+1} - v \\ &= -\frac{1+v^2}{1+v} \end{aligned}$$

$$\Rightarrow \frac{1+v}{1+v^2} dv = -\frac{dx}{x}$$

Integrating, we get

$$\tan^{-1}(v) + \frac{1}{2} \log|1+v^2| = c - \log|x|$$

$$\tan^{-1}\left(\frac{y}{x}\right) + \frac{1}{2} \log|x^2+y^2| = c$$

which is the required solution.

82. We have,

$$2xy \frac{dy}{dx} = x^2 + y^2$$

$$\Rightarrow \frac{dy}{dx} = \frac{x^2+y^2}{2xy} = \frac{1+\left(\frac{y}{x}\right)^2}{2\left(\frac{y}{x}\right)}$$

$$\Rightarrow v + x \frac{dv}{dx} = \frac{1+v^2}{2v}$$

$$\Rightarrow x \frac{dv}{dx} = \frac{1+v^2}{2v} - v = \frac{1-v^2}{2v}$$

$$\Rightarrow \frac{2v dv}{v^2-1} = -\frac{dx}{x}$$

Integrating, we get

$$\log|v^2-1| + \log|x| = \log c$$

$$\Rightarrow (v^2-1)x = c$$

$$\Rightarrow (y^2-x^2) = cx$$

which is the required solution.

83. We have,

$$\frac{dy}{dx} = \frac{y^2-2xy}{x^2-2xy}$$

$$\Rightarrow \frac{dy}{dx} = \frac{\left(\frac{y}{x}\right)^2 - 2\left(\frac{y}{x}\right)}{1 - 2\left(\frac{y}{x}\right)}$$

$$\Rightarrow v + x \frac{dv}{dx} = \frac{v^2-2v}{1-2v}$$

$$\begin{aligned}
\Rightarrow x \frac{dv}{dx} &= \frac{3v^2 - 3v}{1 - 2v} \\
\Rightarrow \frac{1 - 2v}{3v(v - 1)} dv &= \frac{dx}{x} \\
\Rightarrow \frac{1}{3} \left(\frac{1}{v(v - 1)} - \frac{2}{(v - 1)} \right) dv &= \frac{dx}{x} \\
\Rightarrow \frac{1}{3} \left(\frac{1}{v - 1} - \frac{1}{v} - \frac{2}{(v - 1)} \right) dv &= \frac{dx}{x} \\
\Rightarrow -\frac{1}{3} \left(\frac{1}{v - 1} + \frac{1}{v} \right) dv &= \frac{dx}{x}
\end{aligned}$$

Integrating, we get

$$\begin{aligned}
-\frac{1}{3} \log |v(v - 1)| &= \log |x| \\
\Rightarrow -\frac{1}{3} \log \left| \frac{y}{x} \left(\frac{y}{x} - 1 \right) \right| &= \log |x|
\end{aligned}$$

which is the required solution.

84. Given differential equation is

$$\begin{aligned}
x^2 y \, dx - (x^3 + y^3) \, dy &= 0 \\
\Rightarrow \frac{dy}{dx} &= \frac{x^2 y}{x^3 + y^3} \\
&= \frac{\frac{y}{x}}{1 + \left(\frac{y}{x}\right)^3} \\
\Rightarrow v + x \frac{dv}{dx} &= \frac{v}{1 + v^3}, \quad \left(\text{Let } v = \frac{y}{x} \right) \\
\Rightarrow x \frac{dv}{dx} &= \frac{v}{1 + v^3} - v \\
&= \frac{v - v - v^4}{1 + v^3} \\
&= -\frac{v^4}{1 + v^3} \\
\Rightarrow \left(\frac{1 + v^3}{v^4} \right) dv &= -\frac{dx}{x}
\end{aligned}$$

Integrating, we get

$$\begin{aligned}
-\frac{1}{3v^3} + \log |v| &= c - \log |x| \\
\Rightarrow -\frac{x^3}{3y^3} + \log |y| &= c
\end{aligned}$$

which is the required solution

85. Given differential equation is

$$\begin{aligned}
(x^3 - 3xy^2) \, dx &= (y^3 - 3x^2y) \, dy \\
\Rightarrow \frac{dy}{dx} &= \frac{x^3 - 3xy^2}{y^3 - 3x^2y}
\end{aligned}$$

$$\begin{aligned}
&= \frac{1 - 3\left(\frac{y}{x}\right)^2}{\left(\frac{y}{x}\right)^3 - 3\left(\frac{y}{x}\right)} \\
\Rightarrow v + x \frac{dv}{dx} &= \frac{1 - 3v^2}{v^3 - 3v}, \quad \left(\text{Let } v = \frac{y}{x} \right) \\
\Rightarrow x \frac{dv}{dx} &= \frac{1 - 3v^2}{v^3 - 3v} - v \\
&= \frac{1 - 3v^2 - v^4 + 3v^2}{v^3 - 3v} \\
&= \frac{1 - v^4}{v^3 - 3v}
\end{aligned}$$

$$\begin{aligned}
\Rightarrow \left(\frac{v^3 - 3v}{1 - v^4} \right) dv &= \frac{dx}{x} \\
\Rightarrow \left(\frac{v^3}{1 - v^4} \right) dv - \left(\frac{3v}{1 - v^4} \right) dv &= \frac{dx}{x}
\end{aligned}$$

Integrating, we get

$$\Rightarrow \left(\frac{v^3}{1 - v^4} \right) dv - \left(\frac{3v}{1 - v^4} \right) dv = \frac{dx}{x}$$

Integrating, we get

$$\begin{aligned}
-\frac{1}{4} \log |1 - v^4| + \frac{3}{4} \log \left| \frac{v^2 - 1}{v^2 + 1} \right| &= \log |x| + \log c \\
\Rightarrow -\frac{1}{4} \log \left| 1 - \left(\frac{y}{x} \right)^4 \right| + \frac{3}{4} \log \left| \frac{\left(\frac{y}{x} \right)^2 - 1}{\left(\frac{y}{x} \right)^2 + 1} \right| &= \log |x|
\end{aligned}$$

which is the required solution.

86. The given differential equation is

$$\begin{aligned}
\frac{dy}{dx} &= \frac{y}{x} - \sqrt{\frac{y^2}{x^2} - 1} \\
\Rightarrow v + x \frac{dv}{dx} &= v - \sqrt{v^2 - 1} \quad \left(\text{Let } \frac{y}{x} = v \right) \\
\Rightarrow x \frac{dv}{dx} &= -\sqrt{v^2 - 1} \\
\Rightarrow \frac{dv}{\sqrt{v^2 - 1}} &= -\frac{dx}{x}
\end{aligned}$$

Integrating, we get

$$\begin{aligned}
\log |v + \sqrt{v^2 - 1}| &= \log c - \log x \\
\Rightarrow \log |v + \sqrt{v^2 - 1}| &= \log \left| \frac{c}{x} \right| \\
\Rightarrow (v + \sqrt{v^2 - 1}) &= \left(\frac{c}{x} \right) \\
\Rightarrow \left(\frac{y}{x} + \sqrt{\left(\frac{y}{x} \right)^2 - 1} \right) &= \left(\frac{c}{x} \right)
\end{aligned}$$

$$\Rightarrow (y + \sqrt{y^2 - x^2}) = c$$

which is the required solution.

87. The given differential equation is

$$\Rightarrow \frac{dy}{dx} = \frac{y}{x} + \sin\left(\frac{y}{x}\right)$$

$$\Rightarrow v + x \frac{dv}{dx} = v + \sin(v) \quad \left(\text{Let } \frac{y}{x} = u\right)$$

$$\Rightarrow x \frac{dv}{dx} = \sin(v)$$

$$\Rightarrow \frac{dv}{\sin(v)} = \frac{dx}{x}$$

Integrating, we get

$$\Rightarrow \log \left| \tan\left(\frac{v}{2}\right) \right| = \log c + \log |x|$$

$$\Rightarrow \tan\left(\frac{y}{2x}\right) = cx$$

which is the required solution.

88. The given differential equation is

$$x \frac{dy}{dx} = y - x \tan\left(\frac{y}{x}\right)$$

$$\Rightarrow \frac{dy}{dx} = \left(\frac{y}{x}\right) - \tan\left(\frac{y}{x}\right)$$

$$\Rightarrow v + x \frac{dv}{dx} = v - \tan(v) \quad \left(\text{Let } \frac{y}{x} = u\right)$$

$$\Rightarrow x \frac{dv}{dx} = -\tan(v)$$

$$\Rightarrow \frac{dv}{\tan(v)} = -\frac{dx}{x}$$

Integrating, we get

$$\Rightarrow \log \left| \tan\left(\frac{\pi}{4} + \frac{v}{2}\right) \right| = \log c - \log |x|$$

$$\Rightarrow \log \left| \tan\left(\frac{\pi}{4} + \frac{v}{2}\right) \right| = \log \left| \frac{c}{x} \right|$$

$$\Rightarrow \tan\left(\frac{\pi}{4} + \frac{y}{2x}\right) = \left(\frac{c}{x}\right)$$

which is the required solution.

89. The given differential equation is

$$(1 + e^{\frac{x}{y}}) dx + e^{\frac{x}{y}} \left(1 - \frac{x}{y}\right) dy = 0$$

$$\Rightarrow \frac{dx}{dy} = \frac{\left(\frac{x}{y} - 1\right) e^{\frac{x}{y}}}{(1 + e^{\frac{x}{y}})}$$

$$\Rightarrow v + x \frac{dv}{dy} = \frac{(v - 1)e^v}{1 + e^v}, v = \frac{x}{y}$$

$$\Rightarrow x \frac{dv}{dy} = \frac{(v - 1)e^v}{1 + e^v} - v$$

$$= \frac{ve^v - e^v - v - ve^v}{1 + e^v}$$

$$= \frac{-e^v - v}{1 + e^v}$$

$$\Rightarrow \left(\frac{1 + e^v}{v + e^v}\right) dv = -\frac{dy}{y}$$

Integrating, we get

$$\log |v + e^v| = \log c - \log |y|$$

$$\Rightarrow \left(\frac{x}{y} + e^{\frac{x}{y}}\right) = \left(\frac{c}{y}\right)$$

$$\Rightarrow (x + y e^{\frac{x}{y}}) = c$$

which is the required solution.

90. The given differential equation is

$$\frac{dy}{dx} = \frac{x + 2y + 3}{2x + 3y + 4}$$

$$\text{Let } x = X + a \text{ and } y = Y + \beta$$

$$\text{Thus, } \frac{dY}{dX} = \frac{(X + \alpha) + 2(Y + \beta) + 3}{2(X + \alpha) + 3(Y + \beta) + 4}$$

$$\frac{dY}{dX} = \frac{(X + 2Y) + (\alpha + 2\beta + 3)}{(2X + 3Y) + (2\alpha + 3\beta + 4)} \quad \dots(i)$$

Let us choose α and β in such a way that

$$\alpha + 2\beta + 3 = 0 \text{ and } 2\alpha + 3\beta + 4 = 0$$

Solving, we get

$$\alpha = 1, \beta = -2.$$

Equation (i) reduces to

$$\frac{dY}{dX} = \frac{X + 2Y}{2X + 3Y} \quad \dots(ii)$$

which is a homogeneous equation.

$$\text{Put } \frac{Y}{X} = V$$

$$\Rightarrow \frac{dY}{dX} = V + X \frac{dV}{dX}$$

Equation (ii) reduces to

$$V + X \frac{dV}{dX} = \frac{1 + 2V}{2 + 3V}$$

$$\Rightarrow \frac{2 + 3V}{1 - 3V^2} dV = \frac{dX}{X}$$

Integrating, we get

$$\int \frac{2 + 3V}{1 - 3V^2} dV = \int \frac{dX}{X}$$

$$\Rightarrow \frac{1}{\sqrt{3}} \log \left| \frac{1 + \sqrt{3}V}{1 - \sqrt{3}V} \right| - \frac{1}{2} \log |(1 - 3V^2)|$$

$$= \log C + \log |X|$$

$$\Rightarrow \frac{1}{\sqrt{3}} \log \left| \frac{X + \sqrt{3}Y}{X - \sqrt{3}Y} \right| - \frac{1}{2} \log |(X^2 - 3Y^2)| = \log c$$

$$\Rightarrow \frac{1}{\sqrt{3}} \log \left| \frac{(x-1) + \sqrt{3}(y+2)}{(x-1) - \sqrt{3}(y+2)} \right| - \frac{1}{2} \log |(x^2 - 3y^2 - 2x - 12y - 11)| = \log c$$

which is the required solution of the given differential equation.

91. The given differential equation is

$$\frac{dy}{dx} = \frac{(x+y)^2}{(x+2)(y-2)} \quad \dots(i)$$

Let $X = x + 2$ and $Y = y + 2$

Equation (i) reduces to

$$\frac{dY}{dX} = \frac{(X-2+Y+2)^2}{XY} = \frac{(X+Y)^2}{XY} \quad \dots(ii)$$

Let $Y = VX \Rightarrow \frac{dY}{dX} = V + X \frac{dV}{dX}$

Equation (ii) reduces to

$$V + X \frac{dV}{dX} = \frac{(1+V)^2}{V}$$

$$\Rightarrow X \frac{dV}{dX} = \frac{(1+V)^2}{V} - V$$

$$\Rightarrow X \frac{dV}{dX} = \frac{(1+2V)}{V}$$

$$\Rightarrow \frac{VdV}{(2V+1)} = \frac{dX}{X}$$

Integrating, we get

$$\int \frac{VdV}{(2V+1)} = \int \frac{dX}{X}$$

$$\Rightarrow \frac{1}{2} \int \left(1 - \frac{1}{(2V+1)} \right) = \int \frac{dX}{X}$$

$$\Rightarrow \frac{1}{2} (V - \log(2V+1)) = \log |X| + c$$

$$\Rightarrow \frac{1}{2} \left(\frac{Y}{X} - \log \left| 2 \frac{Y}{X} + 1 \right| \right) = \log |X| + c$$

$$\Rightarrow \frac{1}{2} \left(\left(\frac{y-2}{x+2} \right) - \log \left| 2 \left(\frac{y-2}{x+2} \right) + 1 \right| \right)$$

$$= \log |x+2| + c$$

which is the required solution of the given differential equation.

97. The given differential equation is

$$ydx - xdy + \ln x dx = 0$$

$$\Rightarrow y - x \frac{dy}{dx} + \ln x = 0$$

$$\Rightarrow x \frac{dy}{dx} - y = \ln x$$

$$\Rightarrow \frac{dy}{dx} - \frac{y}{x} = \frac{\ln x}{x} \quad \dots(i)$$

which is a linear differential equation.

$$IF = e^{-\int \frac{dx}{x}} = e^{-\log x} = \frac{1}{x}$$

Multiplying both sides of Eq. (i) by IF and integrating, we get

$$y \cdot (IF) = \int Q \cdot (IF) dx + c$$

$$\Rightarrow y \cdot \frac{1}{x} = \int \frac{\ln x}{x^2} dx + c$$

$$\Rightarrow \frac{y}{x} = -\left(\frac{\log x}{x} + \frac{1}{x} \right) + c$$

$$\Rightarrow y = -(\log x + 1) + c$$

which is the required solution of the given differential equation.

98. The differential equation is

$$x^2 \frac{dy}{dx} - 3xy = 4x^4 + 2x^2$$

$$\Rightarrow \frac{dy}{dx} - 3 \frac{y}{x} = 4x^2 + 2 \quad \dots(i)$$

which is a linear differential equation.

$$IF = e^{-3 \int \frac{dx}{x}} = e^{-3 \log x} = \frac{1}{x^3}$$

Multiplying both sides of Eq. (i) by IF and integrating, we get

$$y \cdot (IF) = \int Q \cdot (IF) dx + c$$

$$\Rightarrow y \cdot \frac{1}{x^3} = \int \left(\frac{4x^2 + 2}{x^3} \right) dx + c$$

$$\Rightarrow \frac{y}{x^3} = \int \left(\frac{4}{x} + \frac{2}{x^3} \right) dx + c$$

$$= \left(4 \log x - \frac{1}{x^2} \right) + c$$

which is the required solution of the given differential equation.

99. The given differential equation is

$$\frac{dy}{dx} = y \tan x - \sin x$$

$$\Rightarrow \frac{dy}{dx} + (-\tan x)y = -\sin x \quad \dots(i)$$

which is a linear differential equation.

$$IF = e^{-\int \tan x dx} = e^{\log(\cos x)} = \cos x$$

Multiplying both sides of Eq. (i) by I.F and integrating, we get

$$\begin{aligned} y \cdot (IF) &= \int Q \cdot (IF) dx + c \\ \Rightarrow y(\cos x) &= -\int (\sin x \cos x) dx + c \\ &= -\frac{1}{2} \int \sin 2x dx + c \\ &= \frac{\cos 2x}{4} + c \end{aligned}$$

which is the required solution of the given differential equation.

100. The given differential equation is

$$\frac{dy}{dx} + \frac{y}{x \log x} = \frac{2}{x^2} \quad \dots(i)$$

which is a linear differential equation.

$$IF = e^{\int \frac{dx}{x \log x}} = e^{\log(\log x)} = \log x$$

Multiplying both sides of Eq. (i) by IF and integrating, we get

$$\begin{aligned} y(IF) &= \int Q(IF) dx + c \\ \Rightarrow y \cdot \log x &= \int \frac{2}{x^2} \log x dx + c \\ &= -\frac{2}{x} (1 + \log x) + c \quad \dots(i) \end{aligned}$$

which is the required solution of the given differential equation.

101. The given differential equation is

$$\begin{aligned} x dx &= \left(\frac{x}{y^2} - y \right) dy \\ \Rightarrow x \frac{dx}{dy} + y &= xy^2 \\ \Rightarrow \frac{dy}{dx} + \frac{y}{x} &= y^2 \quad \dots(i) \end{aligned}$$

which is a linear differential equation.

102. The given differential equation is

$$\begin{aligned} (x + 3y + 2) \frac{dy}{dx} &= 1 \\ \Rightarrow \frac{dx}{dy} - x &= 3y + 2 \quad \dots(i) \end{aligned}$$

which is a linear differential equation.

$$IF = e^{-\int dy} = e^{-y}$$

Multiplying both sides of Eq. (i) by IF and integrating, we get

$$\begin{aligned} x \cdot (IF) &= \int Q \cdot (IF) dy + c \\ \Rightarrow x \cdot e^{-y} &= \int (3y + 2) e^{-y} dy + c \\ &= 3(-ye^{-y} + 2e^{-y}) + c \\ \Rightarrow x &= -3(y + 2) + ce^y \end{aligned}$$

which is the required solution of the given differential equation.

103. The given differential equation is

$$\frac{dy}{dx} + 2y = e^{3x} \quad \dots(i)$$

which is a linear differential equation.

$$IF = e^{\int 2 dx} = e^{2x}$$

Multiplying both sides of Eq. (i) by IF and integrating, we get

$$\begin{aligned} y \cdot (IF) &= \int Q \cdot (IF) dx + c \\ \Rightarrow y(e^x) &= \int (e^{2x} \cdot e^x) dx + c \\ &= \int (e^{3x}) dx + c \\ &= \left(\frac{e^{3x}}{3} \right) + c \\ \Rightarrow y &= \left(\frac{e^{3x}}{3} \right) + ce^{-x} \end{aligned}$$

which is the required solution.

104. The given differential equation is

$$\begin{aligned} x \frac{dy}{dx} &= x + y \\ \Rightarrow \frac{dy}{dx} &= 1 + \frac{y}{x} \\ \Rightarrow \frac{dy}{dx} - \frac{y}{x} &= 1 \quad \dots(i) \end{aligned}$$

which is a linear differential equation.

$$\text{Thus, } IF = e^{-\int \frac{dx}{x}} = e^{-\log x} = \frac{1}{x}$$

Multiplying both sides of Eq. (i) by IF and integrating we get

$$\begin{aligned} y \cdot (IF) &= \int Q \cdot (IF) dx + c \\ \Rightarrow \frac{y}{x} &= \int \frac{dx}{x} + c \\ \Rightarrow \frac{y}{x} &= \log |x| + c \end{aligned}$$

which is the required solution.

105. The given differential equation is

$$x \frac{dy}{dx} + y = xe^x$$

$$\Rightarrow \frac{dy}{dx} + \frac{y}{x} = e^x$$

which is a linear differential equation.

$$\text{Thus, IF} = e^{\int \frac{dx}{x}} = e^{\log x} = x$$

Multiplying both sides of Eq. (i) by IF and integrating we get

$$y \cdot (\text{IF}) = \int Q \cdot (\text{IF}) dx + c$$

$$\Rightarrow y \cdot x = \int xe^x dx + c$$

$$\Rightarrow (xy) = e^x(x-1) + c$$

which is the required solution.

106. The given differential equation is

$$x \frac{dy}{dx} + y = x \log x$$

$$\Rightarrow \frac{dy}{dx} + \frac{y}{x} = \log x \quad \dots(i)$$

which is a linear differential equation.

$$\text{Thus, IF} = e^{\int \frac{dx}{x}} = e^{\log x} = x$$

Multiplying both sides of Eq. (i) by IF and integrating we get

$$y \cdot (\text{IF}) = \int x \log x dx + c$$

$$\Rightarrow y \cdot x = \int x \log x dx + c$$

$$= \int x \log x dx - \frac{1}{2} \int x dx + c$$

$$= \left(\frac{x^2}{2} \right) \log x - \frac{x^2}{4} + c$$

which is the required solution.

107. The given differential equation is

$$\frac{dy}{dx} + \frac{y}{x} = x^3 \quad \dots(i)$$

which is a linear differential equation.

$$\text{Thus, IF} = e^{\int \frac{dx}{x}} = e^{\log x} = x$$

Multiplying both sides of Eq. (i) by IF and integrating we get

$$y \cdot (\text{IF}) = \int Q \cdot (\text{IF}) dx + c$$

$$\Rightarrow xy = \int x^4 dx + c$$

$$= \frac{x^5}{5} + c$$

which is the required solution.

108. The given differential equation is

$$\frac{dy}{dx} = y \tan x - 2 \sin x$$

$$\Rightarrow \frac{dy}{dx} + (-\tan x)y = -2 \sin x \quad \dots(i)$$

which is a linear differential equation

$$\text{Thus, IF} = e^{\int -\tan x dx} = e^{-\log(\sec x)} = x$$

Multiplying both sides of Eq. (i) by IF and integrating we get

$$y \cdot (\text{IF}) = \int Q \cdot (\text{IF}) dx + c$$

$$\Rightarrow y \cdot \cos x = \int (-\sin x \cdot \tan x) dx + c$$

$$= -\int \left(\frac{1 - \cos^2 x}{\cos x} \right) dx + c$$

$$= -\int (\sec x - \cos x) dx + c$$

$$= -\log |\sec x + \tan x| + \sin x + c$$

which is the required solution.

109. The given differential equation is

$$\frac{dy}{dx} - \frac{y}{x} = 2x^2 \quad \dots(i)$$

which is a linear differential equation.

$$\text{Thus, I.F.} = e^{-\int \frac{dx}{x}} = e^{-\log x} = \frac{1}{x}$$

Multiplying both sides of Eq. (i) by IF and integrating we get

$$y \cdot (\text{I.F.}) = \int Q \cdot (\text{IF}) dx + c$$

$$\Rightarrow y \cdot \frac{1}{x} = 2 \int x dx + c$$

$$\Rightarrow \frac{y}{x} = x^2 + c$$

which is the required solution.

110. The given differential equation is

$$x \log x \frac{dy}{dx} + y = \frac{2}{x} \log x$$

$$\Rightarrow \frac{dy}{dx} + \frac{y}{x \log x} = \frac{2}{x^2} \quad \dots(i)$$

which is a linear differential equation.

$$\text{Thus, IF} = e^{\int \frac{dx}{x \log x}} = e^{\log(\log x)} = x$$

Multiplying both sides of Eq. (i) by IF and integrating we get

$$y \cdot (\text{IF}) = \int Q \cdot (\text{IF}) dx + c$$

$$\begin{aligned}
\Rightarrow y \cdot \log x &= 2 \int \frac{\log x}{x^2} dx + c \\
&= 2 \int (te^{-t}) dt + c, t = \log x \\
&= -2(te^{-t} - e^{-t}) + c \\
&= -2\left(\frac{\log x - 1}{x}\right) + c
\end{aligned}$$

which is the required solution.

111. The given differential equation is

$$\begin{aligned}
y dx - (x + 2y^2) dy &= 0 \\
\Rightarrow y \frac{dx}{dy} + (x - y^2) &= 0 \\
\Rightarrow y \frac{dx}{dy} + x &= y^2 \\
\Rightarrow \frac{dx}{dy} + \frac{x}{y} &= y \quad \dots(i)
\end{aligned}$$

which is a linear differential equation.

Thus, $IF = e^{\int \frac{dy}{y}} = e^{\log y} = y$

Multiplying both sides of Eq. (i) by IF and integrating we get

$$\begin{aligned}
x \cdot (IF) &= \int y^2 dy + c \\
\Rightarrow x \cdot y &= \int y^2 dy + c \\
&= \frac{y^3}{3} + c
\end{aligned}$$

which is the required solution.

112. The given differential equation is

$$\begin{aligned}
y dx + (x - y^3) dy &= 0 \\
\Rightarrow y \frac{dx}{dy} + (x - y^3) &= 0 \\
\Rightarrow y \frac{dx}{dy} + x &= y^3 \\
\Rightarrow \frac{dx}{dy} + \frac{x}{y} &= y^2 \quad \dots(i)
\end{aligned}$$

which is a linear differential equation

Thus, $IF = e^{\int \frac{dy}{y}} = e^{\log y} = y$

Multiplying both sides of Eq. (i) by IF and integrating we get

$$\begin{aligned}
x \cdot (IF) &= \int Q \cdot (IF) dy + c \\
\Rightarrow x \cdot y &= \int y^3 dy + c \\
&= \frac{y^4}{4} + c
\end{aligned}$$

which is the required solution.

113. The given differential equation is

$$\begin{aligned}
x \frac{dx}{dy} - ay &= x + 1 \\
\Rightarrow \frac{dy}{dx} + \left(-\frac{a}{x}\right)y &= x + \frac{1}{x} \quad \dots(i)
\end{aligned}$$

which is a linear differential equation

Thus, $IF = e^{-a \int \frac{dx}{x}} = e^{-a \log x} = e^{\log(x^{-a})} = \frac{1}{x^a}$

Multiplying both sides of Eq. (i) by IF and integrating we get

$$\begin{aligned}
y \cdot (IF) &= \int Q \cdot (IF) dx + c \\
\Rightarrow \frac{y}{x^a} &= \int (x^{-a} + x^{-a-1}) dx + c \\
&= \left(\frac{x^{-a+1}}{1} - a + \frac{x^{-a}}{-a} + c \right)
\end{aligned}$$

which is the required solution.

114. The given differential equation is

$$\begin{aligned}
(x+1) \frac{dy}{dx} - ny &= e^x (x+1)^{n+1} \\
\Rightarrow \frac{dy}{dx} - \left(\frac{n}{x+1}\right)y &= e^x (x+1)^n \quad \dots(i)
\end{aligned}$$

which is a linear differential equation

Thus, $IF = e^{-n \int \frac{dx}{x+1}} = e^{-n \log(x+1)}$

$$= e^{\log\left(\frac{1}{(x+1)^n}\right)} = \frac{1}{(x+1)^n}$$

Multiplying both sides of Eq. (i) by IF and integrating we get

$$\begin{aligned}
y \cdot (IF) &= \int Q \cdot (IF) dx + c \\
\Rightarrow \frac{y}{(x+1)^n} &= \int e^x dx + c \\
&= e^x + c
\end{aligned}$$

which is the required solution.

115. The given differential equation is

$$\begin{aligned}
(1+y^2) dx &= (\tan^{-1} y - x) dy \\
\Rightarrow \frac{dx}{dy} &= \frac{(\tan^{-1} y - x)}{1+y^2} \\
\Rightarrow \frac{dx}{dy} + \frac{x}{1+y^2} &= \frac{\tan^{-1} y}{1+y^2} \quad \dots(i)
\end{aligned}$$

which is a linear differential equation

Thus, $IF = e^{\int \frac{dy}{1+y^2}} = e^{\tan^{-1}y}$

• Multiplying both sides of Eq. (i) by IF and integrating we get

$$x \cdot (IF) = \int Q \cdot (IF) dy + c$$

$$\Rightarrow x \cdot e^{\tan^{-1}y} = \int \left(\frac{\tan^{-1}y}{1+y^2} \right) e^{\tan^{-1}y} dy$$

$$\Rightarrow x \cdot e^{\tan^{-1}y} = \int t e^t dt + c, t = \tan^{-1}y$$

$$\Rightarrow x \cdot e^{\tan^{-1}y} = e^t(t-1) + c$$

$$\Rightarrow x \cdot e^{\tan^{-1}y} = e^{\tan^{-1}y}(\tan^{-1}y - 1) + c$$

which is the required solution.

116. The given differential equation is

$$(x + 3y + 2) \frac{dx}{dy} = 1$$

$$\Rightarrow \frac{dx}{dy} = (x + 3y + 2)$$

$$\Rightarrow \frac{dx}{dy} - x = (3y + 2) \quad \dots(i)$$

which is a linear differential equation

Thus, $IF = e^{-\int dy} = e^{-y}$

Multiplying both sides of Eq. (i) by IF and integrating we get

$$x \cdot (IF) = \int Q \cdot (IF) dy + c$$

$$\begin{aligned} \Rightarrow x \cdot e^{-y} &= \int \{3(ye^{-y}) + 2(e^{-y})\} dy + c \\ &= 3(-ye^{-y} - e^{-y}) - 2e^{-y} + c \\ &= -3(y+1)e^{-y} - 2e^{-y} + c \end{aligned}$$

$$\Rightarrow x = -3(y+1) - 2 + ce^y$$

which is the required solution.

117. The given differential equation is

$$(1+y^2)dx = (xy + y^3 + y)dy$$

$$\Rightarrow \frac{dx}{dy} = \frac{xy}{y^2+1} + y$$

$$\Rightarrow \frac{dx}{dy} + \left(\frac{y}{y^2+1} \right) x = y \quad \dots(i)$$

which is a linear differential equation

Thus, $IF = e^{-\int \frac{ydy}{y^2+1}} = e^{-\frac{1}{2} \log y^2 + 1} = \frac{1}{\sqrt{y^2+1}}$

Multiplying both sides of Eq. (i) by IF and integrating we get

$$x \cdot (IF) = \int Q \cdot (IF) dy + c$$

$$\Rightarrow \left(\frac{x}{\sqrt{y^2+1}} \right) = \int \frac{ydy}{\sqrt{y^2+1}} + c$$

$$\Rightarrow \left(\frac{x}{\sqrt{y^2+1}} \right) = \sqrt{y^2+1} + c$$

which is the required solution.

118. The given differential equation is

$$\frac{dy}{dx} + \frac{y}{x} \log x = \frac{y}{x^2} (\log y)^2$$

Dividing both the sides by $y(\log y)^2$, we get

$$\frac{1}{y(\log y)^2} \cdot \frac{dy}{dx} + \frac{1}{\log y} \cdot \frac{1}{x} = \frac{1}{x^2}$$

Let $\frac{1}{\log y} = v$

$$\Rightarrow \frac{1}{y(\log y)^2} \cdot \frac{dy}{dx} = -\frac{dv}{dx}$$

$$\Rightarrow -\frac{dv}{dx} + v \cdot \frac{1}{x} = \frac{1}{x^2}$$

$$\Rightarrow \frac{dv}{dx} - \frac{v}{x} = -\frac{1}{x^2}$$

which is a linear differential equation.

Thus, $IF = e^{-\int \frac{1}{x} dx} = e^{-\log x} = e^{\log x^{-1}} = \frac{1}{x}$

Hence the solution is

$$v \cdot (IF) = \int Q \cdot (IF) dx + c$$

$$\Rightarrow v \cdot \left(\frac{1}{x} \right) = \int -\frac{1}{x^2} \cdot \left(\frac{1}{x} \right) dx + c$$

$$\Rightarrow \frac{1}{x \log y} - \frac{1}{2x^2} = c$$

119. The given differential equation is

$$\frac{dy}{dx} + xy = x^3 y^6$$

$$\Rightarrow \frac{1}{y^6} \frac{dy}{dx} + \frac{x}{y^5} = x^3 \quad \dots(i)$$

Let $\frac{1}{y^5} = v \Rightarrow \frac{1}{y^6} \frac{dy}{dx} = -\frac{1}{5} \frac{dv}{dx}$

$$\Rightarrow -\frac{1}{5} \frac{dv}{dx} + vx = x^3$$

$$\Rightarrow \frac{dv}{dx} - 5vx = 5x^3 \quad \dots(ii)$$

which is a linear differential equation.

$$IF = e^{-5 \int x dx} = e^{-\frac{5x^2}{2}}$$

Multiplying both sides of Eq. (ii) by IF and integrating, we get

$$\begin{aligned} v \cdot (\text{IF}) &= \int Q \cdot (\text{IF}) dx + C \\ \Rightarrow v \cdot e^{-\frac{5x^2}{2}} &= -5 \int x^3 \cdot e^{-\frac{5x^2}{2}} dx + C \\ \Rightarrow \frac{e^{-\frac{5x^2}{2}}}{y^5} &= \frac{2}{5} \left(\frac{5x^2}{2} + 1 \right) e^{-\frac{5x^2}{2}} + C \end{aligned}$$

which is the required solution of the given differential equation.

120. The given differential equation is

$$\begin{aligned} \frac{dy}{dx} + \frac{\sin 2y}{x} &= x^3 \cos^2 y \\ \Rightarrow \sec^2 y \frac{dy}{dx} + \frac{2 \tan y}{x} &= x^3 \quad \dots(i) \end{aligned}$$

Let $\tan y = v$

$$\Rightarrow \sec^2 y \frac{dy}{dx} = \frac{dv}{dx}$$

$$\Rightarrow \frac{dv}{dx} + \frac{2v}{x} = x^3 \quad \dots(ii)$$

which is a linear differential equation

$$\text{IF} = e^{2 \int \frac{dx}{x}} = e^{2 \log x} = x^2$$

Multiplying both sides of Eq. (ii) by IF and integrating, we get

$$\begin{aligned} v \cdot (\text{IF}) &= \int Q \cdot (\text{IF}) dx + c \\ \Rightarrow v \cdot x^2 &= \int x^5 dx + c \\ \Rightarrow x^2 \tan y &= \frac{x^6}{6} + c \end{aligned}$$

which is the required solution of the given differential equation.

121. The given differential equation is

$$\begin{aligned} \frac{dy}{dx} - x^3 y^3 + xy &= 0 \\ \Rightarrow \frac{dy}{dx} + xy &= x^3 y^3 \\ \Rightarrow \frac{1}{y^3} \frac{dy}{dx} + xy^{-2} &= x^3 \quad \dots(i) \end{aligned}$$

$$\text{Let } y^{-2} = v \Rightarrow \frac{1}{y^3} \frac{dy}{dx} = -\frac{1}{2} \frac{dv}{dx}$$

$$\begin{aligned} \Rightarrow -\frac{1}{2} \frac{dv}{dx} + vx &= x^3 \\ \Rightarrow \frac{dv}{dx} - 2vx &= -2x^3 \quad \dots(ii) \end{aligned}$$

which is a linear differential equation.

$$\text{IF} = e^{-2 \int x dx} = e^{-x^2}$$

Multiplying both sides of Eq. (ii) by IF and integrating, we get

$$\begin{aligned} v \cdot (\text{IF}) &= \int Q \cdot (\text{IF}) dx + c \\ \Rightarrow v \cdot e^{-x^2} &= \int x^3 e^{-x^2} dx + c \\ \Rightarrow v \cdot e^{-x^2} &= -\frac{1}{2} (x^2 + 1) e^{-x^2} + c \\ \Rightarrow \frac{e^{-x^2}}{y^2} &= -\frac{1}{2} (x^2 + 1) e^{-x^2} + c \end{aligned}$$

which is the required solution of the given differential equation.

122. The given differential equation is

$$\begin{aligned} (1 - x^2) \frac{dy}{dx} + xy &= xy^2 \\ \Rightarrow \frac{dy}{dx} + \frac{xy}{1 - x^2} &= \frac{xy^2}{1 - x^2} \\ \Rightarrow \frac{1}{y^2} \frac{dy}{dx} + \frac{x}{y} (1 - x^2) &= \frac{x}{1 - x^2} \quad \dots(i) \end{aligned}$$

$$\text{Let } \frac{1}{y} = v \Rightarrow \frac{1}{y^2} \frac{dy}{dx} = -\frac{dv}{dx}$$

$$\Rightarrow \frac{dv}{dx} - \frac{vx}{1 - x^2} = \frac{x}{1 - x^2} \quad \dots(ii)$$

which is a linear differential equation.

$$\text{IF} = e^{-\int \frac{x}{1-x^2} dx} = e^{\frac{1}{2} \log |1-x^2|} = \sqrt{1-x^2}$$

Multiplying both sides of Eq. (ii) by IF and integrating, we get,

$$\begin{aligned} v \cdot (\text{IF}) &= \int Q \cdot (\text{IF}) dx + C \\ \Rightarrow v \sqrt{1-x^2} &= -\int \frac{x \sqrt{1-x^2}}{1-x^2} dx \\ &= -\int \frac{x}{\sqrt{1-x^2}} dx \\ &= \sqrt{1-x^2} + C \\ &= \sqrt{1-x^2} + C \end{aligned}$$

which is the required solution of the given differential equation.

123. The given differential equation is

$$x \frac{dy}{dx} + y = x^3 y^6$$

$$\Rightarrow \frac{dy}{dx} + \frac{y}{x} = x^2 y^6$$

$$\Rightarrow \frac{1}{y^6} \frac{dy}{dx} + \frac{y^{-5}}{x} = x^2 \quad \dots(i)$$

Put $y^{-5} = v$

$$\Rightarrow \frac{-5}{y^6} \frac{dy}{dx} = \frac{dv}{dx}$$

$$\Rightarrow \frac{1}{y^6} \frac{dy}{dx} = -\frac{1}{5} \frac{dv}{dx}$$

$$\Rightarrow -\frac{1}{5} \frac{dv}{dx} + \frac{v}{x} = x^2$$

$$\Rightarrow \frac{dv}{dx} - \frac{5v}{x} = -5x^2 \quad \dots(ii)$$

which is a linear differential equation

Thus, $IF = e^{-5 \int \frac{dx}{x}} = e^{-5 \log x} = \frac{1}{x^5}$

Multiplying both sides of Eq. (ii) by IF and integrating, we get

$$v \cdot (IF) = \int Q \cdot (IF) dx + c$$

$$\Rightarrow \frac{1}{(xy)^5} = -5 \int \frac{dx}{x^3} + c$$

$$\Rightarrow \frac{1}{(xy)^5} = \frac{5}{2x^2} + c$$

which is the required solution.

124. The given differential equation is

$$(1 + x^2) \frac{dy}{dx} + y = e^{\tan^{-1} x}$$

$$\Rightarrow \frac{dy}{dx} + \frac{y}{(x^2 + 1)} = \frac{e^{\tan^{-1} x}}{(x^2 + 1)} \quad \dots(i)$$

which is a linear differential equation

Thus, $IF = e^{\int \frac{dx}{x^2 + 1}} = e^{\tan^{-1} x}$

Multiplying both sides of Eq. (i) by IF and integrating, we get

$$y \cdot e^{\tan^{-1} x} = \int \frac{(e^{\tan^{-1} x})^2}{(x^2 + 1)} dx + c$$

$$= \int e^{2t} dt + c, \quad t = \tan^{-1} x$$

$$= \int \frac{e^{2t}}{2} + c$$

$$\Rightarrow \log x^2 + c_1 + 2 \tan^{-1} \left(\frac{x}{C_1} \right) + y = c_2$$

$$\Rightarrow y \cdot e^{\tan^{-1} x} = \frac{e^{2 \tan^{-1} x}}{2} + c$$

which is the required solution of the given differential equation.

125. The given differential equation is

$$\frac{dy}{dx} (x^2 y^3 + xy) = 1$$

$$\Rightarrow \frac{dx}{dy} = x^2 y^3 + xy$$

$$\Rightarrow \frac{dx}{dy} - xy = x^2 y^3$$

$$\Rightarrow \frac{1}{x^2} \frac{dx}{dy} - \frac{y}{x} = y^3 \quad \dots(i)$$

Let $v = -\frac{1}{x}$

$$\Rightarrow \frac{dv}{dy} = \frac{1}{x^2} \frac{dx}{dy}$$

$$\Rightarrow \frac{dv}{dy} + vy = y^3 \quad \dots(ii)$$

which is a linear differential equation

Thus, $IF = e^{\int y dy} = e^{\frac{y^2}{2}}$

Multiplying both sides of Eq. (ii) by IF and integrating, we get

$$v \cdot e^{\frac{y^2}{2}} = \int y^3 e^{\frac{y^2}{2}} dy + c$$

$$\Rightarrow -\frac{e^{\frac{y^2}{2}}}{x} = 2 \int y \left(\frac{y^2}{2} \right) e^{\frac{y^2}{2}} dy + c$$

$$\Rightarrow -\frac{e^{\frac{y^2}{2}}}{x} = (y^2 - 2) e^{\frac{y^2}{2}} dy + c$$

which is the required solution.

126. The given differential equation is

$$(y \log - 1) y dx = x dy$$

$$y^2 \log x dx - y dx = x dy$$

$$x \frac{dy}{dx} + y = y^2 \log x$$

$$\frac{dy}{dx} + \frac{y}{x} = \frac{y^2 \log x}{x}$$

$$\frac{1}{y^2} \frac{dy}{dx} + \frac{1}{xy} = \frac{\log x}{x}$$

$$-\frac{dv}{dx} + \frac{v}{x} = \frac{\log x}{x}, \quad \left(\text{Let } v = \frac{1}{y} \right)$$

$$\frac{dv}{dx} - \frac{v}{x} = -\frac{\log x}{x} \quad \dots(i)$$

which is a linear differential equation

Thus, $\text{I.F.} = e^{-\int \frac{1}{x} dx} = e^{-\log x} = \frac{1}{x}$

Multiplying both sides of Eq. (i) IF and integrating, we get

$$\begin{aligned} v \cdot \frac{1}{x} &= -\int \frac{\log x}{x^2} dx \\ \Rightarrow \frac{v}{x} &= -\int t e^{-t} dt, \quad (\text{Let } t = \log x) \\ &= (t + 1)e^{-t} + c \end{aligned}$$

$$\Rightarrow \frac{1}{xy} = \left(\frac{\log x + 1}{x} \right) + c$$

which is the required solution of the given differential equation.

127. The given differential equation is

$$\begin{aligned} \frac{dy}{dx} &= x^3 y^3 = xy \\ \Rightarrow \frac{dy}{dx} + xy &= x^3 y^3 \\ \Rightarrow \frac{1}{y^3} \frac{dy}{dx} + \frac{x}{y^2} &= x^3 y^3 \quad \dots(i) \end{aligned}$$

$$\begin{aligned} \text{Let } \frac{1}{y^2} &= v \\ \Rightarrow -\frac{2}{y^3} \frac{dy}{dx} &= \frac{dv}{dx} \end{aligned}$$

$$\begin{aligned} \Rightarrow -\frac{1}{2} \frac{dv}{dx} + vx &= x^2 \\ \Rightarrow \frac{dv}{dx} - 2vx &= -2x^3 \quad \dots(ii) \end{aligned}$$

which is a linear differential equation

Thus, $\text{IF} = e^{-2 \int x dx} = e^{-x^2}$

Multiplying both sides of Eq. (ii) by IF and integrating, we get

$$\begin{aligned} v(\text{IF}) &= \int Q(\text{IF}) dx + c \\ \Rightarrow v \cdot e^{-x^2} &= \int 2x^3 e^{-x^2} dx + c \\ &= (1 + x^2)e^{-x^2} + c \\ \Rightarrow v &= (1 + x^2) + ce^{x^2} \\ \Rightarrow \frac{1}{y^2} &= (1 + x^2) + ce^{x^2} \end{aligned}$$

which is the required solution of the given differential equation.

128. The given differential equation is

$$\frac{dy}{dx} + \frac{xy}{1-x^2} = x\sqrt{y}$$

$$\Rightarrow \frac{1}{\sqrt{y}} \frac{dy}{dx} + \frac{x\sqrt{y}}{1-x^2} = x \quad \dots(i)$$

Let $\sqrt{y} = v$

$$\Rightarrow \frac{1}{2\sqrt{y}} \frac{dy}{dx} = \frac{dv}{dx}$$

$$\Rightarrow \frac{1}{\sqrt{y}} \frac{dy}{dx} = 2 \frac{dv}{dx}$$

$$\Rightarrow 2 \frac{dv}{dx} + \frac{vx}{1-x^2} = x$$

$$\Rightarrow \frac{dv}{dx} + \left(\frac{x}{2(1-x^2)} \right) v = \frac{x}{2} \quad \dots(ii)$$

which is a linear differential equation.

Thus, $\text{IF} = e^{\int \frac{xdx}{2(1-x^2)}} = e^{-\frac{1}{4} \log |1-x^2|} = \frac{1}{\sqrt[4]{1-x^2}}$

Multiplying both sides of Eq. (ii) by IF and integrating, we get

$$\begin{aligned} v \cdot \frac{1}{\sqrt[4]{1-x^2}} &= \frac{1}{2} \int \frac{xdx}{\sqrt[4]{1-x^2}} + c \\ \Rightarrow \frac{\sqrt{y}}{\sqrt[4]{1-x^2}} &= \frac{(1-x^2)^{3/4}}{3} + c \end{aligned}$$

which is the required solution.

129. The given differential equation is

$$\begin{aligned} \frac{dy}{dx} + \frac{y}{x} \log y &= \frac{y}{x^2} (\log y)^2 \\ \Rightarrow \frac{1}{y(\log y)^2} \frac{dy}{dx} + \frac{1}{x(\log y)} &= \frac{1}{x^2} \quad \dots(i) \end{aligned}$$

Let $\frac{1}{(\log y)} = v$

$$\Rightarrow -\frac{1}{y(\log y)^2} \frac{dy}{dx} = \frac{dv}{dx}$$

$$\Rightarrow \frac{1}{y(\log y)^2} \frac{dy}{dx} = -\frac{dv}{dx}$$

$$\Rightarrow -\frac{dv}{dx} + \frac{v}{x} = \frac{1}{x^2}$$

$$\Rightarrow \frac{dv}{dx} - \frac{v}{x} = -\frac{1}{x^2} \quad \dots(ii)$$

which is a linear differential equation.

Thus, $\text{IF} = e^{-\int \frac{dx}{x}} = e^{-\log x} = \frac{1}{x}$

Multiplying both sides of Eq. (ii) by IF and integrating, we get

$$\begin{aligned}\frac{v}{x} &= -\int \frac{dx}{x^3} + c \\ &= \frac{1}{2x^2} + c \\ \Rightarrow \frac{1}{x(\log y)} &= \frac{1}{2x^2} + c\end{aligned}$$

which is the required solution.

130. The given differential equation is

$$\begin{aligned}\frac{dy}{dx} + \frac{1}{x} \sin^2 y &= x^3 \cdot \cos^2 y \\ \Rightarrow \sec^2 y \frac{dy}{dx} + \frac{2 \tan y}{x} &= x^3 \quad \dots(i)\end{aligned}$$

Let $\tan y = v$

$$\begin{aligned}\Rightarrow \sec^2 y \frac{dy}{dx} &= \frac{dv}{dx} \\ \Rightarrow \frac{dv}{dx} + \frac{2v}{x} &= x^3 \quad \dots(ii)\end{aligned}$$

which is a linear differential equation

$$\text{Thus, IF} = e^{2\int \frac{dx}{x}} = e^{2\log x} = x^2$$

Multiplying both sides of Eq. (ii) by IF and integrating, we get

$$\begin{aligned}\Rightarrow v \cdot x^2 &= \int x^5 dx + c \\ &= \frac{x^6}{6} + c \\ \Rightarrow (\tan y)x^2 &= \frac{x^6}{6} + c\end{aligned}$$

which is the required solution of the given differential equation.

131. The given differential equation is

$$\begin{aligned}(xy^2 - e^{1/x^3})dx - x^2 y dy &= 0 \\ \Rightarrow x^2 y \frac{dy}{dx} &= xy^2 - e^{1/x^3} \\ \Rightarrow y \frac{dy}{dx} &= \frac{y^2}{x} - \frac{e^{1/x^3}}{x^2} \\ -\frac{y^2}{x} &= \frac{e^{1/x^3}}{x^2} \quad \dots(i)\end{aligned}$$

Let $y^2 = v$

$$\Rightarrow y \frac{dy}{dx} = \frac{1}{2} \frac{dv}{dx}$$

$$\begin{aligned}\Rightarrow \frac{1}{2} \frac{dv}{dx} - \frac{v}{x} &= -\frac{e^{1/x^3}}{x^2} \\ \Rightarrow \frac{dv}{dx} - \frac{2v}{x} &= -\frac{2e^{1/x^3}}{x^2} \quad \dots(ii)\end{aligned}$$

which is a linear differential equation

$$\text{Thus, IF} = e^{-2\int \frac{dx}{x}} = e^{-2\log x} = \frac{1}{x^2}$$

Multiplying both sides of Eq. (ii) by IF and integrating, we get

$$\begin{aligned}\frac{v}{x^2} &= -2 \int \frac{e^{1/x^3}}{x^4} dx + c \\ \Rightarrow \left(\frac{y}{x}\right)^2 &= \frac{2}{3} e^{1/x^3} + c\end{aligned}$$

which is the required solution.

132. The given differential equation is

$$\begin{aligned}\frac{dy}{dx} + x(x+y) &= x^3(x+y)^3 - 1 \\ \frac{1}{(x+y)^3} \left(\frac{dy+dx}{dx} \right) + \frac{x}{(x+y)^2} &= x^3 \\ \frac{1}{(x+y)^3} \left(\frac{d(x+y)}{dx} \right) + \frac{x}{(x+y)^2} &= x^3 \quad \dots(i)\end{aligned}$$

$$\text{put } \frac{1}{(x+y)^2} = v$$

$$\Rightarrow -\frac{2}{(x+y)^3} \frac{d(x+y)}{dx} = \frac{dv}{dx}$$

$$\Rightarrow \frac{1}{(x+y)^3} \frac{d(x+y)}{dx} = -\frac{1}{2} \frac{dv}{dx}$$

$$\Rightarrow -\frac{1}{2} \frac{dv}{dx} + vx = x^3$$

$$\Rightarrow \frac{dv}{dx} - 2xv = -2x^3$$

which is a linear differential equation

$$\text{Thus, IF} = e^{-2\int x dx} = e^{-x^2}$$

Multiplying both sides of Eq. (ii) by IF and integrating, we get

$$\begin{aligned}v \cdot e^{-x^2} &= -2 \int x^3 e^{-x^2} dx + c \\ &= (x^2 + 1)e^{-x^2} + c\end{aligned}$$

$$\Rightarrow v = (x^2 + 1) + ce^{x^2}$$

$$\Rightarrow \frac{1}{(x^2 + 1)} = (x^2 + 1) + ce^{x^2}$$

which is the required solution of the given differential equation.

133. The given differential equation is

$$x dx + y dy = x dy - y dx$$

$$\Rightarrow \frac{1}{2} d(x^2 + y^2) = x dy - y dx$$

$$\Rightarrow d(x^2 + y^2) = 2(x dy - y dx)$$

$$\Rightarrow \frac{d(x^2 + y^2)}{(x^2 + y^2)} = \frac{2(x dy - y dx)}{(x^2 + y^2)}$$

$$= 2 \frac{d\left(\frac{y}{x}\right)}{1 + \left(\frac{y}{x}\right)^2}$$

Integrating, we get

$$\log |x^2 + y^2| = 2 \tan^{-1}\left(\frac{y}{x}\right) + c$$

which is the required solution of the given differential equation.

134. The given differential equation is

$$x dy - y dx = x^4 dx$$

$$\Rightarrow d\left(\frac{y}{x}\right) = x^2 dx$$

Integrating, we get

$$\Rightarrow \frac{y}{x} = \frac{x^3}{3} + c$$

which is the required solution of the given differential equation.

135. The given differential equation is

$$x dy + y dx = \sin y dy$$

$$\Rightarrow d(xy) = \sin y dy$$

Integrating, we get

$$\Rightarrow xy = c - \cos y$$

which is the required solution of the given differential equation.

136. The given differential equation is

$$x dy + y dx + y^2(x dy - y dx) = 0.$$

$$\Rightarrow d(xy) = y^2(y dx - x dy)$$

$$\Rightarrow \frac{d(xy)}{x^2 y^2} = \frac{y^2(y dx - x dy)}{x^2 y^2}$$

$$\Rightarrow d\left(-\frac{1}{xy}\right) = -d\left(\frac{x}{y}\right)$$

Integrating, we get

$$\Rightarrow -\frac{1}{xy} = -\frac{y}{x} - c$$

$$\Rightarrow \frac{1}{xy} = \frac{y}{x} + c$$

which is the required solution of the given differential equation.

137. The given differential equation is

$$x dy + y dx + xy^2 dx - x^2 y dy = 0.$$

$$\Rightarrow d(xy) = xy(x dx - y dy)$$

$$\Rightarrow \frac{d(xy)}{x^2 y^2} = \frac{(x dy - y dx)}{xy}$$

$$\Rightarrow d\left(-\frac{1}{xy}\right) = \left(\frac{dy}{y} - \frac{dx}{x}\right)$$

Integrating, we get

$$\Rightarrow -\frac{1}{xy} = \log\left(\frac{y}{x}\right) + c$$

$$\Rightarrow \frac{1}{xy} + \log\left(\frac{y}{x}\right) + c = 0$$

which is the required solution of the given differential equation.

138. The given differential equation is

$$(4x - 3y) dx + (2y - 3x) dy = 0$$

$$\Rightarrow 4x dx + 2y dy - 3(x dy + y dx) = 0$$

$$\Rightarrow 4x dx + 2y dy - 3d(xy) = 0$$

Integrating, we get

$$\Rightarrow 2x^2 + y^2 - 3xy = c$$

which is the required solution of the given differential equation.

139. The given differential equation is

$$\left(\sin y + y \sin x + \frac{1}{x}\right) dx$$

$$+ \left(x \cos y - \cos x + \frac{1}{y}\right) dy = 0$$

$$\Rightarrow (\sin y dx + x \cos y dy) - (\cos x dy - y \sin x dx)$$

$$+ \left(\frac{dx}{x} + \frac{dy}{y}\right) = 0$$

$$\Rightarrow d(x \sin y) - d(y \cos x) + \left(\frac{dx}{x} + \frac{dy}{y}\right) = 0$$

Integrating, we get

$$\Rightarrow x \sin y - y \cos x + \log(xy) = c$$

which is the required solution of the given differential equation.

140. The given differential equation is

$$\frac{x dx + y dy}{\sqrt{x^2 + y^2}} = \frac{y dx - x dy}{x^2}$$

$$\Rightarrow \frac{d(x^2 + y^2)}{\sqrt{x^2 + y^2}} = 2d\left(\frac{y}{x}\right)$$

$$\Rightarrow d(\sqrt{x^2 + y^2}) = 2d\left(\frac{y}{x}\right)$$

Integrating, we get

$$\Rightarrow (\sqrt{x^2 + y^2}) = 2\left(\frac{y}{x}\right) + c$$

which is the required solution of the given differential equation.

141. The given differential equation is

$$\left(\frac{\sin 2x}{y} + x\right)dx + \left(y - \frac{\sin^2 x}{y^2}\right)dy = 0$$

$$\Rightarrow (x dy + y dy) + \left(\frac{\sin 2x}{y} dx - \frac{\sin^2 x}{y^2} dy\right) = 0$$

$$\Rightarrow d(xy) + d\left(\frac{\sin^2 x}{y}\right) = 0$$

Integrating, we get

$$\Rightarrow xy + \frac{\sin^2 x}{y} = c$$

which is the required solution of the given differential equation.

142. The given differential equation is

$$\frac{x + y \frac{dy}{dx}}{y - x \frac{dy}{dx}} = x^2 + 2y^2 + \frac{y^4}{x^2}$$

$$\Rightarrow \frac{x dx + y dy}{y dx - x dy} = \frac{x^4 + 2x^2 y^2 + y^4}{x^2}$$

$$= \frac{(x^2 + y^2)^2}{x^2}$$

$$\Rightarrow \frac{x dy + y dx}{(x^2 + y^2)^2} = \frac{y dx - x dy}{x^2}$$

$$\Rightarrow \frac{d(x^2 + y^2)}{(x^2 + y^2)^2} = -2d\left(\frac{y}{x}\right),$$

Integrating, we get

$$\frac{1}{(x^2 + y^2)} = c - \frac{y}{x}$$

$$\Rightarrow \frac{y}{x} - \frac{1}{(x^2 + y^2)} = c$$

which is the required solution of the given differential equation.

143. The given differential equation is

$$x + y \frac{dy}{dx} = \frac{a^2 \left(x \frac{dy}{dx} - y\right)}{x^2 + y^2} \quad \dots(i)$$

Let $x = r \cos \theta$ and $y = r \sin \theta$,

$$\Rightarrow x^2 + y^2 = r^2, \tan \theta = \frac{y}{x}$$

$$\text{and } x dx + y dy = r dr, x dy - y dx = r^2 d\theta$$

$$\Rightarrow r dr = \frac{a^2 \cdot r^2 d\theta}{r^2}$$

Integrating, we get

$$\int r dr = \int a^2 \cdot d\theta$$

$$\Rightarrow \frac{r^2}{2} = a^2 \theta + c$$

$$\Rightarrow \frac{(x^2 + y^2)}{2} = a^2 \tan^{-1}\left(\frac{y}{x}\right) + c$$

$$\Rightarrow (x^2 + y^2) = 2a^2 \tan^{-1}\left(\frac{y}{x}\right) + k$$

which is the required solution.

144. The given differential equation is

$$x dx + y dy = (x^2 + y^2) y dy$$

$$\Rightarrow \frac{x dx + y dy}{x^2 + y^2} = y dy$$

$$\Rightarrow \frac{1}{2} d(\log(x^2 + y^2)) = y dy$$

Integrating, we get

$$\frac{1}{2} (\log(x^2 + y^2)) = \frac{y^2}{2} + \frac{c}{2}$$

$$\Rightarrow \log(x^2 + y^2) = y^2 + c$$

which is the required solution.

145. The given differential equation is

$$(x + y)dx + (x - y)dy = 0$$

$$\Rightarrow (x dx - y dy) + (x dy - y dx) = 0$$

$$(x dx - y dy) + d(xy) = 0$$

Integrating, we get

$$\frac{x^2}{2} - \frac{y^2}{2} + xy = c$$

which is the required solution.

146. The given differential equation is

$$y dx + x(x - 1)dy = 0$$

$$\Rightarrow y dx - x dy = -x^2 dy$$

$$\Rightarrow x dy - y dx = x^2 dy$$

$$\Rightarrow \frac{x dy - y dx}{x^2} = dy$$

$$\Rightarrow d\left(\frac{y}{x}\right) = dy$$

Integrating, we get

$$\left(\frac{y}{x}\right) = y + c$$

which is the required solution.

147. The given differential equation is

$$y dx - x(1 - xy)dy = 0$$

$$\Rightarrow \frac{y dx - x dy}{x^2} = y dy$$

$$\Rightarrow d\left(\frac{y}{x}\right) = y dy$$

Integrating, we get

$$\left(\frac{y}{x}\right) = \frac{y^2}{2} + c$$

which is the required solution.

148. The given differential equation is

$$(x + y)(dx - dy) = (dx + dy)$$

$$\Rightarrow (x + y) d(x - y) = d(x + y)$$

$$\Rightarrow d(x - y) = \frac{d(x + y)}{(x + y)}$$

Integrating, we get

$$(x - y) = \log|x + y|$$

which is the required solution.

149. The given differential equation is

$$dx + dy = x dy + y dx$$

$$\Rightarrow d(x + y) = d(xy)$$

Integrating, we get

$$(x + y) = xy + c$$

which is the required solution.

150. The given differential equation is

$$x dy - y dx = (x^2 + y^2) dx$$

$$\Rightarrow \frac{x dy - y dx}{(x^2 + y^2)} = dx$$

$$\Rightarrow \frac{y dx - x dy}{(x^2 + y^2)} = -dx$$

$$\Rightarrow \frac{y dx - x dy}{x^2} = -dx$$

$$\Rightarrow \frac{d\left(\frac{y}{x}\right)}{\left(1 + \left(\frac{y}{x}\right)^2\right)} = -dx$$

$$\Rightarrow \frac{d\left(\frac{x}{y}\right)}{\left(1 + \left(\frac{y}{x}\right)^2\right)} = -dx$$

Integrating, we get

$$\tan^{-1}\left(\frac{y}{x}\right) = c - x$$

which is the required solution.

151. The given differential equation is

$$\left(\frac{1}{x} + \frac{1}{y}\right)(x dy + y dx) = dx + dy$$

$$\Rightarrow \left(\frac{x + y}{xy}\right)(x dy + y dx) = d(x + y)$$

$$\Rightarrow \left(\frac{x + y}{xy}\right)d(xy) = d(x + y)$$

$$\Rightarrow \frac{d(xy)}{xy} = \frac{d(x + y)}{(x + y)}$$

Integrating, we get

$$\log|xy| = \log|x + y| + \log c$$

$$\Rightarrow (xy) = c(x + y)$$

which is the required solution.

152. The given differential equation is

$$(x dy + y dx)\sqrt{x^2 + y^2} = x^2 y dx + xy^2 dy$$

$$\Rightarrow \frac{x dy + y dx}{xy} = \frac{x dx + y dy}{\sqrt{x^2 + y^2}}$$

$$\Rightarrow \frac{d(xy)}{xy} = \frac{1}{2} \frac{d(x^2 + y^2)}{\sqrt{x^2 + y^2}}$$

$$= \frac{d(\sqrt{x^2 + y^2})}{\sqrt{x^2 + y^2}}$$

Integrating, we get

$$\log |xy| = \sqrt{x^2 + y^2} + c$$

which is the required solution.

153. The given differential equation is

$$(x - y^3)dy = y(dx + x^2dy)$$

$$\Rightarrow xdy - y^3dy = ydx + x^2ydy$$

$$\Rightarrow \frac{xdy - ydx}{y} = y^2dy + x^2dy$$

$$\Rightarrow \frac{xdy - ydx}{(x^2 + y^2)} = ydy$$

$$\Rightarrow \frac{d\left(\frac{x}{y}\right)}{\left(1 + \left(\frac{y}{x}\right)^2\right)} = ydy$$

Integrating, we get

$$\tan^{-1}\left(\frac{y}{x}\right) = \frac{y^2}{2} + c$$

which is the required solution.

154. The given differential equation is

$$(x + 2y)dy + ydx = 0$$

$$\Rightarrow (xdy + ydx) + 2ydy = 0$$

$$\Rightarrow d(xy) + 2ydy = 0$$

Integrating, we get

$$(xy) + y^2 = c$$

which is the required solution.

155. The given differential equation is

$$(x + \sin y)dy + ydx = 0$$

$$\Rightarrow (xdy + ydx) + \sin ydy = 0$$

$$\Rightarrow d(xy) + \sin ydy = 0$$

Integrating, we get

$$(xy) - \cos y = c$$

which is the required solution.

156. The given differential equation is

$$e^y dx + (xe^y - 2y)dy = 0$$

$$\Rightarrow (e^y dx + xe^y dy) = 2ydy$$

$$\Rightarrow d(xe^y) = 2ydy$$

Integrating, we get

$$(xe^y) = \frac{y^2}{2} + c$$

which is the required solution.

157. The given differential equation is

$$2ydy + (\cos x \cdot \cot y - y^2)dx$$

$$= (2xy + \sin x \cdot \operatorname{cosec}^2 y)dy$$

$$\Rightarrow 2ydy - d(xy^2)$$

$$-(\sin x \operatorname{cosec}^2 y dy - \cos x \cot y dx) = 0$$

$$\Rightarrow 2ydy - d(xy^2) + d(\sin x \cot y) = 0$$

Integrating, we get

$$y^2 - xy^2 + (\sin x \cot y) = c$$

which is the required solution.

158. The given differential equation is

$$\frac{dy}{dx} = \frac{2x - 3y + 1}{3x - 2y - 2}$$

$$\Rightarrow 3xdy - 2(y + 1)dy = 2xdx - 3ydx + dx$$

$$\Rightarrow 3(xdy + ydx) - 2(y + 1)dy = (2x + 1)dx$$

$$\Rightarrow 3d(xy) - 2(y + 1)dy = (2x + 1)dx$$

Integrating, we get

$$3(xy) - 2\left(\frac{y^2}{2} + y\right) = \left(\frac{x^2}{2} + x\right) + c$$

which is the required solution.

159. The given differential equation is

$$x(dy + dx) = y(dx - dy)$$

$$\Rightarrow \frac{xdy - ydx}{(x^2 + y^2)} = \frac{xdx + ydy}{(x^2 + y^2)}$$

$$\Rightarrow \frac{xdy - ydx}{y^2} = \frac{1}{2} \frac{d(x^2 + y^2)}{(x^2 + y^2)}$$

$$\Rightarrow \frac{d\left(\frac{x}{y}\right)}{1 + \left(\frac{x}{y}\right)^2} = \frac{1}{2} \frac{d(x^2 + y^2)}{(x^2 + y^2)}$$

Integrating, we get

$$\tan^{-1}\left(\frac{x}{y}\right) = \frac{1}{2} \log |(x^2 + y^2)| + c$$

which is the required solution.

160. The given differential equation is

$$x(\log x - \tan^{-1} x) dy + y dx = \left(\frac{xy}{1+x^2}\right) dx$$

$$\Rightarrow x(\log x - \tan^{-1} x) \frac{dy}{dx} + y = \frac{xy}{1+x^2}$$

$$\Rightarrow x(\log x - \tan^{-1} x) \frac{dy}{dx} = \frac{xy}{1+x^2} - y$$

$$\Rightarrow x(\log x - \tan^{-1} x) \frac{dy}{dx} = y \left(\frac{x}{1+x^2} - 1 \right)$$

$$\Rightarrow (\log x - \tan^{-1} x) \frac{dy}{dx} = y \left(\frac{x}{1+x^2} - \frac{1}{x} \right)$$

$$\Rightarrow \frac{d}{dx} \{y(\log x - \tan^{-1} x)\} = 0$$

Integrating, we get

$$y(\tan^{-1} x - \log x) = C$$

which is the required solution.

161. The given differential equation is

$$x(dy - x dx) + y dx = 0$$

$$\Rightarrow x dy + y dx = x^2 dx$$

$$\Rightarrow d(xy) = x^2 dx$$

Integrating, we get

$$xy = \frac{x^3}{3} + c$$

which is the required solution.

162. The given differential equation is

$$\Rightarrow \frac{y - x \frac{dy}{dx}}{y + x \frac{dy}{dx}} = \frac{1}{x^2} + \frac{1}{y^2}$$

$$\Rightarrow \frac{y dx - x dy}{y dx + x dy} = \frac{x^2 + y^2}{(xy)^2}$$

$$\Rightarrow \frac{y dx - x dy}{(x^2 + y^2)} = \frac{y dx + x dy}{(xy)^2}$$

$$\Rightarrow \frac{\frac{y dx - x dy}{y^2}}{\left(1 + \left(\frac{x}{y}\right)^2\right)} = \frac{d(xy)}{(xy)^2}$$

$$\Rightarrow \frac{d\left(\frac{x}{y}\right)}{1 + \left(\frac{x}{y}\right)^2} = \frac{d(xy)}{(xy)^2}$$

Integrating, we get

$$\tan^{-1}\left(\frac{x}{y}\right) + \frac{1}{xy} = c$$

which is the required solution.

163. The given differential equation is

$$x^3 x dy - x^2 y^2 dx = x^4 dy + x^3 y dx$$

$$\Rightarrow \frac{x^2 y (x dy - y dx)}{x^3} = (x dy + y dx)$$

$$\Rightarrow xy \left(\frac{x dy - y dx}{x^2} \right) = d(xy)$$

$$\Rightarrow d\left(-\frac{y}{x}\right) = \frac{d(xy)}{(xy)}$$

Integrating, we get

$$\log |xy| - \frac{y}{x} = c$$

which is the required solution.

164. The given differential equation is

$$dx + x(y dx + x dy) = e^{-xy} dx$$

$$\Rightarrow dx + x d(xy) = e^{-xy} dx$$

$$\Rightarrow (1 - e^{-xy}) dx + x d(xy) = 0$$

$$\Rightarrow \frac{dx}{x} + \frac{d(xy)}{(1 - e^{-xy})} = 0$$

Integrating, we get

$$\log |x| + \log |e^{xy} - 1| = \log c$$

$$\Rightarrow x(e^{xy} - 1) = c$$

which is the required solution.

165. The given differential equation is

$$y(y^2 dx + (x^2 dy - xy dx)) = e^{\frac{x}{y}} dy$$

$$\Rightarrow y^3 dx + xy(x dy - y dx) = e^{\frac{x}{y}} dy$$

$$\Rightarrow y^3 dx + xy^3 \left(\frac{x dy - y dx}{y^2} \right) = e^{\frac{x}{y}} dy$$

$$\Rightarrow y^3 dx + xy^3 d(-xy) = e^{-\frac{x}{y}} dy$$

166. The given differential equation is

$$ydx - xdy + xy^2 dx = 0$$

$$\Rightarrow xdy - ydx = xy^2 dx$$

$$\Rightarrow \frac{xdy - ydx}{x^2} = xdx$$

$$\Rightarrow d\left(\frac{x}{y}\right) = xdx$$

Integrating, we get

$$\left(\frac{x}{y}\right) = \frac{x^2}{2} + c$$

which is the required solution.

167. The given differential equation is

$$2\left(x - y\frac{dy}{dx}\right)(x^2 + y^2) = (x^2 - y^2)\left(y - x\frac{dy}{dx}\right)$$

$$\Rightarrow \frac{2\left(x - y\frac{dy}{dx}\right)}{(x^2 - y^2)} = \frac{\left(y - x\frac{dy}{dx}\right)}{(x^2 + y^2)}$$

$$\Rightarrow \frac{2(xdx - ydy)}{(x^2 - y^2)} = \frac{(ydx - xdy)}{(x^2 + y^2)}$$

$$\Rightarrow \frac{d(x^2 - y^2)}{(x^2 - y^2)} = \frac{\left(\frac{ydx - xdy}{y^2}\right)}{\left(1 + \left(\frac{x}{y}\right)^2\right)}$$

$$= \frac{d\left(\frac{x}{y}\right)}{\left(1 + \left(\frac{x}{y}\right)^2\right)}$$

Integrating, we get

$$\log|x^2 - y^2| = \tan^{-1}\left(\frac{x}{y}\right) + c$$

which is the required solution.

168. The given differential equation is

$$(x^2 + y^2 + a^2)y dy = (x^2 - y^2 - a^2)x dx$$

$$\Rightarrow x^2 y dy + y^2 x dx + (y^3 + a^2 y) dy = (x^3 - a^2 x) dx$$

$$\Rightarrow xy(xdy + ydx) + (y^3 + a^2 y) dy = (x^3 - a^2 x) dx$$

$$\Rightarrow xy d(xy) + (y^3 + a^2 y) dy = (x^3 - a^2 x) dx$$

Integrating, we get

$$\frac{(xy)^2}{2} + \frac{y^4}{2} + \frac{a^2 y^2}{2} = \frac{x^4}{4} - \frac{a^2 x^2}{2} + c$$

which is the required solution.

169. The given differential equation is

$$(1 + xy)ydx + x(1 - xy)dy = 0$$

$$\Rightarrow ydx + xdy + xy(ydx - xdy) = 0$$

$$\Rightarrow d(xy) + xy(ydx - xdy) = 0$$

$$\Rightarrow \frac{d(xy)}{(xy)} + (ydx - xdy) = 0$$

$$\Rightarrow \frac{d(xy)}{(xy)} = (xdy - ydx)$$

$$\Rightarrow \frac{d(xy)}{(xy)^2} = \left(\frac{xdy - ydx}{xy}\right) = \frac{dy}{y} - \frac{dx}{x}$$

Integrating, we get

$$-\frac{1}{xy} = \log\left|\frac{y}{x}\right| + c$$

which is the required solution.

170. The given differential equation is

$$\frac{xdy}{x^2 + y^2} = \left(\frac{y}{x^2 + y^2} - 1\right) dx$$

$$\Rightarrow xdy = ydx - (x^2 + y^2) dx$$

$$\Rightarrow xdy - ydx = -(x^2 + y^2) dx$$

$$\Rightarrow \frac{xdy - ydx}{(x^2 + y^2)} = -dx$$

$$\Rightarrow \frac{\frac{xdy - ydx}{x^2}}{\left(1 + \left(\frac{y}{x}\right)^2\right)} = -dx$$

$$\Rightarrow \frac{d\left(\frac{x}{y}\right)}{\left(1 + \left(\frac{y}{x}\right)^2\right)} = -dx$$

Integrating, we get

$$\tan^{-1}\left(\frac{y}{x}\right) = c - x$$

which is the required solution.

171. The given differential equation is

$$\frac{x + y \frac{dy}{dx}}{y - x \frac{dy}{dx}} = x^2 + 2y^2 + \frac{y^4}{x^2}$$

$$\Rightarrow \frac{xdx + ydy}{ydx - xdy} = \frac{(x^2 + y^2)^2}{x^2}$$

$$\Rightarrow \frac{1}{2} \frac{d(x^2 + y^2)}{(x^2 + y^2)^2} = \frac{ydx - xdy}{x^2}$$

$$\Rightarrow \frac{1}{2} \frac{d(x^2 + y^2)}{(x^2 + y^2)^2} + d\left(\frac{y}{x}\right) = 0$$

Integrating, we get

$$\frac{y}{x} - \frac{1}{2(x^2 + y^2)} = c$$

which is the required solution.

172. The given differential equation is

$$xdx + ydy = m(xdy - ydx)$$

$$\Rightarrow \frac{xdx + ydy}{(x^2 + y^2)} = \frac{m(xdy - ydx)}{(x^2 + y^2)}$$

$$\Rightarrow \frac{1}{2} \frac{d(x^2 + y^2)}{(x^2 + y^2)} = \frac{md\left(\frac{y}{x}\right)}{\left(1 + \left(\frac{y}{x}\right)^2\right)}$$

Integrating, we get

$$\tan^{-1}\left(\frac{y}{x}\right) + \frac{1}{2(x^2 + y^2)} = c$$

which is the required solution.

173. The given differential equation is

$$\frac{x + y \frac{dy}{dx}}{y - x \frac{dy}{dx}} = \frac{x \sin^2(x^2 + y^2)}{y^3}$$

$$\Rightarrow \frac{xdx + ydy}{ydx - xdy} = \frac{x \sin^2(x^2 + y^2)}{y^3}$$

$$\Rightarrow \frac{xdx + ydy}{\sin^2(x^2 + y^2)} = \frac{x(ydx + xdy)}{y^3}$$

$$\Rightarrow \frac{1}{2} \frac{d(x^2 + y^2)}{\sin^2(x^2 + y^2)} + \frac{x}{y} d\left(\frac{y}{x}\right) = 0$$

Integrating, we get

$$\frac{1}{2} \left(\frac{x}{y}\right)^2 + \frac{1}{2} \cot(x^2 + y^2) = c$$

which is the required solution.

174. The given differential equation is

$$x + y \frac{dy}{dx} = \frac{a^2 \left(x \frac{dy}{dx} - y\right)}{x^2 + y^2}$$

$$\Rightarrow xdx + ydy = \frac{a^2(xdy - ydx)}{(x^2 + y^2)}$$

$$= \frac{a^2 \left(\frac{xdy - ydx}{y^2}\right)}{\left(1 + \left(\frac{x}{y}\right)^2\right)}$$

$$= \frac{a^2 d\left(\frac{x}{y}\right)}{\left(1 + \left(\frac{x}{y}\right)^2\right)}$$

$$\Rightarrow \frac{1}{2} d(x^2 + y^2) = \frac{a^2 d\left(\frac{x}{y}\right)}{\left(1 + \left(\frac{x}{y}\right)^2\right)}$$

Integrating, we get

$$\Rightarrow \frac{1}{2} (x^2 + y^2) = \tan^{-1}\left(\frac{x}{y}\right) + c$$

which is the required solution.

175. The given differential equation is

$$\frac{xdx + ydy}{x dy - y dx} = \sqrt{\frac{a^2 - x^2 - y^2}{x^2 + y^2}} \quad \dots(i)$$

Let $x = r \cos \theta$, $y = r \sin \theta$

$$\Rightarrow x^2 + y^2 = r^2, \frac{y}{x} = \tan \theta$$

$$xdx + ydy = r dr, \frac{xdy - ydx}{x^2} = \sec^2 \theta d\theta$$

$$\Rightarrow \frac{r dr}{r^2 d\theta} = \sqrt{\frac{1 - r^2}{r^2}}$$

$$\Rightarrow \frac{dr}{r d\theta} = \sqrt{\frac{1 - r^2}{r^2}}$$

$$\Rightarrow \frac{dr}{d\theta} = \sqrt{1 - r^2}$$

$$\Rightarrow \frac{dr}{\sqrt{1 - r^2}} = d\theta$$

Integrating, we get

$$\sin^{-1}(r) = \theta + c$$

$$\Rightarrow \sin^{-1}(\sqrt{x^2 + y^2}) = \tan^{-1}\left(\frac{y}{x}\right) + c$$

which is the required solution.

176. The given differential equation is

$$\frac{x dx - y dy}{x dy - y dx} = \sqrt{\frac{1+x^2-y^2}{x^2+y^2}} \quad \dots(i)$$

Let $x = r \sec \theta$, $y = r \tan \theta$

$$\Rightarrow x^2 - y^2 = r^2$$

$$\Rightarrow x dx - y dy = r dr, \quad x dy - y dx = r^2 \sec \theta d\theta$$

$$\Rightarrow \frac{r dr}{r^2 \sec \theta d\theta} = \sqrt{\frac{1+r^2}{r}}$$

$$\Rightarrow \frac{dr}{\sqrt{1+r^2}} = \sec \theta d\theta$$

Integrating, we get

$$\log |r + \sqrt{r^2 + 1}| = \log |\sec \theta + \tan \theta| + \log c$$

$$\Rightarrow (r + \sqrt{r^2 + 1}) = c(\sec \theta + \tan \theta)$$

$$\Rightarrow (\sqrt{x^2 - y^2} + \sqrt{(x^2 - y^2) + 1}) = c \left(\frac{x+y}{\sqrt{x^2 - y^2}} \right)$$

which is the required solution.

177. The given differential equation is

$$\sin\left(\frac{x}{y}\right)(y dx - x dy) = xy^3(x dy + y dx)$$

$$\Rightarrow \sin\left(\frac{x}{y}\right) \left(\frac{y dx - x dy}{y^2} \right) = xy d(xy)$$

$$\Rightarrow \sin\left(\frac{x}{y}\right) d\left(\frac{x}{y}\right) = xy d(xy)$$

Integrating, we get

$$-\cos\left(\frac{x}{y}\right) = \frac{(xy)^2}{2} + c$$

which is the required solution.

178. The given differential equation is

$$x dy - y dx + y^2 = y^2(x dy + y dx) = 0$$

$$\Rightarrow y^2 d(xy) = y dx - x dy$$

$$\Rightarrow d(xy) = \frac{y dx - x dy}{y^2}$$

$$= d\left(\frac{x}{y}\right)$$

Integrating, we get

$$\Rightarrow \left(xy - \frac{x}{y}\right) = c$$

which is the required solution.

179. The given differential equation is

$$(4x^3 + e^x \sin y) dx + e^x \cos y dy = 0$$

$$\Rightarrow (4x^3) dx + (e^x \sin y dx + e^x \cos y dy) = 0$$

$$\Rightarrow (4x^3) dx + d(e^x \sin y) = 0$$

Integrating, we get

$$x^4 + e^x \sin y = c$$

which is the required solution.

180. Let the equation be

$$y = mx,$$

where m is an arbitrary constant ...(i)

Differentiating w.r.t. x , we get,

$$\frac{dy}{dx} = m \quad \dots(ii)$$

Eliminating m , between Eqs (i) and (ii), we get

$$\frac{dy}{dx} = \frac{y}{x} \quad \dots(iii)$$

which is the differential equation of a family of lines.

Now replacing $\frac{dy}{dx}$ by $-\frac{dx}{dy}$ in Eq. (iii), we get,

$$-\frac{dx}{dy} = \frac{x}{y}$$

$$\Rightarrow \frac{dy}{y} = -\frac{dx}{x}$$

Integrating, we get

$$\int \frac{dy}{y} = -\int \frac{dx}{x}$$

$$\Rightarrow \log |y| = \log |c| - \log |x| = \log \left| \frac{c}{x} \right|$$

$$\Rightarrow xy = c$$

which is required orthogonal trajectories.

181. The given family of curves are

$$x^2 + y^2 = a^2$$

Differentiating w.r.t. x , we get

$$2x + 2y \frac{dy}{dx} = 0$$

$$\Rightarrow x + y \frac{dy}{dx} = 0 \quad \dots(i)$$

$$\text{Replacing } \frac{dy}{dx} \text{ by } \frac{\frac{dy}{dx} - \tan 45^\circ}{1 + \frac{dy}{dx} \tan 45^\circ} =$$

$$\frac{\frac{dy}{dx} - 1}{1 + \frac{dy}{dx}}$$

in Eq. (i), we get

$$x + y \left(\frac{\frac{dy}{dx} - 1}{1 + \frac{dy}{dx}} \right) = 0$$

$$x\left(1 + \frac{dy}{dx}\right) + y\left(\frac{dy}{dx} - 1\right) = 0$$

$$\Rightarrow (x + y) \frac{dy}{dx} = (y - x)$$

$$\Rightarrow \frac{dy}{dx} = \frac{y - x}{y + x} \quad \dots(ii)$$

which is a homogeneous differential equation.

$$\text{Put } y = vx \Rightarrow \frac{dy}{dx} = v + x \frac{dv}{dx}$$

$$\Rightarrow v + x \frac{dv}{dx} = \frac{v - 1}{v + 1}$$

$$\Rightarrow x \frac{dv}{dx} = \frac{v - 1}{v + 1} - v = \frac{v - 1 - v^2 - v}{v + 1}$$

$$\Rightarrow \left(\frac{v + 1}{v^2 + 1}\right) dv = -\frac{dx}{x}$$

Integrating, we get

$$\frac{1}{2} \log |v^2 + 1| + \tan^{-1} v = c - \log x$$

$$\Rightarrow \log |x^2 + y^2| + \tan^{-1}\left(\frac{y}{x}\right) = c$$

which is the required trajectories of the given family of curves.

182. The given curve is

$$x^2 + y^2 = 1 \quad \dots(i)$$

$$\Rightarrow 2ax + 2y \frac{dy}{dx} = 0$$

$$\Rightarrow ax + y \frac{dy}{dx} = 0 \quad \dots(ii)$$

Solving Eqs (i) and (ii), we get

$$xy \frac{dy}{dx} + y^2 = 1$$

$$\Rightarrow xy \frac{dy}{dx} = 1 - y^2$$

$$\Rightarrow \frac{dy}{dx} = \frac{1 - y^2}{xy}$$

Replace $\frac{dy}{dx}$ by $-\frac{dx}{dy}$, we get

$$\Rightarrow -\frac{dx}{dy} = \frac{1 - y^2}{xy}$$

$$\Rightarrow x dx = \left(y - \frac{1}{y}\right) dy$$

$$\Rightarrow x dx - y dy + \frac{dy}{y} = 0$$

Integrating, we get

$$\Rightarrow \frac{x^2}{2} - \frac{y^2}{2} + \log |y| = c$$

$$\Rightarrow (x^2 - y^2) + 2 \log |y| = c$$

which is the required orthogonal trajectories.

183. Given curve is

$$x^2 + y^2 - ay = 0$$

$$\Rightarrow 2x + 2y \frac{dy}{dx} - a \frac{dy}{dx} = 0$$

$$\Rightarrow a = \frac{2x + 2y \frac{dy}{dx}}{\frac{dy}{dx}}$$

Eliminating a between (i) and (ii), we get

$$(x^2 + y^2) - \left(\frac{2x + 2y \frac{dy}{dx}}{\frac{dy}{dx}}\right) y = 0$$

$$\Rightarrow (x^2 + y^2) \frac{dy}{dx} - \left(2x + 2y \frac{dy}{dx}\right) y = 0$$

$$\Rightarrow (x^2 + y^2 - 2y^2) \frac{dy}{dx} = 2xy$$

$$\Rightarrow (x^2 - y^2) \frac{dy}{dx} = 2xy$$

$$\Rightarrow \frac{dy}{dx} = \frac{2xy}{(x^2 - y^2)}$$

Replace dy/dx by $-dx/dy$, we get

$$-\frac{dx}{dy} = \frac{2xy}{(x^2 - y^2)}$$

$$\Rightarrow \frac{dy}{dx} = \frac{y^2 - x^2}{2xy} = \frac{\left(\frac{y}{x}\right)^2 - 1}{2\left(\frac{y}{x}\right)}$$

$$\Rightarrow v + x \frac{dv}{dx} = \frac{v^2 - 1}{2v}, \quad v = \frac{y}{x}$$

$$\Rightarrow x \frac{dv}{dx} = \frac{v^2 - 1}{2v} - v = -\frac{v^2 - 1}{2v}$$

$$\Rightarrow \frac{2v dv}{(v^2 + 1)} = -\frac{dx}{x}$$

Integrating, we get

$$\log |v^2 + 1| + \log |x| = \log c$$

$$\Rightarrow x(v^2 + 1) = c$$

$$\Rightarrow x^2 + y^2 - cx = 0$$

which is the required orthogonal trajectories.

184. The given curve is

$$y^2 = 4ax \quad \dots(i)$$

$$\Rightarrow 2y \frac{dy}{dx} = 4a$$

$$\Rightarrow a = \frac{y}{2} \frac{dy}{dx} \quad \dots(ii)$$

Eliminating a from Eqs (i) and (ii), we get

$$y^2 = 4x \left(\frac{y}{2} \frac{dy}{dx} \right)$$

$$\Rightarrow y = 2x \frac{dy}{dx}$$

Replacing dy/dx by $-dx/dy$, we get

$$y = 2x \times -\frac{dy}{dx}$$

$$\Rightarrow y dy + 2x dx = 0$$

Integrating, we get

$$\frac{y^2}{2} + x^2 = c$$

which is the required orthogonal trajectory.

185. Given curve is

$$xy = c^2$$

$$\Rightarrow x \frac{dy}{dx} + y = 0$$

$$\Rightarrow \frac{dy}{dx} = -\frac{y}{x}$$

Replacing dy/dx by $-dx/dy$, we get

$$-\frac{dx}{dy} = -\frac{y}{x}$$

$$\Rightarrow x dx - y dy = 0$$

$$\Rightarrow 2x dx - 2y dy = 0$$

Integrating, we get

$$x^2 - y^2 = a^2$$

which is the required orthogonal trajectories.

186. The given curve is

$$x^2 - y^2 = c^2.$$

$$\Rightarrow 2x - 2y \frac{dy}{dx} = 0$$

$$\Rightarrow \frac{dy}{dx} = \frac{x}{y}$$

Replacing dy/dx by $-dx/dy$, we get

$$-\frac{dx}{dy} = \frac{x}{y}$$

$$\Rightarrow \frac{dx}{x} + \frac{dy}{y} = 0$$

Integrating, we get

$$\log|x| + \log|y| = \log(c^2)$$

$$\Rightarrow xy = c^2$$

which is the required orthogonal trajectories.

187. The given equation is

$$(x + y + p)(2x + p) = 0$$

$$\Rightarrow (x + y + p) = 0 \quad \dots(i)$$

$$\text{and } (2x + p) = 0 \quad \dots(ii)$$

From Eq. (i), we get,

$$\frac{dy}{dx} + y + x = 0$$

$$\Rightarrow \frac{dy}{dx} + y = -x$$

which is a linear differential equation.

$$\begin{aligned} \Rightarrow y \cdot e^x &= -\int x e^x dx + c_1 \\ &= -e^x(x - 1) + c_1 \end{aligned}$$

$$\Rightarrow y = -(x - 1) + c_1 e^{-x} \quad \dots(iii)$$

Also, from Eq. (ii), we get,

$$\frac{dy}{dx} + 2x = 0$$

$$\Rightarrow dy = -2x dx$$

$$\Rightarrow y = c_2 - x^2 \quad \dots(iv)$$

Hence, the required solution is

$$(y + (x - 1) - c_1 e^{-x})(y + x^2 - c_2) = 0.$$

188. The given equation is

$$p^2 - p(e^x + e^{-x}) + 1 = 0 \quad \dots(i)$$

$$\Rightarrow p^2 - p \left(e^x + \frac{1}{e^x} \right) + 1 = 0$$

$$\Rightarrow p^2 e^x - p(e^{2x} + 1) + e^x = 0$$

$$\Rightarrow p^2 e^x - p e^{2x} - p + e^x = 0$$

$$\Rightarrow p e^x (p - e^x) - 1(p - e^x) = 0$$

$$\Rightarrow (p - e^x)(p e^x - 1) = 0$$

$$\Rightarrow (p - e^x) = 0 \quad \dots(ii)$$

$$\text{and } (p e^x - 1) = 0 \quad \dots(iii)$$

From Eq. (i), we get,

$$\frac{dy}{dx} = e^x$$

$$\Rightarrow dy = e^x dx$$

Integrating, we get

$$\Rightarrow y = e^x + c_1 \quad \dots(iv)$$

From Eq. (iii), we get,

$$\frac{dy}{dx} \cdot e^x - 1 = 0$$

$$\Rightarrow e^x dy = dx$$

$$\Rightarrow dy = e^{-x} dx$$

Integrating, we get

$$\Rightarrow y = c_2 - e^{-x}$$

Hence, the general solution of Eq. (i) is

$$(x - e^{-x} - c)(y - e^{-x} - c) = 0.$$

189. The given equation is

$$p^2 + 2py \cot x = y^2 \quad \dots(i)$$

$$\Rightarrow (p + y \cot x)^2 = y^2(1 + \cot^2 x)$$

$$\Rightarrow (p + y \cot x)^2 = y^2 \operatorname{cosec}^2 x$$

$$\Rightarrow (p + y \cot x) = \pm y \operatorname{cosec} x$$

$$\Rightarrow p = y(\operatorname{cosec} x - \cot x) \quad \dots(ii)$$

$$\text{and } p = -y(\operatorname{cosec} x - \cot x) \quad \dots(iii)$$

From Eq. (ii), we get,

$$\frac{dy}{dx} = y(\operatorname{cosec} x - \cot x)$$

$$\Rightarrow \frac{dy}{y} = (\operatorname{cosec} x - \cot x) dx$$

$$\Rightarrow \frac{dy}{y} = \left(\frac{1 - \cos x}{\sin x} \right) dx$$

$$\Rightarrow \frac{dy}{y} = \tan\left(\frac{x}{2}\right) dx$$

Integrating, we get

$$\log |y| = 2 \log \left| \sec\left(\frac{x}{2}\right) \right| + 2 \log(c_1)$$

$$\Rightarrow y = c_1^2 \left(\sec^2\left(\frac{x}{2}\right) \right)$$

From Eq. (iii), we get

$$\frac{dy}{dx} = -y(\operatorname{cosec} x + \cot x)$$

$$\Rightarrow \frac{dy}{y} = -(\operatorname{cosec} x + \cot x) dx$$

Integrating, we get

$$\Rightarrow \frac{dy}{y} = -\cot\left(\frac{x}{2}\right) dx$$

$$\Rightarrow \log y = 2 \log c_2 - 2 \log \left(\sin\left(\frac{x}{2}\right) \right)$$

$$\Rightarrow y \left(\sin^2\left(\frac{x}{2}\right) \right) = c_2$$

Hence, the required solution is

$$\left(y - c \left(\sec^2\left(\frac{x}{2}\right) \right) \right) \left(y \sin^2\left(\frac{x}{2}\right) - c \right) = 0.$$

190. The given differential equation is

$$p^2 + px - xy - y^2 = 0$$

$$\Rightarrow (p^2 - y^2) + x(p - y) = 0$$

$$\Rightarrow (p - y)(p + x + y) = 0$$

$$\Rightarrow (p - y) = 0, (p + x + y) = 0$$

$$\text{When } (p - y) = 0$$

$$\Rightarrow \frac{dy}{dx} = y$$

$$\Rightarrow \frac{dy}{y} = dx$$

Integrating, we get

$$\log |y| = x + c_1$$

$$\text{When } (p + x + y) = 0,$$

$$\Rightarrow \frac{dy}{dx} + (x + y) = 0$$

$$\Rightarrow \frac{dy}{dx} + y = -x$$

which is a linear differential equation

$$\text{Thus, IF} = e^{\int dx} = e^x$$

Thus, the solution is

$$y \cdot e^x = -\int x e^x dx + c$$

$$\Rightarrow y \cdot e^x = -(x - 1)e^x + c_2$$

Hence, the required solution is

$$(\log |y| - x c_1)(y \cdot e^x + (x - 1)e^x - c_2) = 0$$

191. The given differential equation is

$$p^2 y + (x - y)p - x = 0$$

$$\Rightarrow p^2 y + px - py - x = 0$$

$$\Rightarrow py(p - 1) + x(p - 1) = 0$$

$$\Rightarrow (py + x)(p - 1) = 0$$

$$\Rightarrow (py + x) = 0, (p - 1) = 0$$

$$\text{When } (py + x) = 0$$

$$\Rightarrow \frac{y dy}{dx} + x = 0$$

$$\Rightarrow y dy + x dx = 0$$

Integrating, we get

$$\Rightarrow \frac{x^2}{2} + \frac{y^2}{2} = \frac{c^2}{2}$$

$$\Rightarrow x^2 + y^2 = c^2$$

$$\text{When } (p - 1) = 0$$

$$p = 1$$

$$\Rightarrow \frac{dy}{dx} = 1$$

$$\Rightarrow dy = dx$$

Integrating, we get

$$y = x + c_1$$

Hence, the solution is

$$(x^2 + y^2 - c^2)(y - x - c_1) = 0$$

192. The given differential equation is

$$(p^2 - 1)xy = (x^2 - y^2)p$$

$$\Rightarrow p^2xy - xy - px^2 + py^2 = 0$$

$$\Rightarrow p^2xy - px^2 + py^2 - xy = 0$$

$$\Rightarrow px(py - x) + y(py - x) = 0$$

$$\Rightarrow (px + y)(py - x) = 0$$

$$\Rightarrow (px + y) = 0, (py - x) = 0$$

When $(px + y) = 0$

$$\Rightarrow \left(\frac{dy}{dx}x + y \right) = 0$$

$$\Rightarrow \frac{dx}{x} + \frac{dy}{y} = 0$$

Integrating, we get

$$\log x + \log y = \log c$$

$$\Rightarrow xy = c$$

When $(py - x) = 0$

$$\Rightarrow \left(y \frac{dy}{dx} - x \right) = 0$$

$$\Rightarrow xdx - ydy = 0$$

Integrating, we get

$$\frac{x^2}{2} - \frac{y^2}{2} = \frac{c_1^2}{2}$$

$$\Rightarrow x^2 - y^2 = c_1^2$$

Hence, the required solution is

$$(xy - c)(x^2 - y^2 - c_1^2) = 0$$

193. The given differential equation is

$$xyp^2 - (x^2 + y^2)p + xy = 0$$

$$\Rightarrow xyp^2 - x^2p - y^2p + xy = 0$$

$$\Rightarrow px(py - x) - y(py - x) = 0$$

$$\Rightarrow (px - y)(py - x) = 0$$

$$\Rightarrow (px - y) = 0, (py - x) = 0$$

When $(px - y) = 0$

$$\Rightarrow \left(\frac{dy}{dx}x - y \right) = 0$$

$$\Rightarrow \frac{dx}{x} - \frac{dy}{y} = 0$$

Integrating, we get

$$\log x - \log y = \log c$$

$$\Rightarrow x = cy$$

When $(py - x) = 0$

$$\Rightarrow \left(y \frac{dy}{dx} - x \right) = 0$$

$$\Rightarrow xdx - ydy = 0$$

Integrating, we get

$$x^2 - y^2 = a^2$$

Hence, the required solution is

$$(xy - c)(x^2 - y^2 - a^2) = 0$$

194. The given equation can be written as

$$x = yp + a \quad \dots(i)$$

Differentiating w.r.t. y, we get

$$\frac{dx}{dy} = p + y \frac{dp}{dy}$$

$$\Rightarrow \frac{1}{p} = p + y \frac{dp}{dy}$$

$$\Rightarrow \left(\frac{1}{p} - p \right) = y \frac{dp}{dy}$$

$$\Rightarrow \left(\frac{p}{1 - p^2} \right) dp = \frac{dy}{y}$$

$$\Rightarrow \left(\frac{-2p}{1 - p^2} \right) dp = \frac{-2dy}{y}$$

Integrating, we get

$$\log |1 - p^2| = \log c - 2 \log |y|$$

$$\Rightarrow \log |1 - p^2| = \log \left| \frac{c}{y^2} \right|$$

$$\Rightarrow (1 - p^2) = \frac{c}{y^2}$$

$$\Rightarrow p^2 = \left(1 - \frac{c}{y^2} \right) \quad \dots(ii)$$

Eliminating p between Eqs (i) and (ii), we get

$$x = a \pm y \sqrt{1 - \frac{c}{y^2}}$$

which is the required solution of Eq. (i)

195. The given equation can be written as

$$x = \frac{1}{2} \left(\frac{y}{p} - y^2 p^2 \right) \quad \dots(i)$$

Differentiating w.r.t. y, we get

$$\frac{dx}{dy} = \frac{1}{2} \left(\frac{1}{p} - \frac{y}{p^2} \frac{dp}{dy} - 2yp^2 - 2py^2 \frac{dp}{dy} \right)$$

$$\Rightarrow \frac{1}{p} = \frac{1}{2} \left(\frac{1}{p} - \frac{y}{p^2} \frac{dp}{dy} - 2yp^2 - 2py^2 \frac{dp}{dy} \right)$$

$$\Rightarrow 2p = \left(p - y \frac{dp}{dy} - 2yp^4 - 2p^3 y^2 \frac{dp}{dy} \right)$$

$$\Rightarrow p + 2yp^4 = - \left(y \frac{dp}{dy} - 2p^3 y^2 \frac{dp}{dy} \right)$$

$$\Rightarrow p(1 + 2yp^3) = -y \frac{dp}{dy} (1 + 2p^3 y)$$

$$\Rightarrow (1 + 2yp^3) \left(p + y \frac{dp}{dy} \right) = 0$$

$$\Rightarrow \left(p + y \frac{dp}{dy} \right) = 0$$

$$\Rightarrow d(py) = 0$$

Integrating, we get

$$py = c$$

$$\Rightarrow p = \frac{y}{c} \quad \dots(ii)$$

Eliminating p between Eq. (i) and (ii), we get

$$y = 2\frac{c}{y}x + \frac{c^3}{y^3} \cdot y^2$$

$$\Rightarrow y^2 = 2cx + c^3$$

which is the required solution.

196. The given differential equation is

$$y^2 \log y = xyp + p^2 \quad \dots(i)$$

$$\Rightarrow x = \frac{y^2 \log y - p^2}{yp}$$

Differentiating w.r.t. y , we get

$$\frac{dy}{dx} = \frac{yp \left(2y \log y + y - 2p \frac{dp}{dy} \right) - (y^2 \log y - p^2) \left(p + y \frac{dp}{dy} \right)}{(yp)^2}$$

$$\Rightarrow y^2 p = yp \left(2y \log y + y - 2p \frac{dp}{dy} \right) - (y^2 \log y - p^2) \left(p + y \frac{dp}{dy} \right)$$

$$\Rightarrow y \frac{dp}{dy} (y^2 \log y - p^2) = p(y^2 \log y - p^2)$$

$$\Rightarrow y \frac{dp}{dy} = p$$

$$\Rightarrow \frac{dy}{p} = \frac{dp}{p}$$

Integrating, we get

$$\Rightarrow \log p = \log y + \log c$$

$$\Rightarrow p = cy \quad \dots(ii)$$

Eliminating p between Eqs (i) and (ii), we get

$$\log y = cx + c^2$$

which is the required solution of (i)

197. The given differential equation is

$$p^2 y + 2px = y \quad \dots(i)$$

$$\Rightarrow 2px = y - p^2 y$$

$$\Rightarrow x = \frac{y - p^2 y}{2p}$$

Differentiating, w.r.t. y , we get

$$\frac{dx}{dy} = \frac{2p \left(1 - p^2 - 2py \frac{dp}{dy} \right) - 2(y - p^2 y) \frac{dp}{dy}}{4p^2}$$

$$\Rightarrow \frac{1}{p} = \frac{p \left(1 - p^2 - 2py \frac{dp}{dy} \right) - (y - p^2 y) \frac{dp}{dy}}{2p^2}$$

$$\Rightarrow 2p = p - p^3 - 2p^2 y \frac{dp}{dy} - y \frac{dp}{dy} + p^2 y \frac{dp}{dy}$$

$$\Rightarrow p = -p^3 - p^2 y \frac{dp}{dy} - y \frac{dp}{dy}$$

$$\Rightarrow p(p^2 + 1) = -(p^2 + 1)y \frac{dp}{dy}$$

$$\Rightarrow p = -y \frac{dp}{dy}$$

$$\Rightarrow \frac{dp}{p} + \frac{dy}{y} = 0$$

$$\Rightarrow py = c \quad \dots(ii)$$

Eliminating p between Eqs (i) and (ii), we get

$$y^2 + 2cx = c^2$$

which is the required solution.

$$198. c(4x - c)^2 = 64y$$

199. The given differential equation is

$$yp = 2p^2 x + y^2 p^4 \quad \dots(i)$$

$$\Rightarrow x = \frac{y - y^2 p^3}{2p}$$

Differentiating, w.r.t. y , we get

$$\Rightarrow \frac{dx}{dy} = \frac{1}{2} \frac{p \left(1 - 2yp^3 - 3y^2 p^2 \frac{dp}{dy} \right) - (y - y^2 p^3) \frac{dp}{dy}}{p^2}$$

$$\Rightarrow p + 2yp^4 + 2y^2 p^3 \frac{dp}{dy} + y \frac{dp}{dy} = 0$$

$$\Rightarrow p(1 + 2yp^3) + y \frac{dp}{dy} (1 + 2yp^3) = 0$$

$$\Rightarrow \left(p + y \frac{dp}{dy} \right) (1 + 2yp^3) = 0$$

$$\Rightarrow \left(p + y \frac{dp}{dy} \right) = 0$$

$$\Rightarrow \frac{dp}{p} + \frac{dy}{y} = 0$$

$$\Rightarrow py = c \quad \dots(ii)$$

Eliminating p between Eqs (i) and (ii), we get

$$y^2 = 2cx + c^3$$

which is the required solution.

200. The given differential equation is

$$xp^2 - yp - p + 1 = 0 \quad \dots(i)$$

$$\Rightarrow x = \frac{yp + p - 1}{p^2}$$

Differentiating, w.r.t. y , we get

$$\frac{dx}{dy} = \frac{p^2 \left(p + y \frac{dp}{dy} - \frac{dp}{dy} \right) - 2(yp + p - 1)p \frac{dp}{dy}}{p^4}$$

$$\Rightarrow p^3 = p^3 + p^2 y \frac{dp}{dy} - p^2 \frac{dp}{dy} - 2p^2 y \frac{dp}{dy} - 2p^2 \frac{dp}{dy} + 2p \frac{dp}{dy}$$

$$\Rightarrow p^2 y \frac{dp}{dy} - p^2 \frac{dp}{dy} - 2p^2 y \frac{dp}{dy} - 2p^2 \frac{dp}{dy} + 2p \frac{dp}{dy} = 0$$

$$\Rightarrow \frac{dp}{dy} = 0$$

$$\Rightarrow dp = 0$$

$$\Rightarrow p = c \quad \dots(ii)$$

Eliminating p between Eqs (i) and (ii) we get

$$yc = c^2x - c + 1$$

which is the required solution.

201. The given equation is $y = (1 + p)x + ap^2$ $\dots(i)$

Differentiating w.r.t. x , we get

$$\Rightarrow \frac{dy}{dx} = (1 + p) \cdot 1 + x \frac{dp}{dx} + 2ap \frac{dp}{dx}$$

$$\Rightarrow p = (1 + p) \cdot 1 + x \frac{dp}{dx} + 2ap \frac{dp}{dx}$$

$$\Rightarrow (x + 2ap) \frac{dp}{dx} = -1$$

$$\Rightarrow \frac{dx}{dp} = -(x + 2ap)$$

$$\Rightarrow \frac{dx}{dp} + x = -2ap \quad \dots(ii)$$

which is a linear differential equation.

$$IF = e^{\int dp} = e^p$$

Hence, the required solution of Eq. (ii) is

$$x \cdot e^p = -2a \int p e^p dp = -2ae^p(p - 1) + c$$

$$\Rightarrow x = -2a(p - 1) + ce^{-p} \quad \dots(iii)$$

Eliminating p between Eqs (i) and (iii), we get the required solution of Eq. (i).

202. The given equation is

$$y = yp^2 + 2px$$

$$\Rightarrow y = \frac{2px}{1 - p^2}$$

Differentiating w.r.t. x , we get

$$\frac{2dp}{p(p^2 - 1)} = \frac{dx}{x}$$

$$\Rightarrow \left(\frac{1}{p-1} + \frac{1}{p+1} - \frac{2}{p} \right) dp = \frac{dx}{x}$$

Integrating, we get

$$\log|p-1| + \log|p+1| - \log|p^2| = \log|x| + \log c$$

$$\Rightarrow \log \left| \frac{p^2 - 1}{p^2} \right| = \log|cx|$$

$$\Rightarrow \frac{p^2 - 1}{p^2} = cx$$

$$\Rightarrow p^2 - 1 = p^2 cx$$

$$\Rightarrow p^2(1 - cx) = 1$$

$$\Rightarrow p^2 = \left(\frac{1}{1 - cx} \right)$$

$$\Rightarrow p = \sqrt{\frac{1}{1 - cx}} \quad \dots(ii)$$

Eliminating p between Eqs (i) and (ii), we get

$$2x\sqrt{1 - cx} + cxy = 0.$$

which is the required solution.

203. The given differential equation is

$$y = 2px + \tan^{-1}(xp^2) \quad \dots(i)$$

Differentiating w.r.t. x , we get

$$\frac{dy}{dx} = 2p + 2x \frac{dp}{dx} + \frac{p^2 + 2px \frac{dp}{dx}}{1 + (xp^2)^2}$$

$$\Rightarrow \left(-p - 2x \frac{dp}{dx} \right) (1 + p^4 x^2) = p^2 + 2px \frac{dp}{dx}$$

$$\Rightarrow -p - 2x \frac{dp}{dx} - p^5 x^2 - 2x^3 p^4 \frac{dp}{dx} = p^2 + 2px \frac{dp}{dx}$$

$$\Rightarrow -p - p^5 x^2 - p^2 = + 2x \frac{dp}{dx} + 2x \frac{dp}{dx} + 2x^3 p^4 \frac{dp}{dx}$$

$$\Rightarrow -p(1 + p^4 x^2 - p) = + 2x(p + 1 + x^2 p^4) \frac{dp}{dx}$$

$$\Rightarrow -p = 2x \frac{dp}{dx}$$

$$\Rightarrow \frac{2dp}{p} + \frac{dx}{x} = 0$$

$$\Rightarrow 2 \log p + \log x = \log c$$

$$\Rightarrow p^2 x = c \quad \dots(ii)$$

Eliminating p between Eqs (i) and (iii), we get the required solution of Eq. (i).

$$y = 2\left(\sqrt{\frac{c}{x}}\right)x + \tan^{-1}(c)$$

$$\Rightarrow y = 2\sqrt{cx} + \tan^{-1}(c)$$

which is the required solution.

204. The given differential equation is

$$x^2 p^4 + 2xp = y$$

$$\Rightarrow y = p^4 x^2 + 2px \quad \dots(i)$$

$$\Rightarrow \frac{dy}{dx} = 2xp^4 + 4x^2 p^3 \frac{dp}{dx} + 2p + 2x \frac{dp}{dx}$$

$$\Rightarrow p = 2xp^4 + 4x^2 p^3 \frac{dp}{dx} + 2p + 2x \frac{dp}{dx}$$

$$\Rightarrow -2xp^4 - p = (4x^2 p^3 + 2x) \frac{dp}{dx}$$

$$\Rightarrow 2x(2xp^3 + 1) \frac{dp}{dx} = -p(2xp^3 + 1)$$

$$\Rightarrow 2x \frac{dp}{dx} = -p$$

$$\Rightarrow 2 \frac{dp}{p} = -\frac{dx}{x}$$

Integrating, we get

$$2 \log |p| = \log c - \log |x|$$

$$\Rightarrow 2 \log |p| = \log \left| \frac{c}{x} \right|$$

$$p^2 = \frac{x}{c} \quad \dots(ii)$$

Eliminating p between Eqs (i) and (ii), we get

$$y = 2\sqrt{cx} + c^2$$

which is the required solution.

205. The given differential equation is

$$y + px = x^4 p^2 \quad \dots(i)$$

Differentiating w.r.t. x , we get

$$\frac{dy}{dx} + p + x \frac{dp}{dx} = 2px^4 \frac{dp}{dx} + 4x^3 p^2$$

$$\Rightarrow p + p + x \frac{dp}{dx} = 2px^4 \frac{dp}{dx} + 4x^3 p^2$$

$$\Rightarrow (x - 2px^4) \frac{dp}{dx} = 4x^3 p^2 - 2p$$

$$\Rightarrow x(1 - 2px^3) \frac{dp}{dx} = 2p(2x^3 p - 1)$$

$$\Rightarrow x(1 - 2px^3) \frac{dp}{dx} = -2p(1 - 2x^3 p)$$

$$\Rightarrow x \frac{dp}{dx} = -2p$$

$$\Rightarrow \frac{dp}{p} + \frac{2dx}{x} = 0$$

Integrating, we get

$$\log p + 2 \log x = \log c$$

$$\Rightarrow px^2 = c \quad \dots(ii)$$

Eliminating p between Eqs (i) and (iii), we get the required solution of Eq. (i)

$$y + \frac{c}{x} + c^2$$

which is the required solution.

206. The given differential equation is

$$y = p \sin p + \cos p \quad \dots(i)$$

$$\Rightarrow \frac{dy}{dx} = \sin p \frac{dp}{dx} + p \cos p \frac{dp}{dx} - \sin p \frac{dp}{dx}$$

$$\Rightarrow p = p \cos p \frac{dp}{dx}$$

$$\Rightarrow \cos p \frac{dp}{dx} = 1$$

$$\Rightarrow \cos p dp = dx$$

Integrating, we get

$$\sin p = x + c \quad \dots(ii)$$

Eliminating p from Eqs (i) and (ii), we get

$$y = (x + c) \sin^{-1}(x + c) + \sqrt{1 - (x + c)^2}$$

which is the required solution.

207. The given differential equation can be written as

$$y = Px - \frac{2}{P} - 1 \quad \dots(i)$$

Differentiating w.r.t. x , we get

$$\frac{dP}{dx} = P + x \frac{dP}{dx} + \frac{2}{P^2} \frac{dP}{dx}$$

$$\Rightarrow P = P + x \frac{dP}{dx} + \frac{2}{p^2} \frac{dP}{dx}$$

$$\Rightarrow \left(x + \frac{2}{p^2}\right) \frac{dP}{dx} = 0$$

$$\Rightarrow \frac{dP}{dx} = 0 \text{ or } \left(x + \frac{2}{p^2}\right) = 0$$

$$\Rightarrow P = c \quad \dots(\text{ii})$$

$$\text{or } x = -\frac{2}{p^2} \quad \dots(\text{iii})$$

Eliminating p between Eqs (i) and (ii), we get

$$y = cx - \frac{2}{c} - 1$$

which is the genral solution of Eq. (i).

Eliminating p between Eqs (i) and (iii), we get

$$(y + 1)^2 + 8x = 0$$

which is the singular solution of Eq. (i).

208. The given differential equation is

$$P^3x - P^2y - 1 = 0$$

$$\Rightarrow P^2y = P^3x - 1$$

$$\Rightarrow y = Px - \frac{1}{P^2}$$

which is a Clairaut differential equation.

Hence, the solution is

$$y = cx - \frac{1}{c^2}$$

209. The given differential equation is

$$(y + 1)P - xP^2 + 2 = 0$$

$$\Rightarrow (y + 1)P = P^2x - 2$$

$$\Rightarrow (y + 1) = Px - \frac{2}{P}$$

$$\Rightarrow y = Px - \left(\frac{2}{P} + 1\right)$$

which is a Clairaut differential equation.

Hence, the solution is

$$y = cx - \left(\frac{2}{c} + 1\right)$$

210. The given differential equation is

$$\sin y \cos Px - \cos y \sin Px - P = 0$$

$$\Rightarrow \sin(y - Px) = P$$

$$\Rightarrow (y - Px) = \sin^{-1}(P)$$

$$\Rightarrow y = Px + \sin^{-1}(P)$$

which is a Clairaut differential equation.

Hence, the solution is

$$y = cx + \sin^{-1}(c)$$

211. The given differential equation is

$$(x - a)P^2 + (x - y)P - y = 0$$

$$\Rightarrow P^2x + Px - aP^2 = (P + 1)y$$

$$\Rightarrow (P + 1)y = P(P + 1)x - aP^2$$

$$\Rightarrow y = Px - \frac{aP^2}{(P + 1)}$$

which is a Clairaut differential equation.

Hence, the solution is

$$y = cx - \frac{ac^2}{(c + 1)}$$

212. The given differential equation is

$$y = Px + a\sqrt{1 + P^2}$$

which is a Clairaut differential equation.

Hence, the solution is

$$y = cx + a\sqrt{1 + c^2}$$

213. The given differential equation is

$$y = p(x - b) + \frac{a}{p}$$

$$\Rightarrow y = px + \left(\frac{a}{p} - bp\right)$$

which is a Clairaut differential equation.

Hence, the solution is

$$y = cx + \left(\frac{a}{c} - bc\right)$$

214. The given differential equation is

$$\frac{d^2y}{dx^2} = x + \sin x$$

$$\Rightarrow \frac{d}{dx} \left(\frac{dy}{dx} \right) = x + \sin x$$

Integrating, we get

$$\frac{dy}{dx} = \frac{x^2}{2} - \cos x + c_1$$

Again integrating, we get

$$\Rightarrow y = \frac{x^3}{6} - \sin x + c_1x + c_2$$

which is the required solution.

215. The given differential equation is

$$\frac{d^2y}{dx^2} = e^{2x} + e^x + 2014$$

$$\Rightarrow \frac{d}{dx} \left(\frac{dy}{dx} \right) = e^{2x} + e^x + 2014$$

Integrating, we get

$$\frac{dy}{dx} = \frac{e^{2x}}{2} + e^x + 2014 + c_1$$

Again integrating, we get

$$y = \frac{e^{2x}}{4} + e^x + 1007x^2 + c_1x + c_2$$

which is the required solution.

216. The given differential equation is

$$\frac{d^2y}{dx^2} = \sin^2x$$

$$\Rightarrow \frac{d}{dx} \left(\frac{dy}{dx} \right) = \sin^2x$$

Integrating, we get

$$\left(\frac{dy}{dx} \right) = \frac{1}{2} \int 2 \sin^2x dx + c_1$$

$$\Rightarrow \left(\frac{dy}{dx} \right) = \frac{1}{2} \left(x - \frac{2 \sin 2x}{2} \right) + c_1$$

Again Integrating, we get

$$y = \frac{1}{2} \left(\frac{x^2}{2} + \frac{\cos 2x}{4} \right) + c_1x + c_2$$

which is the required solution.

217. The given differential equation is

$$\frac{d^2y}{dx^2} = \cos^3x$$

$$\Rightarrow \frac{d}{dx} \left(\frac{dy}{dx} \right) = \cos^3x$$

Integrating, we get

$$\left(\frac{dy}{dx} \right) = \int \cos^3x dx$$

$$\Rightarrow \left(\frac{dy}{dx} \right) = \int \cos^3x dx = \left(x - \frac{\sin^3x}{3} \right) + c_1$$

Again Integrating, we get

$$y = \int \left(x - \frac{\sin^3x}{3} \right) dx + c_1x + c_2$$

$$\Rightarrow y = \frac{x^2}{2} + \frac{1}{3} \left(x + \frac{\cos^3x}{3} \right) + c_1x + c_2$$

which is the required solution.

218. The given differential equation is

$$\frac{d^2y}{dx^2} = \frac{1}{\sin^2x \cos^2x}$$

$$\Rightarrow \frac{d}{dx} \left(\frac{dy}{dx} \right) = \frac{\sin^2x + \cos^2x}{\sin^2x \cos^2x}$$

$$\Rightarrow \frac{d}{dx} \left(\frac{dy}{dx} \right) = \sec^2x + \operatorname{cosec}^2x$$

Integrating, we get

$$\left(\frac{dy}{dx} \right) = \tan x - \cot x + c_1$$

Again Integrating, we get

$$\Rightarrow y = \log(\cos x) - \log(\sin x) + c_1x + c_2$$

$$\Rightarrow y = \log(\cot x) + c_1x + c_2$$

which is the required solution.

219. The given differential equation is

$$\begin{aligned} \frac{d^2y}{dx^2} &= \sin^4x + \cos^4x \\ &= 1 - 2\sin^2x \cos^2x \\ &= 1 - \frac{1}{2} \sin^2x \\ &= 1 - \frac{1}{4} (1 - \cos 4x) \\ &= \frac{3}{4} + \frac{1}{4} \cos 4x \end{aligned}$$

$$\Rightarrow \frac{d}{dx} \left(\frac{dy}{dx} \right) = \frac{3}{4} + \frac{1}{4} \cos 4x$$

Integrating, we get

$$\left(\frac{dy}{dx} \right) = \frac{3x}{4} + \frac{1}{16} \sin 4x + c_1$$

Again Integrating, we get

$$y = \frac{3x^2}{8} + \frac{\cos(4x)}{64} + c_1x + c_2$$

which is the required solution.

220. The given differential equation is

$$\frac{d^2y}{dx^2} = xe^x$$

$$\Rightarrow \frac{d}{dx} \left(\frac{dy}{dx} \right) = xe^x$$

Integrating, we get

$$\Rightarrow \left(\frac{dy}{dx} \right) = (x-1)e^x + c_1$$

Again integrating, we get

$$\Rightarrow y = (x-2)e^x + c_1x + c_2$$

which is the required solution.

221. The given differential equation is

$$\frac{d^2y}{dx^2} + y = 0$$

$$\Rightarrow p \frac{dp}{dy} + y = 0$$

$$\Rightarrow p \frac{dp}{dy} = -y$$

$$\Rightarrow 2p dp = -2y dy$$

On integrating, we get

$$\Rightarrow p^2 = c_1^2 - y^2$$

$$\Rightarrow \left(\frac{dy}{dx}\right)^2 = c_1^2 - y^2$$

$$\Rightarrow \frac{dy}{dx} = \pm \sqrt{c_1^2 - y^2}$$

$$\Rightarrow \frac{dy}{\sqrt{c_1^2 - y^2}} = \pm dx$$

Integrating, we get

$$\sin^{-1}\left(\frac{y}{c_1}\right) = c_1 \pm x$$

which is the required solution.

222. The given differential equation is

$$\frac{d^2y}{dx^2} = \frac{1}{y^3}$$

$$\Rightarrow p \frac{dp}{dy} = \frac{1}{y^3} \quad \left(\text{Let } p = \frac{dy}{dx}\right)$$

$$\Rightarrow p dp = -\frac{dy}{y^3}$$

Integrating, we get

$$\frac{p^2}{2} = \frac{1}{2y^2} + \frac{c_1}{2}$$

$$\Rightarrow p^2 = \frac{c_1 y^2 + 1}{y^2}$$

$$\Rightarrow \left(\frac{dy}{dx}\right)^2 = \frac{c_1 y^2 + 1}{y^2}$$

$$\Rightarrow \frac{dy}{dx} = \pm \sqrt{\frac{c_1 y^2 + 1}{y^2}}$$

$$\Rightarrow \frac{y}{\sqrt{c_1 y^2 + 1}} + 1 dy = \pm dx$$

Again integrating, we get

$$\frac{\sqrt{c_1 y^2 + 1}}{c_1} = c_2 \pm x$$

which is the required solution.

223. The given differential equation is

$$\frac{d^2y}{dx^2} = \frac{1}{4\sqrt{y}}$$

$$\Rightarrow \frac{d}{dx}\left(\frac{dy}{dx}\right) = \frac{1}{4\sqrt{y}}$$

$$\Rightarrow \frac{d}{dy}\left(\frac{dy}{dx}\right) \frac{dy}{dx} = \frac{1}{4\sqrt{y}}$$

$$\Rightarrow p \frac{dp}{dy} = \frac{1}{4\sqrt{y}}$$

$$\Rightarrow 4p dp = \frac{dy}{\sqrt{y}}$$

Integrating, we get

$$2p^2 = 2\sqrt{y} + 2c_1$$

$$\Rightarrow p^2 = \sqrt{y} + c_1$$

$$\Rightarrow \left(\frac{dy}{dx}\right)^2 = \sqrt{y} + c_1$$

$$\Rightarrow \left(\frac{dy}{dx}\right) = +\sqrt{\sqrt{y} + c_1}$$

$$\Rightarrow \frac{dy}{\sqrt{\sqrt{y} + c_1}} = \pm dx$$

Again integrating, we get

$$\begin{aligned} c_2 \pm x &= \int \frac{dy}{\sqrt{\sqrt{y} + c_1}} \\ &= \int \frac{2t dt}{\sqrt{t + c_1}}, \text{ where } y = t^2 \\ &= \int \frac{2\{(t + c_1) - c_1\} dt}{\sqrt{t + c_1}} \\ &= 2 \int (\sqrt{t + c_1} - c_1(t + c_1)^{-1/2}) dt \\ &= 4 \left[\frac{1}{3} (t + c_1)^{3/2} - c_1 \sqrt{t + c_1} \right] + c_2 \\ &= 4 \left[\frac{1}{3} (\sqrt{y} + c_1)^{3/2} - c_1 \sqrt{\sqrt{y} + c_1} \right] + c_2 \end{aligned}$$

which is the required solution.

224. The given differential equation is

$$a^2 \frac{d^2y}{dx^2} - y = 0$$

$$\Rightarrow \frac{d^2y}{dx^2} = \frac{y}{a^2}$$

$$\Rightarrow p \frac{dp}{dy} = \frac{y}{a^2}$$

$$\Rightarrow p dp = \frac{y dy}{a^2}$$

Integrating, we get

$$p^2 a^2 = y^2 + c_1^2$$

$$\Rightarrow p^2 = \frac{y^2 + c_1^2}{a^2}$$

$$\Rightarrow \left(\frac{dy}{dx}\right)^2 = \frac{y^2 + c_1^2}{a^2}$$

$$\Rightarrow \left(\frac{dy}{dx}\right) = \pm \sqrt{\frac{y^2 + c_1^2}{a^2}}$$

$$\Rightarrow \frac{dy}{\sqrt{y^2 + c_1^2}} = \pm \frac{dx}{a^2}$$

Again integrating, we get

$$\log|y + \sqrt{y^2 + c_1^2}| = c_2 \pm \frac{x}{a^2}$$

which is the required solution.

225. The given differential equation is

$$\frac{d^2y}{dx^2} = e^{2y}$$

$$\Rightarrow p \frac{dp}{dy} = e^{2y}$$

$$\Rightarrow p dp = e^{2y} dy$$

Integrating, we get

$$\frac{p^2}{2} = \frac{e^{2y}}{2} + \frac{c_1}{2}$$

$$\Rightarrow p^2 = e^{2y} + c_1$$

$$\Rightarrow \left(\frac{dy}{dx}\right)^2 = e^{2y} + c_1$$

$$\Rightarrow \left(\frac{dy}{dx}\right) = \pm \sqrt{e^{2y} + c_1}$$

$$\Rightarrow \frac{dy}{\sqrt{e^{2y} + c_1}} = \pm dx$$

Again integrating, we get

$$\begin{aligned} c_2 \pm x &= \int \frac{dy}{\sqrt{e^{2y} + c_1}} \\ &= \int \frac{e^{-y} dy}{\sqrt{1 + c_1 e^{-2y}}} \\ &= - \int \frac{dt}{\sqrt{1 + c_1 t^2}}, t = e^{-y} \\ &= - \frac{1}{\sqrt{c_1}} \int \frac{dt}{\sqrt{t^2 + \left(\frac{1}{\sqrt{c_1}}\right)^2}} \\ &= - \frac{1}{\sqrt{c_1}} \log \left| t + \sqrt{t^2 + \frac{1}{\sqrt{c_1}}} \right| \\ &= - \frac{1}{\sqrt{c_1}} \log \left| e^{-y} + \sqrt{e^{-2y} + \frac{1}{\sqrt{c_1}}} \right| \end{aligned}$$

226. The given differential equation is

$$2 \frac{d^2y}{dx^2} = 3y^2$$

$$\Rightarrow 2p \frac{dp}{dy} = 3y^2$$

$$\Rightarrow 2p dp = 3y^2 dy$$

Integrating, we get

$$p^2 = y^3 + c_1$$

$$\Rightarrow \left(\frac{dy}{dx}\right)^2 = y^3 + c_1$$

$$\Rightarrow \left(\frac{dy}{dx}\right) = \pm \sqrt{y^3 + c_1}$$

When $x = -2, y = -1$ and $\left(\frac{dy}{dx}\right)_{x=-2} = -1$

$$\therefore c_1 = 2$$

Thus, $\left(\frac{dy}{dx}\right) = \pm \sqrt{y^3 + 2}$

$$\Rightarrow \frac{dy}{\sqrt{y^3 + 2}} = \pm dx$$

227. The given differential equation is

$$y^3 \left(\frac{d^2y}{dx^2}\right) = -1$$

$$\Rightarrow \frac{d^2y}{dx^2} = -\frac{1}{y^3}$$

$$\Rightarrow 2p \frac{dp}{dy} = -\frac{1}{y^3}$$

$$\Rightarrow 2p dp = -\frac{dy}{y^3}$$

Integrating, we get

$$p^2 = -\frac{1}{2y^2} + c_1$$

$$\Rightarrow \left(\frac{dy}{dx}\right)^2 = -\frac{1}{2y^2} + c_1$$

$$\Rightarrow \left(\frac{dy}{dx}\right)^2 = \frac{2c_1 y^2 - 1}{2y^2}$$

When $x = -1, y = -1, \left(\frac{dy}{dx}\right)_{x=-1} = 0$

$$\therefore c_1 = \frac{1}{2}$$

$$\therefore \left(\frac{dy}{dx}\right) = \pm \sqrt{\frac{y^2 - 1}{2y^2}}$$

$$\Rightarrow \frac{y dy}{\sqrt{y^2 - 1}} = \pm \frac{dx}{2}$$

$$\Rightarrow \sqrt{y^2 - 1} = c_2 \pm \frac{x}{2}$$

which is the required solution.

228. The given differential equation is

$$x \frac{d^2 y}{dx^2} + \frac{dy}{dx} + x = 0 \quad \dots(i)$$

$$\text{Let } \frac{dy}{dx} = p \Rightarrow \frac{d^2 y}{dx^2} = \frac{dp}{dx}$$

$$\Rightarrow x \frac{dp}{dx} + p + x = 0$$

$$\Rightarrow x dp + p dx + x dx = 0$$

$$\Rightarrow d(xp) + x dx = 0$$

Integrating, we get

$$xp = -\frac{x^2}{2} + c_1$$

$$\Rightarrow x \frac{dy}{dx} = -\frac{x^2}{2} + c_1$$

$$\Rightarrow \frac{dy}{dx} = -\frac{x}{2} + \frac{c_1}{x}$$

$$\Rightarrow dy = \left(-\frac{x}{2} + \frac{c_1}{x}\right) dx$$

Again Integrating, we get

$$y = -\frac{x^2}{4} + c_1 \log|x| + c_2$$

which is the required solution.

229. The given differential equation is

$$\frac{d^2 y}{dx^2} = \frac{1}{x} \frac{dy}{dx} + x$$

$$\Rightarrow \frac{dp}{dx} = \frac{p}{x} + x$$

$$\Rightarrow x dp = p dx + x^2 dx$$

$$\Rightarrow \frac{x dp - p dx}{x^2} = dx$$

$$\Rightarrow d\left(\frac{p}{x}\right) = dx$$

Integrating, we get

$$\frac{p}{x} = \frac{x^2}{2} + c_1$$

$$\Rightarrow p = \frac{x^3}{2} + c_1 x$$

$$\Rightarrow \frac{dy}{dx} = \frac{x^3}{2} + c_1 x$$

$$\Rightarrow dy = \left(\frac{x^3}{2} + c_1 x\right) dx$$

Again integrating, we get

$$y = \left(\frac{x^4}{6} + \frac{c_1 x^2}{2}\right) + c_2$$

which is the required solution.

230. The given differential equation is

$$y \frac{d^2 y}{dx^2} + \left(\frac{dy}{dx}\right)^2 = 1$$

$$\Rightarrow yp \frac{dp}{dy} + p^2 = 1$$

$$\Rightarrow yp \frac{dp}{dy} = 1 - p^2$$

$$\Rightarrow \frac{p}{1 - p^2} dp = \frac{dy}{y}$$

Integrating, we get

$$-\frac{1}{2} \log|1 - p^2| = \log y + \log c_1$$

$$\Rightarrow \frac{1}{1 - p^2} = (c_1 y)^2$$

$$\Rightarrow 1 - p^2 = \frac{1}{c_1^2 y^2}$$

$$\Rightarrow p^2 - 1 = \frac{1}{c_1^2 y^2}$$

$$\Rightarrow \frac{dy}{dx} = \sqrt{1 - \frac{1}{c_1^2 y^2}}$$

$$\Rightarrow \frac{c_1 y}{\sqrt{c_1^2 y^2 - 1}} dy = dx$$

Again integrating, we get

$$\frac{\sqrt{c_1^2 y^2 - 1}}{c_1^2} = x + c_2$$

which is the required solution.

231. The given differential equation is

$$y \frac{d^2 y}{dx^2} - \left(\frac{dy}{dx}\right)^2 = y^2 \left(\frac{dy}{dx}\right)$$

$$\Rightarrow py \frac{dp}{dy} - p^2 = y^2 p$$

$$\Rightarrow y \frac{dp}{dy} - p = y^2$$

$$\Rightarrow y dp - p dy = y^2 dy$$

$$\Rightarrow \frac{y dp - p dy}{y^2} = dy$$

$$\Rightarrow d\left(\frac{p}{y}\right) = dy$$

Integrating, we get

$$\begin{aligned}\frac{p}{y} &= y + c_1 \\ \Rightarrow p &= y^2 + c_1 y \\ \Rightarrow \frac{dy}{dx} &= y^2 + c_1 y \\ \Rightarrow \frac{dy}{y^2 + c_1 y} &= dx \\ \Rightarrow \frac{dy}{y(y + c_1)} &= dx \\ \Rightarrow \left(\frac{1}{y} - \frac{1}{y + c_1} \right) dy &= dx\end{aligned}$$

Again integrating, we get

$$\log \left| \frac{y}{y + c_1} \right| = x + c_2$$

which is the required solution.

232. Given differential equation is

$$\begin{aligned}(x + a) \frac{d^2 y}{dx^2} + x \left(\frac{dy}{dx} \right)^2 &= \left(\frac{dy}{dx} \right) \\ \Rightarrow (x + a) \frac{dp}{dx} + xp^2 &= p \\ \Rightarrow \frac{dp}{dx} + \frac{xp^2}{(x + a)} &= \frac{p}{x + a} \\ \Rightarrow \frac{dp}{dx} - \frac{p}{x + a} &= -\frac{xp^2}{(x + a)} \\ \Rightarrow \frac{1}{p^2} \frac{dp}{dx} - \frac{1}{p(x + a)} &= -\frac{x}{(x + a)} \\ \text{Put } -\frac{1}{p} &= v \\ \Rightarrow \frac{dv}{dx} + \frac{v}{(x + a)} &= \frac{x}{(x + a)}\end{aligned}$$

which is a linear differential equation.

$$\therefore \text{IF} = e^{\int \frac{dx}{x+a}} = e^{\log|x+a|} = (x + a)$$

$$\begin{aligned}\Rightarrow v(x + a) &= \int x dx \\ \Rightarrow v(x + a) &= \frac{x^2}{2 + c} \\ \Rightarrow -\frac{1}{p}(x + a) &= \frac{x^2}{2} + c \\ \Rightarrow \frac{2(x + a)}{x^2 + C_1} dx + dy &= 0 \\ \Rightarrow \frac{2x}{x^2 + C_1} dx + \frac{2a}{x^2 + C_1} dx + dy &= 0\end{aligned}$$

Integrating, we get

$$\log|x^2 + c_1| + 2a \tan^{-1} \left(\frac{x}{c_1} \right) + y = c_2$$

which is the required solution.

233. Given differential equation is

$$\begin{aligned}x \frac{d^2 y}{dx^2} - \frac{1}{4} \left(\frac{dy}{dx} \right)^2 &= \left(\frac{dy}{dx} \right) \\ \Rightarrow x \frac{dp}{dx} - \frac{1}{4} p^2 &= p \\ \Rightarrow x \frac{dp}{dx} = \frac{p^2}{4} + p &= p \left(\frac{p}{4} + 1 \right) \\ \Rightarrow \frac{4 dp}{p(p + 4)} &= \frac{dx}{x} \\ \Rightarrow \left(\frac{1}{p} - \frac{1}{p + 4} \right) dp &= \frac{dx}{x}\end{aligned}$$

Integrating, we get

$$\begin{aligned}\log \left| \frac{p}{p + 4} \right| &= \log|x| + \log c_1 \\ \Rightarrow \frac{p}{p + 4} &= c_1 x \\ \Rightarrow \frac{p + 4}{p} &= \frac{1}{c_1 x} \\ \Rightarrow 1 + \frac{4}{p} &= \frac{1}{c_1 x} \\ \Rightarrow \frac{4}{p} &= \frac{1}{c_1 x} - 1 \\ \Rightarrow \frac{4}{p} &= \frac{1 - c_1 x}{c_1 x} \\ \Rightarrow \frac{p}{4} &= \frac{c_1 x}{1 - c_1 x} \\ \Rightarrow dy &= \frac{4c_1 x}{1 - c_1 x} dx\end{aligned}$$

Again integrating, we get

$$y = -4x + \frac{1}{c_1} \log|c_1 x - 1| + c_2$$

which is the required solution.

234. Given differential equation is

$$\begin{aligned}y \frac{d^2 y}{dx^2} &= 1 - \left(\frac{dy}{dx} \right)^2 \\ \Rightarrow py \frac{dp}{dx} &= 1 - p^2 \\ \Rightarrow \frac{2p dp}{(p^2 - 1)} &= -2 \frac{dy}{y}\end{aligned}$$

Integrating, we get

$$\log|p^2 - 1| = \log C_1 - \log y^2$$

$$\Rightarrow \log|p^2 - 1| = \log \left| \frac{C_1}{y^2} \right|$$

$$\Rightarrow (p^2 - 1) = \frac{C_1}{y^2}$$

$$\Rightarrow p^2 = \frac{C_1}{y^2} + 1$$

$$= \frac{C_1 + y^2}{y^2}$$

$$\Rightarrow p = \pm \sqrt{\frac{C_1 + y^2}{y^2}}$$

$$\Rightarrow \frac{dy}{dx} = \pm \sqrt{\frac{C_1 + y^2}{y^2}}$$

$$\Rightarrow \frac{y dy}{\sqrt{y^2 + C_1}} = \pm dx$$

$$\Rightarrow \sqrt{y^2 + C_1} = C_2 \pm x$$

which is the required solution.

235. Given differential equation is

$$\frac{d^2y}{dx^2} = 2y \left(\frac{dy}{dx} \right)$$

$$\Rightarrow \frac{dp}{dx} = 2yp$$

$$\Rightarrow p \frac{dp}{dy} = 2yp$$

$$\Rightarrow \frac{dp}{dy} = 2y$$

$$\Rightarrow dp = 2y dy$$

Integrating, we get

$$p = y^2 + C_1$$

$$\Rightarrow \frac{dy}{dx} = y^2 + C_1$$

$$\Rightarrow \frac{dy}{(y^2 + C_1)} = dx$$

$$\Rightarrow \frac{1}{\sqrt{C_1}} \tan^{-1} \left(\frac{y}{\sqrt{C_1}} \right) = x + C_2$$

which is the required solution.

236. Given differential equation is

$$\Rightarrow y \frac{d^2y}{dx^2} - \left(\frac{dy}{dx} \right)^2 = y^2 \left(\frac{dy}{dx} \right)$$

$$\Rightarrow yp \frac{dp}{dy} - p^2 = y^2 p$$

$$\Rightarrow y \frac{dp}{dy} - p = y^2$$

$$\Rightarrow \frac{dp}{dy} - \frac{p}{y} = y \quad \dots(i)$$

which is a linear differential equation.

$$\therefore \text{IF} = e^{-\int \frac{dy}{y}} = e^{-\log y} = \frac{1}{y}$$

Multiplying both sides of Eq. (i) by IF and integrating, we get

$$\frac{p}{y} = \int dy + c_1$$

$$\frac{p}{y} = y + c_1$$

$$p = y^2 + c_1 y$$

$$\frac{dy}{dx} = y^2 + c_1 y$$

$$\frac{dy}{y(y + c_1)} = dx$$

Integrating, we get

$$\frac{1}{c_1} \log \left| \frac{y}{y + c_1} \right| = x + c_2$$

which is the required solution.

237. Given differential equation is

$$\frac{d^2y}{dx^2} + \frac{1}{x} \left(\frac{dy}{dx} \right) - \frac{a}{x^2} = 0$$

$$\Rightarrow x^2 \frac{d^2y}{dx^2} + x \left(\frac{dy}{dx} \right) = a$$

$$\Rightarrow x^2 \frac{dp}{dx} + xp = a$$

$$\Rightarrow x dp + p dx = \frac{a dx}{x}$$

$$\Rightarrow d(xp) = \frac{a dx}{x}$$

Integrating, we get

$$xp = a \log |x| + c_1$$

$$\Rightarrow x \frac{dy}{dx} = a \log |x| + c_1$$

$$\Rightarrow dy = \left(\frac{a \log |x|}{x} + \frac{c_1}{x} \right) dx$$

Again integrating, we get

$$y = a \frac{(\log |x|)^2}{2} + C_1 \log |x| + C_2$$

which is the required solution.

238. Given differential equation is

$$\begin{aligned}
 & y \frac{d^2 y}{dx^2} - y \left(\frac{dy}{dx} \right) \ln y = \left(\frac{dy}{dx} \right)^2 \\
 \Rightarrow & y \frac{dp}{dx} - yp \ln y = p^2 \\
 \Rightarrow & \frac{1}{p^2} \frac{dp}{dx} + \frac{\ln y}{-p} = \frac{1}{y} \\
 \Rightarrow & yp \frac{dp}{dy} - yp \ln y = p^2 \\
 \Rightarrow & y \frac{dp}{dy} - y \ln y = p \\
 \Rightarrow & \frac{dp}{dy} - \ln y = \frac{p}{y} \\
 \Rightarrow & \frac{dp}{dy} - \frac{p}{y} = \ln y \quad \dots(i)
 \end{aligned}$$

which is a linear differential equation.

$$\therefore \text{IF} = e^{-\int \frac{dy}{y}} = e^{-\log y} = \frac{1}{y}$$

Multiplying both sides of Eq. (i) by IF and integrating, we get

$$\begin{aligned}
 & \frac{p}{y} = \int \frac{\log y}{y} dy + c_1 \\
 \Rightarrow & \frac{p}{y} = \frac{(\log y)^2}{2} + c_1 \\
 \Rightarrow & \frac{dy}{y \left(\frac{(\log y)^2}{2} + 2c_1 \right)} = dx \\
 \Rightarrow & \frac{2 dt}{(t^2 + 2c_1)} = dx, \quad (\text{Let } \log y = t)
 \end{aligned}$$

Integrating, we get

$$\begin{aligned}
 \Rightarrow & \frac{2}{\sqrt{2c_1}} \tan^{-1} \left(\frac{t}{\sqrt{2c_1}} \right) = x + c_2 \\
 \Rightarrow & \sqrt{\frac{2}{c_1}} \tan^{-1} \left(\frac{\log y}{\sqrt{2c_1}} \right) = x + c_2
 \end{aligned}$$

which is the required solution.

239. The slope of the curve

$$\begin{aligned}
 & = \text{The slope of the tangent} \\
 & = \tan \psi = \frac{dy}{dx} = \frac{1}{2y} \\
 \Rightarrow & 2y dy = dx \\
 \Rightarrow & y^2 = x + c
 \end{aligned}$$

which is passing through (4, 3), so $c = 5$

Hence the equation of the curve is

$$y^2 = x + 5.$$

240. Given

$$\begin{aligned}
 & \frac{dy}{dx} = y + 2x \\
 \Rightarrow & \frac{dy}{dx} + (-y) = 2x
 \end{aligned}$$

which is a linear differential equation.

$$\text{IF} = e^{-\int dx} = e^{-x}$$

Hence, the solution is

$$\begin{aligned}
 & y \cdot e^{-x} = 2 \int x e^{-x} dx + c \\
 \Rightarrow & y \cdot e^{-x} = -2e^{-x} (x + 1) + c
 \end{aligned}$$

which is passing through (0, 0) then $c = 0$.

Hence, the required equation of the curve is

$$y \cdot e^{-x} = -2e^{-x} (x + 1)$$

241. Given

$$\begin{aligned}
 & \frac{dy}{dx} = \frac{x^4 + 2xy - 1}{x^2 + 1} \\
 & = x^2 - 1 + \frac{2xy}{x^2 + 1} \\
 \Rightarrow & \frac{dy}{dx} - \frac{2x}{x^2 + 1} y = x^2 - 1
 \end{aligned}$$

which is a linear differential equation.

$$\text{IF} = e^{-\int \frac{2x}{x^2 + 1} dx} = e^{-\log(x^2 + 1)} = \frac{1}{x^2 + 1}$$

Hence, the solution is

$$y \cdot \frac{1}{x^2 + 1} = x - 2 \tan^{-1} x + c$$

which passes through (0, 0), then $c = 0$.

Hence, the equation of the curve is

$$y = (x^2 + 1)(x - 2 \tan^{-1} x)$$

242. Given

$$\frac{dy}{dx} = \frac{(x+1)^2 + y - 3}{(x+1)} \quad \dots(i)$$

$$\text{Let } x+1 = X, y-3 = Y$$

$$\Rightarrow \frac{dy}{dx} = \frac{dY}{dX}$$

$$\Rightarrow \frac{dY}{dX} = X + \frac{Y}{X}$$

$$\Rightarrow \frac{dY}{dX} + \left(-\frac{1}{X} \right) Y = X$$

which is a linear differential equation.

$$\therefore \text{IF} = e^{-\int \frac{1}{X} dX} = e^{-\log X} = \frac{1}{X}$$

Hence, the solution is

$$Y \cdot \frac{1}{X} = \int \left(X \cdot \frac{1}{X} \right) dX + c = X + c$$

$$\Rightarrow \frac{Y}{X} = X + c$$

$$\Rightarrow \frac{y-3}{x+1} = (x+1) + c$$

which is passing through (2, 0) then $c = -4$

Hence, the required equation of the curve is

$$\frac{y-3}{x+1} = (x+1) - 4$$

$$\Rightarrow y-3 = (x+1)^2 - 4(x+1)$$

$$\Rightarrow y = x^2 - 2x$$

243. The equation of the tangent at any point (x, y) on the curve is

$$Y - y = \frac{dy}{dx}(X - x)$$

It meets x-axis at $A\left(x - y \frac{dy}{dx}, 0\right)$

and y-axis at $B\left(0, y - x \frac{dy}{dx}\right)$

$\therefore$ Mid-point of AB

$$= \left(\frac{1}{2} \left(x - y \frac{dy}{dx} \right), \frac{1}{2} \left(y - x \frac{dy}{dx} \right) \right)$$

It is given that,

$$\text{and } \frac{1}{2} \left(x - y \frac{dy}{dx} \right) = x,$$

$$\frac{1}{2} \left(y - x \frac{dy}{dx} \right) = y$$

$$\text{Thus, } x \frac{dy}{dx} = -y$$

$$\Rightarrow \frac{dy}{y} = -\frac{dx}{x}$$

$$\Rightarrow \frac{dy}{y} + \frac{dx}{x} = 0$$

$$\Rightarrow \log y + \log x = \log c$$

$$\Rightarrow \log(yx) = \log c$$

$$\Rightarrow xy = c$$

which is passing through (1, 1), then $c = 1$.

Hence, the equation of the curve is

$$xy = 1.$$

244. The equation of the normal at the point (x, y) is

$$Y - y = -\frac{dx}{dy}(X - x) \quad \dots(i)$$

The distance of perpendicular from the origin to the normal (i) is

$$= \frac{\left| y + x \frac{dy}{dx} \right|}{\sqrt{1 + \left(\frac{dy}{dx} \right)^2}}$$

Also the distance between P and x-axis is |y|.

$$\text{Thus, } \frac{\left| y + x \frac{dy}{dx} \right|}{\sqrt{1 + \left(\frac{dy}{dx} \right)^2}} = |y|$$

$$\Rightarrow y^2 + \left(\frac{dx}{dy} \right)^2 x^2 + 2xy \frac{dy}{dx} = y^2 \left(1 + \left(\frac{dy}{dx} \right)^2 \right)$$

$$\Rightarrow (x^2 - y^2) \left(\frac{dx}{dy} \right)^2 + 2xy \frac{dx}{dy} = 0$$

$$\Rightarrow \frac{dx}{dy} \left((x^2 - y^2) \left(\frac{dx}{dy} \right) + 2xy \right) = 0$$

$$\Rightarrow \frac{dx}{dy} = 0, \left(\frac{dx}{dy} \right) = \frac{y^2 - x^2}{2xy}$$

$$\text{Now, } \frac{dx}{dy} = 0 \text{ gives } x = k$$

which is passing through (1, 1), so $k = 1$.

Thus, the equation of the curve is $x = 1$

$$\text{Also, } \frac{dx}{dy} = 0 = \frac{y^2 - x^2}{2xy}$$

which is a homogeneous differential equation.

$$\text{Let } y = vx \Rightarrow \frac{dy}{dx} = v + x \frac{dv}{dx}$$

$$\Rightarrow v + x \frac{dv}{dx} = \frac{v^2 - 1}{2v}$$

$$\Rightarrow x \frac{dv}{dx} = \frac{v^2 - 1}{2v} - v = -\frac{v^2 + 1}{2v}$$

$$\Rightarrow \frac{2v}{v^2 + 1} dv = -\frac{dx}{x}$$

$$\Rightarrow \log |v^2 + 1| = \log c - \log x$$

$$\Rightarrow |v^2 + 1| = \frac{c}{x}$$

$$\Rightarrow x^2 + y^2 = cx$$

which also passes through (1, 1), so $c = 2$.

Hence, the equation of the curve is $x^2 + y^2 = 2x$.

245. The Equation of the tangent at any point (x, y) is

$$Y - y = \frac{dy}{dx}(X - x)$$

It meets the x-axis at $\left(x - y \frac{dy}{dx}, 0 \right)$

$$\text{Given } AP = 1$$

$$\Rightarrow AP^2 = 1$$

$$\Rightarrow \left(-y \frac{dx}{dy}\right)^2 + y^2 = 1$$

$$\Rightarrow \left(\frac{dx}{dy}\right)^2 = \frac{1-y^2}{y^2}$$

$$\Rightarrow \frac{dy}{dx} = \pm \frac{y}{\sqrt{1-y^2}}$$

$$\Rightarrow \frac{\sqrt{1-y^2}}{y} dy = \pm dx$$

Integrating, we get

$$\log \left| \frac{y}{1+\sqrt{1-y^2}} \right| + \sqrt{1-y^2} = c \pm x$$

which is the required equation of the curve.

246. The slope of the line segment joining the points (x, y) and $(-4, -3)$ is

$$\frac{y+3}{x+4}$$

Thus, $\frac{dy}{dx} = 2 \left(\frac{y+3}{x+4} \right)$

$$\Rightarrow \frac{dy}{y+3} = \frac{2 \cdot dx}{x+4}$$

Integrating, we get

$$\log|y+3| = \log c + 2 \log(x+4)$$

$$\Rightarrow (y+3) = c(x+4)^2$$

which is passing through $(-2, 1)$, so $c = 1$

Hence, the required equation of the curve is

$$(y+3) = (x+4)^2$$

247. The equation of the normal at P is

$$Y - y = \frac{dx}{dy}(X - x)$$

Thus, the points A and B are $(x, 0)$ and

$$\left(x + y \frac{dy}{dx}, 0\right)$$

Given condition is

$$x + y \frac{dy}{dx} - x = \frac{x+y}{2}$$

$$\frac{dy}{dx} = \frac{x+y}{2y}$$

which is a homogeneous equation.

$$\text{Put } y = vx \Rightarrow \frac{dy}{dx} = v + x \frac{dv}{dx}$$

$$\Rightarrow v + x \frac{dv}{dx} = \frac{1+v}{2v}$$

$$\Rightarrow x \frac{dv}{dx} = \frac{1+v}{2v} - v = \frac{1+v-2v^2}{2v}$$

$$\Rightarrow \frac{2v}{1+v-2v^2} dv = \frac{dx}{x}$$

$$\Rightarrow \frac{2v}{2v^2-v-1} dv = -\frac{dx}{x}$$

$$\Rightarrow \frac{(4v-1)+1}{2v^2-v-1} dv = -\frac{2dx}{x}$$

$$\Rightarrow \frac{(4v-1)dv}{(2v^2-v-1)} + \frac{dv}{(2v^2-v-1)} = -\frac{2dx}{x}$$

$$\Rightarrow \log|2v^2-v-1| + \frac{1}{3} \log \left| \frac{2v-2}{2v-1} \right| = c - \log x$$

$$\Rightarrow \log \left| \frac{2y^2}{x^2} - \frac{y}{x} - 1 \right| + \frac{1}{3} \log \left| \frac{2y-2x}{2y-x} \right| = c - \log x$$

$$\Rightarrow \log|2y^2 - xy - x^2| + \frac{1}{3} \log \left| \frac{2y-2x}{2y-x} \right| - \log x = c$$

which is the required equation of the curve.

248. The equation of the normal at P is

$$Y - y = -\frac{dx}{dy}(X - x)$$

It meets the x -axis at $A \left(x + y \frac{dy}{dx}, 0 \right)$

Given condition is

$$OP^2 = OA^2$$

$$\Rightarrow x^2 + y^2 = y^2 \left(\frac{dy}{dx} \right)^2 + y^2$$

$$\Rightarrow \frac{dy}{dx} = \pm \frac{x}{y}$$

Taking positive sign,

$$\frac{dy}{dx} = \frac{x}{y}$$

$$\Rightarrow y dy = x dx$$

$$\Rightarrow \frac{x^2}{2} - \frac{y^2}{2} = \frac{c^2}{2}$$

$$\Rightarrow x^2 - y^2 = c^2,$$

which represents a rectangular hyperbola.

Taking negative sign,

$$\frac{dy}{dx} = -\frac{x}{y}$$

$$\Rightarrow y dy = -x dx$$

$$\Rightarrow \frac{x^2}{2} + \frac{y^2}{2} = \frac{a^2}{2}$$

$$\Rightarrow x^2 + y^2 = a^2.$$

which represents a circle.